

ATC (Additional Terms and Conditions)

FOR

**Design, Development and Implementation
Of
Artificial Intelligence (AI) Based Detailed Project Report
Appraisal Platform**



**(A Government of India Enterprise)
RITES Bhawan
Plot No. 1, Sector 29, Gurgaon, Haryana - 122001**

Tender No.: - RITES/ITES/APR/2026

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1. Invitation for Proposal

1.1.Acronyms

QCBS	Quality and Cost-Based Selection
DSC	Digital Signature Certificates
EMD	Earnest Money Deposit
ECS	Electronic Credit System
EFT	Electronic Fund transfer
GST	Goods Services and Tax
OEM	Original Equipment Manufacturer
NEFT	National electronic funds transfer
SLA	Service Level Agreement
TOF	Tender Offer Form

1.2.Tender Summary

RITES invites e-bids through GeM for Appointment of Service Provider for **Design, Development and Implementation of Artificial Intelligence (AI) Based Detailed Project Report Appraisal Platform.**

Requirement/eligibility criteria/Terms and conditions of the contract have been clarified in the tender document, additional documents (if any) and Annexures 1 to 11. Tender document (including additional documents and Annexures) is available online at GeM portal as well as on the RITES Ltd website i.e., www.rites.com. Bidders are advised to read tender document and related Corrigendum, if any, (uploaded on website and on GeM portal) and check their eligibility before participating in the bid.

The interested bidders should upload their bids along with duly signed scanned copies of all the relevant certificates, documents etc. in support of their technical & price bids on the www.gem.gov.in

1.3.Bid Schedule and Address

Sr.No.	Description	Detailed Information
1	Name of Project	Design, Development and Implementation Of AI Based Detailed Project Report Appraisal Platform.
2	Tender Reference Number	RITES/ITES/APR/2026
3	Tender Sales Start Date & Time	As per GEM Portal
4	Bidding End Date & Time (on-line submission)	As per GEM Portal
5	Bid Opening Date & Time	As per GEM Portal
6	Name and Address for Communication	Mr. Tanuj Sukhija (JGM/IT) RITES Ltd, RITES Bhawan, Plot No. 1, Sector 29, Gurugram, Haryana 122001 Email: tanujsukhija@rites.com
7	Bid Related Queries	1. Through online queries over GeM portal 2. Pre bid meeting: Date, time and venue as mentioned for this bid over GeM portal. 3. The Pre-Bid Meeting shall be conducted in hybrid mode. The online meeting link for participation will be published on the RITES website. 4. Over email : Mr. Tanuj Sukhija (JGM/IT) Email: tanujsukhija@rites.com Any query by bidders can be raised till : bid published date + 5 days (refer GEM for exact dates)
8	Cost of Tender Document (Non Refundable)	NA
9	EMD/Bid Security	Rs. 18,00,000 /- (Rupees Eighteen Lakhs only) in the form as mentioned under clause 5.9 (vii)

1.4.Important Information

- i. It is advised to study the Tender document carefully. Submission of bid shall be deemed to have been done after careful study and examination of the tender document with full understanding of its implications.
- ii. The cost of tender document is not applicable.
- iii. The response to this Tender should be full and complete in all respects. The bids without EMD (or exemption documents) or Incomplete or partial bid shall be rejected. Bidder must quote for all the items asked for in this Tender.
- iv. Bidder shall bear all costs associated with the preparation and submission of the bid, including cost of presentation for the purposes of clarification of the bid, if so desired by RITES. RITES will in no case be responsible or liable for those costs, regardless of the conduct or outcome of the bidding process.
- v. **No bidding through joint-ventures or consortium is allowed.**
- vi. RITES reserves the right to change the dates mentioned in table at para 1.3 above. Changes/Amendments/Corrigendum if any, related to the tender document will be posted on GeM Portal. Bidders in their own interest must check the website before submitting the bids.
- vii. The Bidder shall qualify the **Eligibility criteria** as given in **Section 4** of tender document and the bid document submitted should have all the supporting documents as mentioned in the tender document. The checklist of documents to be uploaded along with the bid is provided at Annexure-09 and those to be submitted in Original are provided at Annexure-08.
Documents/information complete in all respects, in support of meeting the Qualification Criteria should be submitted in one go. Submission of additional documents shall not be permitted. Only clarifications and filling of gaps/missing information in the submitted documents, may be permitted. The documents to be furnished by the Bidder to prove that he/she is satisfying the qualification Criteria laid down should all be in the Bidder's name, except in cases where though the name has changed, the owners continued to remain the same and in cases of amalgamation of entities.
- viii. **In case of any discrepancy in dates and times mentioned in tender document/ website then dates and times mentioned in GeM portal shall prevail.**
- ix. RITES reserves the right to accept or reject the bid without assigning any reason thereof.
- x. The present contract is for the Design, Development and Implementation Of AI Based Detailed Project Report Appraisal Platform.

- xi. Bids should be submitted **online** in the prescribed format given at GeM portal.
No other mode of bid submission is acceptable.

GGM/ITS

rites bhawan, shikhar
Plot No. 1, Sector 29, Gurgaon
Haryana 122001

2. General Information

2.1.About RITES Ltd.

RITES Ltd., a Government of India Enterprise was established in 1974, under the aegis of Indian Railways. RITES Ltd. is incorporated in India as a Public Limited Company under the Companies Act, 1956 and is governed by a Board of Directors which includes persons of eminence from various sectors of engineering and management.

RITES Ltd., is a multi-disciplinary consultancy organization in the field of transport, infrastructure and related technologies. It provides a comprehensive array of services under a single roof and believes in transfer of technology to client organizations. In overseas projects, RITES Ltd. actively pursues and develops cooperative links with local consultants / firms, as means of maximum utilization of local resources and as an effective instrument of sharing its expertise. RITES Ltd enjoys Navratna status and has ISO 9001:2008 Certification for its entire range of services.

With work experience in more than sixty countries, RITES Ltd. is today established as a leading Consultancy Organization in the international arena and recognized by multi-lateral funding agencies. RITES Ltd. now stands as one of the largest technical consulting firms in India having presence in a wide range of sectors and offering a variety of services under a single roof. RITES Ltd. client includes government agencies in India and abroad as well as top industrial groups in the country.

2.2.Overview Of The Present Request For Proposal (RFP)

Detailed Project Reports form the foundation of project planning and investment decisions. The quality of a DPR directly influences project cost, timelines, risk exposure, and long-term sustainability. However, current DPR appraisal practices remain largely manual, fragmented, and document-centric, relying on checklists, spreadsheets, and siloed workflows. These approaches often lead to incomplete inputs, inconsistencies across sections, limited traceability, and weak institutional memory.

The AI based DPR APPRAISAL PLATFORM is envisaged to address these challenges by introducing a **structured, integrated, and lifecycle-driven digital environment** for DPR preparation, validation, evaluation, appraisal, and knowledge management.

At its core, the platform transforms DPR appraisal from a **document review exercise** into a **data-driven, intelligence-enabled decision-support system**. It integrates structured data capture, engineering logic, geospatial intelligence, and historical benchmarking within a unified architecture supported by a centralized Data Lake.

The platform shall be designed as a **closed-loop system**, where insights from validation, appraisal, and benchmarking are continuously fed back into DPR preparation. This ensures progressive improvement in DPR quality, consistency, and decision-making over time.

Artificial Intelligence and advanced analytics are to be incorporated as **assistive and advisory components**, supporting anomaly detection, benchmarking, and pattern

recognition, while ensuring that final decisions remain with designated experts and authorities.

The DPR APPRAISAL PLATFORM is not intended to replace expert judgment. Instead, it should enhance transparency, consistency, and efficiency in appraisal processes while enabling organization to transition from **experience-based evaluation to evidence-based, data-driven decision-making**.

3. Scope of Work

The scope of work includes the design, development, deployment, integration, and support of a **AI based DPR Appraisal Platform** to enable structured preparation, validation, appraisal, evaluation and continuous improvement of Detailed Project Reports (DPRs).

The platform shall be designed using virtualization and containerization principles to ensure optimal utilization of compute resources. The solution shall be lightweight, scalable, and optimized at application, database, and AI levels to avoid unnecessary infrastructure overheads.

The user interface shall be responsive and device-agnostic, supporting desktops, tablets (including iPad), and mobile devices.

AI/ML components shall be designed for efficiency and shall not be unduly resource-intensive unless justified and approved by RITES

The platform shall be developed as a scalable, modular, and data-driven system, aligned with the functional and technical requirements specified in **Annexure 12**.

The platform shall be designed to enable future ecosystem participation and integration with emerging national digital platforms without requiring major architectural changes. The platform shall be aligned with principles of Digital Public Infrastructure (DPI), including openness, interoperability, scalability, and vendor neutrality.

3.1. Platform Design and Architecture

The selected bidder shall:

- Design the overall system architecture in line with the principles outlined in Annexure 12
- Develop a modular, API-driven, and scalable platform architecture
- Establish a centralized data architecture, including Data Lake components
- Ensure integration readiness with external systems such as GIS platforms and national infrastructure systems
- The platform shall adopt standard API protocols such as REST / GraphQL / gRPC with OpenAPI-compliant documentation.
- The system shall support event-driven architecture (publish-subscribe / streaming) for real-time data exchange where required.
- The bidder shall provide API documentation, integration specifications, and interoperability compliance artifacts as part of deliverables.

3.2. Development of DPR Preparation and Authoring Capabilities

The bidder shall develop functionalities to:

- Enable structured preparation of DPRs using configurable templates
- Capture project, technical, financial, and site-specific data in structured formats
- Support document management, version control, and submission workflows
- Facilitate reuse of standard data, templates, and reference inputs

3.3. Development of Validation and Appraisal Engine

The bidder shall implement capabilities for:

- Automated completeness checks and validation of DPR inputs
- Benchmarking & risk assessment.
- Rule-based and dependency-driven validation across DPR sections
- Consistency and cross-validation of technical and financial parameters
- Detection of anomalies, inconsistencies, and outliers, gap identification & compliance checks
- Configurable scoring and appraisal frameworks

3.4. Data Management and Data Lake Implementation

The bidder shall:

- Design and implement a Data Lake for storage and management of structured and unstructured data
- Enable ingestion of historical DPRs and related datasets
- Support integration of external datasets including GIS, geotechnical, environmental, and socio-economic data
- Establish metadata, cataloging, and data governance mechanisms
- The platform shall implement canonical data models and standardized schemas to ensure semantic interoperability
- All data exchanges shall be metadata-driven and aligned to approved ontology frameworks
- Schema versioning, data lineage, and traceability shall be maintained across all integrations

3.5. AI Analytics, Decision Support, and Scenario Evaluation

The platform shall support:

- Generation of dashboards, reports, and decision-support dashboards.
- Benchmarking of DPR parameters using historical data
- Scenario analysis and evaluation of alternative project configurations
- Advanced analytics, including anomaly detection and pattern recognition

3.6. Integration with External Systems

The bidder shall ensure integration with:

- GIS platforms for spatial data visualization and analysis
- National platforms such as PM GatiShakti (where applicable)
- BIM and engineering systems (where required)
- Other enterprise systems through API-based integration
- The platform shall support secure authentication and authorization mechanisms including OAuth2.0 / SAML / token-based access
- The system shall expose APIs through a managed API gateway with logging, throttling, and access control

- A sandbox / developer environment shall be provided to enable third-party integrations and testing

3.7. AI and Advanced Analytics Enablement

The platform shall incorporate:

- Artificial Intelligence and advanced analytics for validation, benchmarking, and anomaly detection
- Assistive and explainable AI capabilities to support decision-making
- Mechanisms for continuous learning from historical DPR data

3.8. Deployment, Testing, and Commissioning

The bidder shall:

- Deploy the platform in the designated environment (cloud/on-premise)
- Conduct system integration testing, user acceptance testing (UAT), and performance validation
- Ensure security, scalability, and reliability of the system

3.9. Training, Documentation, and Change Management

The bidder shall:

- Provide training to users, administrators, and stakeholders
- Deliver comprehensive technical and user documentation
- Support adoption and change management processes

3.10. Operations and Maintenance Support

The bidder shall:

- Provide post-deployment support and maintenance for a minimum period as specified in the RFP
- Ensure system availability, issue resolution, and periodic updates
- Maintain all required licenses and subscriptions for the specified duration

3.11. Flexibility and Innovation

Bidders may propose enhancements, alternative approaches, or additional features consistent with the objectives and functional intent described in **Annexure 12**.

Prohibition on Use of AI Wrappers

The proposed DPR Evaluation Tool shall not be a mere “AI wrapper”, i.e., a superficial software layer built only by integrating third-party AI/LLM APIs or publicly available AI tools without substantial proprietary development, customization, domain logic, validation rules, workflow engine, and enterprise-grade controls.

The bidder shall ensure that the core solution is purpose-built for DPR appraisal and shall include bidder-developed/customized capabilities for document ingestion, parsing, data extraction, validation, benchmarking, workflow management, audit trail, reporting, and explainable outputs.

The use of generic AI/LLM APIs as the primary or sole mechanism for DPR evaluation shall not be permitted. Any use of third-party AI models, AI services, APIs, libraries, or platforms shall be clearly disclosed in the technical bid, along with details of data flow, hosting, security, licensing, customization, model governance, auditability, and compliance with applicable Government of India data security requirements.

The solution shall not transmit RITES' DPR data, project documents, drawings, financial data, or other sensitive information to any uncontrolled public AI platform or third-party environment without prior written approval of RITES. Non-disclosure of such dependencies or submission of a solution found to be merely an AI wrapper may lead to rejection of the bid and/or termination of the contract.

4. Eligibility Criteria

4.1 Eligibility Criterion (Fill Details in Annexure-09)

S. NO.	Parameter	Pre qualification Criterion	Document Required
1.	Age of the company	The Bidder should be a Company / Firm / Organisation and must be a registered legal entity in India as on the date of publication of this RFP. The Bidder should be engaged in providing software development services as required under this RFP.	Copy of the Certificate of Incorporation from the office of the Registrar of Companies / Certification of registration of the firm showing details pertaining to place of registration, principal place of business of the firm, nature of business etc. and Memorandum /Articles of Association / Partnership Deed / Proprietorship Deed/ Declaration of Proprietorship etc. as the case may be. (provide document details at D.1 in Annexure-09)
2	Financial Turnover	The Bidder should have a minimum average annual turnover of 4.5 Crores (Rupees Four Crores Fifty Lakhs only), with at least 50% derived from software development contracts , calculated as total certified payments received for contracts in progress or completed within the last 3 Financial years.i.e 2024-25,2023-24 and 2022-23	The Bidder should submit self-attested copy of Auditor's Report along with audited Balance Sheet and Profit and Loss Statement along with Schedules for the relevant Financial Years i.e 2024-25,2023-24 and 2022-23 .Provisional Audit Reports or certified statements will not be accepted. (provide details in Annexure 1: Financial Capability Statements and Annexure 9
3.	Tax registration	Bidder with valid registration regarding GSTIN, PAN	Copy of GST certificate Copy of PAN card

			(provide details in Annexure-09)
4	IT Software Development Experience	The bidder must have successfully completed or <i>substantially completed</i> (at least 80% payments received) at least One Similar work of minimum value ₹2 (Two) Crore* in the last 5 (Five) years. <i>Similar Work shall mean design, development, implementation, configuration and execution of software platforms/applications involving document-centric workflows, project evaluation/appraisal systems, or workflow-based enterprise applications with automation and multi-level approval mechanisms. The work should incorporate or demonstrate experience in advanced technologies such as Artificial Intelligence (AI), Machine Learning (ML), Generative AI (GenAI), Natural Language Processing (NLP), advanced analytics, Document Management Systems (DMS), and use of Large Language Models (LLMs) including prompt engineering.</i> <i>Note: During the RFP Process, Original assignment experience certificates may be called for to substantiate qualifications.</i> <i>1. Assignment experience certificate for completed and Substantial completed consultancy assignment issued by Govt. Organizations/Semi Govt. Organizations of Central or State Govt.; or by Public Sector Undertakings/Autonomous</i>	Copy of the documents as per details at table C in Annexure-09.

		<i>Bodies of Central/State Govt. or their subsidiaries/by Public Ltd. Companies listed in Stock exchange in India or Aboard or subsidiaries of such companies shall only be accepted.</i> <i>2.Substantial Completed Consultancy assignment shall mean whose at least 80% payment (excluding payment for adjustment of price variation (PVC),if any, and excluding GST) of the present value of assignment has been received.</i>	
5	Blacklisting	The bidder should not have been blacklisted by Central/ State Government/ PSU in India at the time of submission of the Bid	Self-declaration by Bidder (Self-Declaration as per Annexure-05) (provide details at D.5 in Annexure-09)
6	Certification	At least one (1) employee must hold a relevant professional certification such as ITIL /Agile/ CISSP/ or certifications in Artificial Intelligence (AI) / Machine Learning (ML) Data Science, or equivalent	Valid Certifications
7	Acceptance	Bidder must accept the terms and conditions of the tender document without any deviation.	The Bidder shall submit the tender offer form duly signed by authorized signatory as per Annexure-04.

*Price Excluding GST/Taxes/

Note:- The terms AI includes all types of AI and its and its evolving forms.

5. Instructions to the Bidder

5.1.Language of Proposals

- i. The proposal and all correspondence and documents shall be written in English. All proposals and accompanying documentation will become the property of RITES LTD and will not be returned.

5.2.Due Diligence

- i. The Bidders are expected to carefully examine all instructions, the Tender document, the terms and specifications stated in this Tender, and if there appears to be any ambiguity, contradictions, inconsistency, gap and/or discrepancy in the Tender document, the Bidders are requested to seek necessary clarifications.
- ii. The bid shall be deemed to have been submitted after careful study and examination of this Tender document. The bid should be precise, complete and in the prescribed format as per the requirement of this Tender document.
- iii. Failure to furnish all information or submission of a bid not responsive to this Tender will be at the Bidders risk and may result in rejection of the bid. Also the grounds for rejection of bid should not be questioned after the final declaration of the successful Bidder.

5.3.Period of Validity of Bids

- i. The bids shall be valid for a period of 90 days from the date of opening of the bid. A bid valid for a shorter period may be rejected as non-responsive. Till completion of the validity period, unless the Bidder formally withdraws his bid in writing, it will be deemed to be valid.
- ii. In exceptional circumstances, at its discretion, RITES Ltd may solicit the Bidder's consent to an extension of the period of validity. The request and the responses thereto shall be made in writing /e-mail. The Bidder may refuse the request without forfeiting his Earnest Money. A Bidder agreeing to the request will not be permitted to modify his Bid but will be required to extend the validity of the Earnest Money for the period of the extension.

5.4.Right to Terminate the Process

- i. RITES Ltd. may terminate the Tender process at any time and without assigning any reason. RITES Ltd. makes no commitments, express or implied, that this process will result in a business transaction with anyone.

5.5.Clarifications On Tender Documents

- i. A prospective bidder requiring any clarification on the Tender Document may notify through queries, online only within the specified period.
- ii. The bidder may seek clarification within the specified period only. The identity of bidder will not be disclosed by the system. RITES Ltd. will clarify the relevant queries of bidders as far as possible. The clarifications given will be visible to all the bidders intending to participate in that tender. The clarifications may be asked from the day of "*Pre Bid Clarification Start Date & Time*" till "*Pre Bid Clarification Receipt End Date & Time*" as specified in table at para 1.3 in

this document.

- iii. RITES Ltd. makes no representation or warranty as to the completeness or accuracy of any response made in good faith.
- iv. Request for clarifications including request for Extension of Time for submission of Bid, if any, must be received not later than 10 (ten) days prior to the deadline for submission of tenders.

5.6.Amendment of Tender Document

- i. Before the deadline for submission of tenders, the Tender Document may be modified by RITES Ltd. by issue of addenda/corrigendum
- ii. Bidders are advised to see the GeM portal for addendum/corrigendum to the tender document which may be uploaded prior to the deadline for submission of tender as finally stipulated.
- iii. Addenda/corrigenda will be binding on Bidders. It will be assumed that the amendments contained in such Addenda/corrigenda had been taken into account by the Bidder while preparing its bid. To give prospective tenderers reasonable time in which to take the addenda/corrigenda into account in preparing their tenders, extension of the deadline for submission of tenders may be given before bid submission end date and time as considered necessary by RITES. Sometimes due to administrative reasons, the deadline for submission of tenders may be extended. All tenderers are advised to see the GeM portal for extension of deadline for submission of tenders.

5.7.Extension of Deadline for submission of Bids

- i. RITES Ltd. may, at its discretion, extend this deadline for the submission of bids by amending the Tender document, in which case all rights and obligations of RITES Ltd. and Bidder previously subject to the deadline will thereafter be subject to the deadline as extended.
- ii. In case the bid due date is extended due to any reason the same will be intimated through GeM portal. All bidders are advised to see the GeM portal for extension of deadline for submission of tenders.

5.8.Preparation of Bids

- i. Detailed tender document may be downloaded from GeM portal prior to the tender sales end date & time.
- ii. After downloading / getting the tender document/schedules, the Bidder should go through them carefully and then submit the documents as asked, otherwise bid will be rejected. It is construed that the bidder has read all the terms and conditions before submitting their offer.

- iii. Bidder is advised to go through the tender document carefully to understand the documents required to be submitted as part of the bid. Bidder may please note the number of covers in which the bid documents have to be submitted, the number of documents – including the names and content of each of the document that need to be submitted. Any deviations from these may lead to rejection of the bid.
- iv. Bidder should take into account any corrigendum published on the tender document before submitting their bids.
- v. Bidders may ensure that all the pages of the documents mentioned must be signed & stamped by authorised signatory and serially numbered.
- vi. Prior to bid submission, bidder should get ready with the documents to be uploaded as part of the bid as indicated in the tender document/schedule (refer Annexure-09 for documents to be uploaded).

5.9.Submission of Bids

- i. The bidders are requested to submit the bids (both technical and financial) through online GeM portal before the deadline for submission of bids (as per Server System Clock displayed on the portal) in the prescribed format. No other mode of submission is accepted.
- ii. Bidder should log into the site well in advance for bid submission so that they can upload the bid in time i.e. before the bid submission closing date & time. Bidder will be responsible for any delay due to other issues. RITES will not be held responsible for any sort of delay or the difficulties faced during online submission of bids by the bidders at the eleventh hour.
- iii. No hard copy of the documents (except those specifically asked for in the tender document) are required to be submitted.
- iv. Bidders to note that the very act of uploading their offers shall be deemed to be a confirmation that they have read all sections and pages of the tender/bid document including terms and conditions without any exception and have understood the entire document and are clear about tender requirements which will be binding upon the bidder.
- v. Each Bidder shall submit only one Bid either as an individual or as a Proprietor in a Proprietary firm or as a Partner in a Partnership firm or as a Director of a Limited Company/Corporation . Any Bidder who has submitted a Bid for a work, shall not be a witness for any other Bidder for the same work. Failure to observe the above stipulations would render all such Tenders submitted as a Bidder and/or as a witness, liable to summary rejection.
- vi. Bidders are requested to note that they should necessarily submit their financial bids in the format provided and no other format is acceptable.

vii. Cost of Tender Document & Earnest Money Deposit (EMD)

- A.** In case of exemption from payment of EMD as a matter of Govt. Policy, the scanned copy of document in support of exemption will have to be uploaded by the bidder during bid submission.
- B.** In case the bidder is registered as a vendor under the category of Micro, Small and Medium Enterprises (MSME), he must state his Udyog Aadhar Memorandum (UAM) number. The scanned copy of document in support of exemption will have to be uploaded by the bidder during bid submission. The onus of proving that the bidder is exempted from payment of EMD lies on the bidder. In this connection, it should be noted that mere opening of bid does not mean that the bid has to be considered by RITES as a valid bid. If later, it is discovered from the uploaded documents that bidder is not exempted from payment of cost of tender and/or EMD, his bid shall be treated as non-responsive.

i. Cost of Tender Document: NA

- ii.** Earnest Money Deposit (EMD) – The bids shall be accompanied by Earnest Money Deposit (EMD) of Rs. 18,00,000 /- (Rupees Eighteen Lakhs only). It shall be in the form :
- As listed in 5.9 (ix) and 5.9 (x) below
 - or
 - Document in support of exemption from payment of EMD and Udyog Aadhar Memorandum (UAM) number.

No other mode of payment will be acceptable.

- C.** Bidders are required to upload scanned copy of acceptable instruments for EMD during on-line submission of Bid. Any Tender not accompanied by scanned copies of the instruments for payment of Earnest Money in an acceptable form (or, if applicable, the scanned copy of documents in support of exemption) shall be rejected by the Employer as non-responsive. These documents shall be deposited in "ORIGINAL" in a sealed envelope and addressed to:

Mr. Tanuj Sukhija, JGM /IT, 3rd Floor, RITES Bhawan, Shikhar ,Plot No.1, Sector 29, Gurugram- 122001, Haryana.

The original should be posted/couriered/given in person to the concerned official, so as to reach him within a week from the date of opening. The details of the DD/any other accepted instrument, physically sent, should tally with the details available in the scanned copy and the data entered during bid submission time. If the date of issue of DD/any other accepted instrument, physically sent, is on or before the bid submission end date, the same shall

also be accepted even if the details are different from the scanned copy uploaded along with the bid. Failing which the bid shall be rejected and the bidder shall be debarred from tendering in RITES Ltd. for a period of 02 (two) years unless the lapse is condoned by the Accepting Authority at the request of the bidder for valid reasons. The envelop should bear the **tender details (Tender no., Tender name etc.)** as mentioned in the tender document. Refer Annexure-08 for details of documents to be submitted in Original.

- viii. Following category of Bidder/Consultants are exempted from payment of earnest money. In case the Bidder falls in these categories, it should furnish/ upload certified copy of its valid registration details:
 - a. Bidder/Consultants who are currently registered and, also, shall continue to remain registered during the Bid validity period as Micro & Small Enterprises (MSEs)
 - b. Start-ups as per definition of DPIIT
 - c. Other category of bidders as specified in GeM GTC .
- ix. The earnest money shall be furnished in one of the following forms:
 - a. Account Payee Demand Draft
 - b. Fixed Deposit Receipt
 - c. Banker's cheque
 - d. Insurance Surety Bond
 - e. Bank Guarantee/e- Bank Guarantee from any of the scheduled commercial banks
- x. The demand draft, fixed deposit receipt or banker's cheque shall be drawn on any scheduled commercial bank in India, in favour of **RITES Ltd.** Unless otherwise specified in TIS, the earnest money shall be valid for a period of forty-five days beyond the final bid validity period . Document for establishing submission or waiver of EMD must be uploaded.

5.10. Bid Security (EMD) refund

- i. The Earnest Money of the Tenderers whose Technical Bid is found not acceptable will be returned without interest soon after scrutiny of Technical Bid has been completed by the Employer subject to provisions of Clause 5.11 (ii). The Earnest Money of the Tenderers whose Technical Bid is found acceptable but who are neither the lowest nor the second lowest will be returned without interest within 07 days of opening of Financial Bid. The Earnest Money of the remaining unsuccessful bidders will be released within seven days of the Accepting Authority's decision on acceptance or otherwise of the tender subject to provisions of Clause 5.11 (ii). The bidder shall submit RTGS/NEFT Mandate Form as per Performa given in Annexure 10, dully filled in for refund of EMD.

5.11. Bid Security (EMD) Forfeiture

The EMD may be forfeited:

- i. if after bid opening, but before expiry of bid validity or issue of Letter of Acceptance, whichever is earlier, any bidder :
 - (a) withdraws his bid;
 - or**
 - (b) makes any modification in the terms and conditions of the tender which are not acceptable to RITES
- ii. In case any information/document which may result in the tenderer's disqualification is concealed by the Tenderer or any statement/information/document furnished by the Tenderer or issued by a Bank/Agency/Third Party and submitted by the tenderer, is subsequently found to be false or fraudulent or repudiated by the said Bank/Agency/Third Party.
- iii. In the case of a successful Bidder, if the Bidder:
 - (a) Fails to accept the Letter of Award /Purchase Order or to furnish Performance Bank Guarantee in accordance with the tender conditions.
 - or**
 - (a) Fails to commence the work without valid reasons within the period as specified after the date of issue of Letter of Acceptance.

In case of forfeiture of Earnest Money as prescribed hereinabove, the bidder shall not be allowed to participate in the re-tendering process of the work.

5.12. Authority To Sign

- i. If the applicant is an individual, he should sign above his full type written name and current address.
- ii. If the applicant is a proprietary firm, the Proprietor should sign above his full type written name and the full name of his firm with its current address.
- iii. If the applicant is a firm in partnership, the Documents should be signed by all the partners of the firm above their full type written names and current addresses. Alternatively, the Documents should be signed by the person holding Power of Attorney for the firm in the Format at Annexure-03.
- iv. If the applicant is a limited Company, or a Corporation, the Documents shall be signed by a duly authorized person holding Power of Attorney for signing the Documents in the Format at Annexure-03.

5.13. Pre- Bid Conference

- i. RITES LTD will conduct a pre-bid conference for all potential Bidders to clarify the doubts or queries of the bidders, if any, as per date & time mentioned under para 1.3 of the tender document.

5.14. Prices

- i. The Bidder shall quote a price for the entire project on a single responsibility basis. The bidder is required to quote his rate for all items. Failure to do so will result in rejection of his bid document.
- ii. The Bidder shall mandatorily submit the Price-Bid (Financial Bid) as per Annexure-07.
- iii. No adjustment of the contract price shall be made on account of variation in costs of labour and materials or any other cost component affecting the total cost in fulfilling the obligations under the contract. The Contract price shall be the only payment, payable by RITES to the contractor for completion of the contractual obligations under the Contract, subject to the terms of payment specified in the Contract.
- iv. The price would be inclusive of all taxes, duties, charges and levies as applicable. Tender evaluation will be done on total price of all items of BOQ taken together along with all taxes and duties etc.
- v. The prices, once offered, must remain firm and must not be subject to escalation for any reason whatsoever within the period of contract.

5.15. Bid Currency

Prices shall be quoted entirely in Indian Rupees.

5.16. Opening and Evaluation of Bids

- i. RITES will open all the bids received on the date, time and venue as specified in table at para 1.3. In the event of the specified date of the opening being declared a holiday by RITES, the bids will be opened at the appointed time and venue on the next working day.
- ii. Opening of bids will be done through online process. RITES reserves the right to postpone or cancel a scheduled bid opening at any time prior to its opening. Information of the same will be displayed on GeM portal.
- iii. Bid opening committee will open the bids online in the presence of bidders or their authorized representatives who chose to attend on opening date and time. Also the bidders can participate online during the bid opening process from their remote end through their dashboard. The bidder's representatives, who are present, shall sign in an attendance register. RITES shall subsequently examine and evaluate the bids in accordance with the provision set out in the tender document.
- iv. RITES shall subsequently examine and evaluate the bids in accordance with the provision set out in the tender document.

- v. It will be the bidder's responsibility to check the status of their Bid on-line regularly after the opening of bid till award of work.
- vi. It may be noted that mere opening of a bid does not mean that the bid has to be considered by RITES as a valid bid. All bids will be evaluated to decide whether the bids are responsive or non- responsive.
- vii. The bids will be evaluated for Eligibility criteria as per Section 4. RITES shall not be responsible for any postal delay in receipt of all original documents including power of attorney and EMD. In case of non receipt of these documents in original within the aforesaid period, the bid will be non-responsive.

5.17. Bid Clarifications

To assist in the examination, evaluation of the bid, RITES Ltd., at its discretion, may seek for clarification(s) from bidders regarding its Bid. The request for clarification and the response shall be in writing and no change in the price or substance of the bid shall be permitted during the process of seeking clarifications.

5.18. Evaluation Methodology

The selection of the consultant will be made using the **Quality and Cost Based Selection (QCBS) methodology, with 70% weightage for the Technical Proposal and 30% for the Financial Proposal**. The evaluation will be carried out by a committee nominated by RITES Ltd. and shall include an assessment and presentation of the bidder's technical proposal, followed by financial evaluation.

In the case of QCBS, the total score is calculated by weighing the technical and financial scores and adding them to obtain a combined QCBS (Technical cum Financial) score, The proposal obtaining the highest total combined score in evaluating quality and cost will be ranked as H-1, followed by the proposals securing lesser marks as H-2, H-3 etc. If two or more bids have the same highest score in the final ranking, the bid with a higher financial score will be H-1.

- a) The Technical Proposals are given an absolute technical score (Ta out of max 100) based on the evaluation criteria in Section VII: Evaluation/ Scoring Criteria. However, to normalise this w.r.t. Financial Score Sf below, a relative Technical Score (St) based on their relative ranking shall be calculated. The highest evaluated Technical Score (Ta-max) is assigned the maximum relative Technical Score (St) of 100 (Hundred). The formula for determining the relative Technical scores (St) of all other Proposals is as follows:

$$St = 100 \times Ta / Ta\text{-max},$$

in which "Ta-max" is the highest evaluated absolute Technical Score, "St" is the relative Technical score calculated, and "Ta" is the absolute Technical Score of the proposal under consideration. This normalisation would avoid any unintended magnification of weightage to the Financial score due to different scales of Technical Scores and Financial Scores.

- b) The Financial Proposals are given cost-score based on the relative ranking of prices, with the lowest evaluated Financial Proposal (Fm) being assigned the maximum financial score (Sf) of 100 (Hundred). The formula for determining the financial scores (Sf) of all other Proposals is as follows:

$$Sf = 100 \times Fm / F,$$

in which "Fm" is the price of the lowest offer, "Sf" is the financial score calculated, and "F" is the price of the proposal under consideration.

- c) The weights given to the Technical (T) and Financial (P) Proposals are specified in AITC: T (the weight given to the Technical Proposal) in %, and P (the weight given to the Financial Proposal) in % (with T + P = 100%)
- d) Proposals would be ranked according to their combined QCBS (weighted technical, St and financial, Sf) scores as follows:

$$S = (St \times T + Sf \times P) / 100.$$

in which "S" is the combined QCBS score, "St" is the relative technical score calculated as per sub-clause a) above and "Sf" is the financial score calculated as per sub-clause b) above.

- e) All scores shall be calculated up to two decimal places only.
- f) A Bidder must get a minimum of 70 marks (out of 100 marks) in the Technical Evaluation (absolute technical score) to proceed to opening of Financial/Price bid. The price bids of bidders scoring the minimum required marks of 70 in the Technical Evaluation Criteria will only be opened.

S.No.	Evaluation Criteria	Sub-Criteria	Max. Marks	Evaluation Basis
1	Average Annual Turn over past 3 years	Rupees 4.5 crores	10	Rs 4.5 crore → 5 marks For turnover above ₹ 4.5 crore → additional 0.25 marks for every ₹1 crore Maximum marks capped at 10

2	Firm’s Relevant Experience	The value of each project should be at least ₹ 2 Crore*. All projects must qualify as ‘Similar Work’ as defined in the RFP. Evaluation Criteria: Note: Only completed or substantially completed projects (minimum 80% completion or payment received) shall be considered for evaluation.	30	<table><tr><th>S. No.</th><th>Number of Similar Projects (≥ ₹2 Crore each)</th><th>Marks</th></tr><tr><td>a)</td><td>≥ 1 and < 3</td><td>10</td></tr><tr><td>b)</td><td>≥ 3 and < 5</td><td>20</td></tr><tr><td>c)</td><td>≥ 5</td><td>30</td></tr></table> (Proportionate marks shall be awarded based on the percentage of completion of such eligible projects, wherever applicable.)	S. No.	Number of Similar Projects (≥ ₹2 Crore each)	Marks	a)	≥ 1 and < 3	10	b)	≥ 3 and < 5	20	c)	≥ 5	30
S. No.	Number of Similar Projects (≥ ₹2 Crore each)	Marks														
a)	≥ 1 and < 3	10														
b)	≥ 3 and < 5	20														
c)	≥ 5	30														
3	Technical Presentation : Approach, Methodology	Technical demonstration Shortlisted bidders may be required to provide a live technical demonstration in an online environment, showcasing relevant implemented systems or working prototypes aligned with the requirements of this RFP. Any demo which includes AI wrapper will be disqualified and score will be marked as 0 (zero) The demonstration shall focus on: Functional capabilities	40	1. Demonstration of relevant functional capabilities aligned with DPR workflows including GIS and alignment = Max score 7.5 2. Demonstration of system architecture, scalability, and technical robustness = Max score 7.5 marks 3. User interface, workflow clarity, and usability = Max score 5 marks 4 :- Live Proof of Concept / Demonstration of AI-Enabled DPR Appraisal Capability = 20 Marks The bidder shall demonstrate its AI/ML capability by processing a sample/legacy DPR. Evaluation shall cover relevance to infrastructure/engineering/DPR appraisal workflows, document parsing, OCR, tabular/financial data extraction, rule/codal												

		- Workflow and user interface - Validation and analytics features - System architecture and scalability aspects - AI/ML Capabilities. Static presentations or slide-based briefings alone shall not be considered sufficient. Bidders are expected to demonstrate working software or prototypes. They may also be requested to show the online admin logins. The bidder shall be required to demonstrate and justify the proposed architecture in terms of resource utilization, including compute, memory, storage, and network efficiency, during technical evaluation and system design approval stages		validation, anomaly/red flag detection, hallucination resistance, explainability, traceability and practical usability of the demonstrated solution.
4	Employee Certifications	1 Certified employee = 4	20	Valid Certificates to be submitted Each employee certified shall be awarded 4 marks with max of 20 marks in this category Each employee will be counted only once, irrespective of the number of certifications held industry certifications

*Price Excluding GST/Taxes

5.19. Contacting RITES Ltd.

Any attempt by a Bidder to influence RITES Ltd. in its bid evaluation or contract award decisions may result in the rejection of the Bidder's Bid.

5.20. Notification of Award

- i. RITES Ltd. will notify successful Bidder in writing or by email that its bid has been accepted and will issue letter of award / Purchase order to this effect.
- ii. The successful Bidder is required to convey acceptance of the contract by returning duly signed & stamped duplicate copy of the award letter/purchase order within 7 days of receipt of communication and commence the work within 10 days of award of contract. In the absence of any formal contact the award letter/purchase order shall constitute a binding contact between RITES and the successful Bidder.

5.21. Performance Bank Guarantee (PBG)

The successful Bidder shall execute a Performance Bank Guarantee (as per format in Annexure-06) for 5% of the contract value, within 30 days of receipt of the notification of award of contract. The PBG shall remain valid for a period of 60 days beyond the expiry of the contract period of one year calculated from the date of award of contract by RITES.

In event of the contract being extended by RITES for additional one year the successful Bidder shall get the PBG extended by one year from its expiry date.

Failure of successful Bidder to comply with the above requirement of submission of PBG shall constitute a sufficient ground for the annulment of the award and forfeiture of the EMD/PBG.

The Contractor shall be fully responsible for all matters arising out of the Performance of the Contract and shall, at his own expense, comply with all laws/acts/enactments/orders/regulations/obligations whatsoever of the Government of India, State Government, Local Body and any Statutory Authority.

In case the Successful Bidder abandons the work or fails to execute the work or work is not satisfactory, the RITES may cancel the work order by giving 30 days written notice and PBG will be forfeited.

6. Terms and Conditions

6.1 Duration of Contract

The development shall be in phases as outlined in the tender document. Total contract duration will be 24 months.

6.2 Team Composition

Refer Annexure 13 for details. The Qualifications of some Key personnel are as follows.

S. No.	Role	Minimum Qualification	Minimum Experience	Key Skill Requirements
1	Artificial Intelligence (AI) Engineer	B.Tech/M.Tech/PhD in AI/ML/CS/Data Science	5+ years	AI model development, LLMs, deep learning, computer vision, deployment
2	Machine Learning Engineer	B.Tech/M.Tech (CS/AI/Data Science)	4+ years	ML model building, training pipelines, model optimization, MLOps
3	Data Engineer	B.Tech/M.Tech (CS/IT/Data Engineering)	4+ years	ETL pipelines, data warehousing, SQL/NoSQL, big data tools
4	Software Engineer	B.Tech/MCA	3+ years	Backend/frontend development, APIs, enterprise applications
5	Data Scientist	B.Tech/M.Tech/MCA/Data Science	5+ years	Statistical analysis, ML models, predictive analytics
6	BI Developer	B.Tech/MCA / Equivalent	3+ years	Dashboarding (Power BI/Tableau), reporting, data visualization
7	Research Scientist	M.Tech/PhD (AI/ML/Data Science/CS)	6+ years	Advanced research, algorithm design, publications preferred
8	Software Architect	B.Tech/M.Tech (CS/IT)	8+ years	System architecture, scalability, cloud/on-prem design
9	Big Data Engineer / Architect	B.Tech/M.Tech (CS/IT/Data)	6+ years	Hadoop/Spark, distributed systems, large-scale data processing
10	NLP Engineer	B.Tech/M.Tech (AI/ML/NLP/CS)	4+ years	NLP, text analytics, semantic search, document parsing

11	UX Designer	Graduate/Postgraduate in Design/IT	3+ years	UI/UX design, prototyping, user research, dashboards
12	Algorithm Developer	B.Tech/M.Tech/PhD (CS/Math/AI)	5+ years	Algorithm design, optimization, computational models
13	AI Data Analyst	Graduate/Postgraduate (Data/Stats/IT)	3+ years	Data analysis, reporting, AI-assisted analytics tools
14	Prompt Engineer	Graduate/Postgraduate (any with AI exposure)	2+ years	LLM prompting, prompt design, AI workflow tuning

6.3 Service Level Agreements

The Service Level Agreements (SLAs) defined in this RFP shall remain applicable during the entire period of the contract, including development, implementation, testing, production deployment, stabilization, warranty, operation and maintenance/support period, as applicable.

Performance SLAs (Latency, Uptime, AI Accuracy)

6.3.1 System Availability (Uptime SLA)

Parameter	Requirement
Application Uptime	≥ 99.5% monthly uptime (excluding scheduled maintenance)
Critical Modules (Login, DPR Access, Validation Engine)	≥ 99.7% uptime
Scheduled Maintenance Window	Max 4 hours/month , with prior approval and notice
Measurement Basis	24x7 monitoring, excluding approved downtime

6.3.2 Response Time & Latency SLA

DPR Processing & AI Latency

The platform shall process DPR documents, including scanned documents, digital PDFs, tables, annexures and supporting files, within reasonable indicative timelines. DPR parsing/OCR/indexing may be targeted within **5–10 minutes per 1,000 pages**, depending on document quality, file size

and OCR complexity. Comprehensive validation and AI-assisted appraisal execution may be targeted within **15–30 minutes for a standard DPR**, depending on document complexity, number of checks, sector-specific rules and AI workload.

For large DPRs, poor-quality scanned documents, complex tables, drawings, annexures, financial models or bulk uploads, additional processing time may be permitted. The system shall provide upload status, queue status, processing status, failed job alerts and completion status to users.

Accuracy, completeness, explainability, traceability and reliability of AI outputs shall be prioritized over processing speed. The Bidder shall clearly indicate the expected processing time, assumptions, dependencies and infrastructure requirements in its technical proposal.

6.3.3 AI Accuracy & Quality SLA

S. No.	Parameter	Version 1 / Draft Tool D+3 Months	Version 2 / Enhanced Tool D+5 Months	Final Version D+7 Months	Stabilization D+9 Months
1	Document Text Extraction Accuracy including OCR and digital PDFs	≥ 85%	≥ 90%	≥ 95%	≥ 95%
2	Tabular, Financial and Numerical Data Extraction Accuracy	≥ 90%	≥ 95%	≥ 98%	≥ 99%
3	DPR Section / Clause Identification Accuracy	≥ 80%	≥ 85%	≥ 90%	≥ 95%
4	Rule-based Validation and Checklist Accuracy	≥ 85%	≥ 90%	≥ 95%	≥ 95%
5	AI-based Anomaly / Inconsistency / Red Flag Detection Accuracy	≥ 75%	≥ 85%	≥ 90%	≥ 95%
6	Cross-reference Checking Accuracy across DPR Sections / Tables / Annexures	≥ 75%	≥ 85%	≥ 90%	≥ 90%

7	DPR Summary / Observation Generation Quality	≥ 75%	≥ 80%	≥ 85%	≥ 90%, subject to human validation
8	Engineering Code / Reference Hallucination Control	≤ 5%	≤ 2%	0% for validated reference library	0%
9	Explainability, Traceability and Reasoning Quality	Basic page/section reference with brief reasoning	Page/section/table-level reference with reasoning	Full source traceability with logical reasoning	Full traceability with reasoning, confidence score and audit trail

In addition to the above accuracy levels, the following requirements shall be mandatory across all versions:

1. All AI-generated outputs shall provide traceable references to the relevant DPR page, section, table, annexure, drawing, rule, benchmark, code or source document.
2. The system shall provide confidence scores and exception flags wherever feasible, particularly for critical outputs and low-confidence results.
3. AI-generated summaries, observations, anomalies, red flags and recommendations shall be treated as decision-support inputs only and shall not be auto-accepted without review and approval by RITES officials/domain experts.
4. The platform shall not generate or rely upon fabricated, non-existent, obsolete, irrelevant or non-compliant engineering codes, standards, circulars, manuals, guidelines or references.

6.3.4 Scalability & Performance Efficiency

- System shall support **horizontal scaling**
- AI/ML workloads shall be **resource-optimized and not GPU-intensive unless justified**

The system shall be capable of handling:

- Large DPR documents ranging from approximately **500 to 5,000 pages**
- Bulk upload and ingestion of multiple DPRs and supporting documents
- Concurrent document parsing, AI processing, validation and appraisal workflows

- Scalable storage, indexing and retrieval of historical DPRs, annexures, tables, drawings and supporting records
- Future expansion in number of users, sectors, DPR volume, AI models, validation rules and knowledge repositories

The Bidder shall clearly specify the proposed architecture, scalability approach, infrastructure sizing assumptions, concurrent-user capacity, simultaneous AI-processing capacity and performance optimization measures in its technical proposal.

6.3.5 Integration Performance SLA

Parameter	Requirement
Platform-owned API Availability	≥ 99% monthly uptime
External API Response Handling	The system shall handle responses from external APIs in a reasonable time, subject to availability/performance of external systems
Data Synchronization	As per integration requirement mutually finalized with RITES
Error Handling & Logging	Mandatory

6.3.6 Incident Management SLA

Severity Level	Description	Response Time	Resolution Time	RCA Timeline
Severity 1 – Critical	Complete system outage, application not accessible, or AI/system generating corrupted or materially incorrect financial, numerical or tabular data impacting appraisal decisions.	≤ 15 minutes	≤ 4 hours	Within 24 hours
Severity 2 – High	Major module unavailable or severely impacted, such as Appraisal Engine, DPR processing, validation engine, document upload/parsing or user access module failure, with no acceptable workaround.	≤ 30 minutes	≤ 8 hours	Within 48 hours

Severity 3 – Medium	Partial feature degradation, intermittent issue, slow performance, non-critical error, or functionality impacted where a reasonable workaround is available.	≤ 2 hours	≤ 2 working days	On request / if required
Severity 4 – Low	Minor issue, cosmetic defect, UI/formatting error, report layout issue, user support query, or non-critical enhancement/change request not affecting core operations.	≤ 6 hours	≤ 5 working days	Not applicable

For Severity 1 and Severity 2 incidents, the Service Provider shall provide timely updates to RITES until closure and submit Root Cause Analysis within the prescribed timelines. Resolution shall include restoration of service, correction of incorrect outputs, data integrity checks, and preventive action wherever applicable.

6.3.7 SLA Measurement & Reporting

The Service Provider shall provide a role-based operational dashboard for monitoring platform usage, DPR processing status, incidents, SLA status and AI processing status. The Service Provider shall also submit a monthly AI Confidence and Hallucination Audit Report covering low-confidence outputs, user corrections, false positives/false negatives, rule failures, missed red flags, hallucinated references, and corrective actions taken.

6.4 Liquidity Damages/ Penalty

The Service Provider shall comply with the timelines, SLA requirements, accuracy requirements and performance obligations specified in this RFP/Agreement. In case of delay, non-performance, repeated service failure, breach of SLA, or failure to meet AI accuracy and quality requirements, RITES may impose liquidated damages/penalty as specified below.

Penalties shall be applicable only for delays, failures or deficiencies attributable to the Service Provider. Delays attributable to RITES, approved change requests, non-availability of required inputs/data/infrastructure from RITES, or force majeure events shall be suitably excluded while calculating penalty.

A. System Availability / Uptime Penalty

Monthly Uptime	Penalty
≥ 99.5%	No penalty
99.00% to 99.49%	2% deduction from applicable monthly support payment

Below 99.00%	5% deduction from applicable monthly support payment
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B. Incident Resolution Penalty

The Service Provider shall resolve incidents within the response and resolution timelines defined under the Severity Level SLA. Penalty shall apply for delay beyond the prescribed resolution time where such delay is attributable to the Service Provider.

Incident Category	Penalty
Severity 1 – Critical	₹10,000 per hour of delay beyond SLA resolution time
Severity 2 – High	₹5,000 per hour of delay beyond SLA resolution time
Severity 3 – Medium	Warning / corrective action plan; penalty may be imposed for repeated breaches
Severity 4 – Low	No monetary penalty; to be tracked for closure

For Severity 1 incidents continuing beyond 24 hours, RITES may withhold the applicable monthly payment, in addition to other remedies available under the Agreement.

C. AI Accuracy / Quality Non-Compliance

In case the AI accuracy or quality levels fall below the prescribed accuracy levels defined for the respective milestone, the Service Provider shall rectify, fine-tune, retrain, reconfigure or improve the system.

For repeated or material non-compliance in critical areas such as financial/numerical extraction, tabular data extraction, red flag detection, anomaly identification, or engineering code/reference accuracy, **RITES may withhold the relevant milestone/quarterly payment until the issue is satisfactorily rectified (as per defined accuracy levels)**

AI-generated outputs shall remain subject to human review and validation; however, repeated generation of materially incorrect, non-traceable, fabricated or misleading outputs shall be treated as a quality deficiency.

D. Delay in Milestone Deliverables

The Service Provider shall complete all deliverables within the timelines specified in the milestone schedule. Delay in submission, deployment or acceptance of milestone deliverables attributable to the Service Provider shall attract liquidated damages as under:

Deliverable Delay	Penalty
Delay for each week or part thereof beyond approved milestone timeline	0.5% of the total contract value, subject to overall penalty cap

E. Maximum Penalty Cap

The total cumulative penalty/liquidated damages under this Agreement shall be capped at **10% of the total contract value**.

If the cumulative penalty reaches or exceeds **10% of the total contract value**, the same may be treated as a material breach of contract, and RITES may initiate further action, including withholding of payments, invocation of performance security, termination of contract, or other remedies available under the Agreement.

6.5 Deliverables

As per clause 6.10 and 6.11

6.6 Roles and Responsibilities

Role of RITES Team

RITES shall act as the project owner, domain authority, and approval body for the Digital Project Tool. RITES shall provide strategic direction, domain expertise, validation frameworks, and governance oversight, while the bidder shall be responsible for technical implementation and data infrastructure.

1. Domain Authority and Validation Ownership

- RITES shall retain full ownership of:
 - DPR validation frameworks
 - Appraisal methodologies
 - Scoring mechanisms
 - Technical standards and evaluation criteria
- All validation logic, rules, and appraisal outputs shall be governed and approved by RITES.

2. Template, Ontology, and Standards Approval

RITES shall:

- Review and approve:
 - DPR templates (structure, sections, mandatory inputs)
 - Ontology and dependency frameworks

- Dictionary, standards, and reference datasets
- Provide domain inputs through Subject Matter Experts (SMEs) to ensure that:
 - engineering logic is accurate
 - dependencies are realistic
 - validation frameworks are context-aware

3. SME Engagement and Domain Inputs

- RITES shall nominate Subject Matter Experts (SMEs) across domains such as:
 - civil, structural, transport, finance, environment, etc.
- SMEs shall:
 - validate templates and rules
 - provide domain-specific inputs
 - review outputs of validation and analytics

4. Design Review and Approval

- RITES shall review and approve:
 - system architecture
 - database design and data models
 - module configurations and workflows
- The bidder shall submit all design artifacts for vetting prior to implementation.

5. Data Governance and Validation Oversight

- RITES shall define:
 - data validation principles
 - data quality thresholds
 - benchmarking logic and interpretation
- RITES shall **not be responsible for bulk data ingestion or structuring**, which shall be undertaken by the bidder.
- RITES shall validate and approve:
 - data usage frameworks
 - benchmarking approaches
 - interpretation of analytical outputs

6. Stakeholder Coordination

- Facilitate coordination with:
 - internal departments
 - external agencies
 - national platforms (e.g., PM GatiShakti)
- Enable access and engagement required for integration and validation

7. Testing, Review, and Acceptance

- Nominate users and SMEs for:
 - User Acceptance Testing (UAT)

- validation of outputs
- Review system outputs and provide:
 - feedback
 - approvals
 - final sign-off

8. Infrastructure and Hosting (Post-UAT)

- Provide or designate infrastructure for deployment post-UAT
- Ensure readiness of hosting environment and access controls

9. Governance and Oversight

- Monitor project progress and milestones
- Conduct periodic review meetings
- Provide strategic and operational guidance

10. Post-Implementation Oversight

- Oversee platform usage, performance, and outcomes
- Guide enhancements and scaling
- Ensure continued alignment with organizational objectives

6.6.2 Roles and Responsibilities of Bidder

The selected bidder shall be responsible for the end-to-end execution of the project, including design, development, deployment, integration, and support of the Digital PARAKH, in accordance with the requirements specified in this RFP and Annexure X.

1. Overall Responsibility

- Act as the single point of responsibility for successful implementation of the platform
- Ensure delivery of all components within the agreed timelines, quality standards, and performance requirements
- Coordinate with all stakeholders, including RITES and other agencies, as required
- Ensure that the **SMEs (of 13 sectors as per Annexure 12 + Cost Finance etc as required)** from Bidder interact with respective SMEs from RITES to understand, scope and facilitate translation and transfer of engineering knowledge from RITES to the bidder

2. Solution Design and Architecture

The bidder shall:

- Prepare detailed system architecture and design documents
- Ensure alignment with the architectural principles and requirements specified in Annexure X
- Design a scalable, modular, secure, and integration-ready system
- Obtain necessary approvals from RITES at key design stages

3. Development and Configuration

The bidder shall:

- Develop and configure all platform modules as per the scope
- Ensure adherence to coding standards, security practices, and performance benchmarks
- Implement configurable templates, validation rules, workflows, and scoring mechanisms
- Ensure flexibility and extensibility of the system

4. Data Management and Migration

The bidder shall:

- Design and implement the Data Lake and data management framework
- Ingest and migrate historical DPR data and related datasets, where required
- Ensure data quality, consistency, and integrity during migration and processing
- Establish data governance, metadata, and cataloging mechanisms

5. Integration Responsibilities

The bidder shall:

- Develop and implement APIs and integration interfaces
- Integrate the platform with:
 - GIS systems
 - National platforms such as PM GatiShakti (where applicable)
 - BIM and engineering systems (if required)
 - Other enterprise systems
- Ensure secure, reliable, and scalable integration

6. AI and Analytics Implementation

The bidder shall:

- Implement AI and analytics components for validation, anomaly detection, and benchmarking
- Ensure that AI outputs are explainable, auditable, and aligned with domain requirements
- Enable continuous improvement through learning from historical data

7. Testing and Quality Assurance

The bidder shall:

- Conduct comprehensive testing, including unit testing, integration testing, system testing, and performance testing
- Support User Acceptance Testing (UAT) and resolve identified issues
- Ensure compliance with functional, technical, and security requirements

8. Deployment and Commissioning

The bidder shall:

- Deploy the solution in the designated infrastructure environment
- Configure and optimize system performance
- Ensure readiness for production use
- Support go-live and stabilization

9. Training and Documentation

The bidder shall:

- Provide training to users, administrators, and technical personnel
- Deliver complete documentation, including:
 - System architecture
 - User manuals
 - Technical manuals
 - API documentation
- Support knowledge transfer to RITES teams

10. Operations and Maintenance

The bidder shall:

- Provide support and maintenance services for the duration specified in the RFP
- Ensure system uptime, performance, and security
- Address defects, issues, and change requests in a timely manner
- Provide periodic updates, patches, and enhancements

11. Licensing and Compliance

The bidder shall:

- Provide and maintain all required licenses, subscriptions, and support for proprietary components for a minimum period of three (3) years
- Clearly identify all licensed components and associated costs
- Ensure compliance with applicable standards, regulations, and security requirements

12. Security and Data Protection

The bidder shall:

- Implement appropriate security controls, including access control, encryption, and monitoring
- Ensure protection of sensitive data and compliance with applicable data protection guidelines
- Maintain audit trails and logging mechanisms

13. Project Management and Reporting

The bidder shall:

- Deploy a qualified project team with defined roles and responsibilities
- Prepare and maintain project plans, timelines, and deliverables
- Submit periodic progress reports to RITES
- Participate in review meetings and provide updates as required

14. Innovation and Continuous Improvement

The bidder shall:

- Propose improvements, optimizations, and enhancements during implementation
- Incorporate feedback from users and stakeholders
- Ensure the platform evolves in line with emerging requirements and technologies

15. Open-source and third-party compliance

The bidder warrants that:

- all third-party components are lawfully used;
- all notices, attribution, and pass-through license obligations will be fulfilled;
- no malicious code or known prohibited components are included.

16. Software Bill of Materials (SBOM)

The bidder shall submit and update the following at implementation milestones and before go-live:

- dependency inventory
- license register
- vulnerability register

All system architecture, database design, data models, and related technical artifacts shall be subject to review and approval by RITES at defined stages of the project.

The bidder shall submit detailed design documents, including architecture diagrams, database schemas, data structures, and integration designs, for vetting by RITES prior to implementation.

Any subsequent changes or deviations from the approved design shall require prior approval from RITES.

Operations, Maintenance, and Support (AMC)

The bidder shall provide Operations and Maintenance (O&M) support for the Digital PARAKH for a period of **fifteen (15) months** from the date of Go-Live / Final Acceptance.

Scope of AMC

During the AMC period, the bidder shall be responsible for:

- Ensuring system availability, performance, and reliability
- Monitoring and maintaining all application components, databases, and integrations
- Providing bug fixes, patches, and updates
- Resolving defects and technical issues within defined timelines
- Supporting users and administrators through helpdesk mechanisms
- Managing backups, recovery, and system health checks
- Ensuring security updates and compliance

Support Levels

The bidder shall provide:

- **L1 Support:** User assistance and issue logging
- **L2 Support:** Application-level troubleshooting and fixes
- **L3 Support:** Code-level fixes and enhancements

All licenses, subscriptions, and support for proprietary components (if any) shall be valid and maintained during the AMC period.

In case of licenses procured for a longer duration (e.g., 3 years), the bidder shall ensure continuity of services without disruption. Expiry of third-party licenses shall

not render the core platform or RITES-owned components unusable; bidder shall ensure continuity or provide equivalent alternatives

The bidder shall demonstrate interoperability through working integrations during technical evaluation and implementation, including:

- Exposure and testing of APIs using standard tools (e.g., Postman/OpenAPI interface)
- Integration with at least one external system or simulated environment
- Demonstration of data exchange using defined schemas and APIs
- Demonstration of event-driven data flow (where applicable)

Failure to demonstrate practical interoperability capabilities may result in rejection or non-acceptance of deliverables.

6.7 Intellectual Property Rights

The intellectual property related to domain frameworks, including templates, ontology, validation logic, and benchmarking methodologies, shall remain exclusively with RITES and shall not be replicated or reused by the bidder in any other project without prior approval. All trained AI/ML models, model weights, training datasets (excluding bidder's pre-existing datasets), and derived intelligence outputs developed under this project shall be the exclusive property of RITES.

6.7.1. Ownership of Platform and Deliverables

- All **project-specific deliverables/custom developments** vest in RITES;
- Bidder retains ownership of **pre-existing IP/background materials**;
- Notwithstanding the foregoing, ownership of bidder's pre-existing intellectual property, tools, frameworks, utilities, and reusable components shall remain with the bidder; however, to the extent any such background intellectual property is embedded in, necessary for, or used in connection with the solution, the bidder grants RITES a perpetual, irrevocable, royalty-free, non-exclusive, transferable license to use, operate, maintain, modify, and enhance the solution without restriction/ any vendor lock-in.
- All intellectual property rights in the software, source code, configurations, workflows, documentation, and other deliverables developed under this project shall vest exclusively with RITES. The bidder shall have absolutely no claim, commercial or otherwise, to the custom developments, logic, or outputs of this project.
- The bidder shall ensure that the platform is developed in a manner that allows full ownership, control, and future extensibility by RITES.

6.7.2. Ownership of Data and Knowledge Assets

- All data, including DPR data, historical datasets, and derived datasets, shall be the exclusive property of RITES.
- All knowledge assets created under the platform, including but not limited to:
 - DPR templates
 - validation rules and frameworks
 - ontology and dependency structures
 - benchmarking datasets

- scoring and appraisal logic

shall be owned exclusively by RITES.

6.7.3. Open Architecture and Vendor Neutrality

The bidder shall ensure that:

- The platform is based on open standards and interoperable architecture
- RITES is not dependent on the bidder for future enhancements, maintenance, or scaling
- All configurations, code, and documentation are accessible and usable by RITES or any third party engaged by RITES

6.7.4. Use of Open-Source Components

- The bidder is encouraged to use open-source technologies and frameworks where feasible
- Any open-source components used shall comply with applicable licensing terms
- The bidder shall provide details of all open-source components and associated licenses

6.7.5. Third-Party / Proprietary Components

- Any proprietary or licensed components used in the solution shall be clearly identified
- The bidder shall provide valid licenses for such components for the period specified in the RFP
- Use of proprietary components shall not restrict RITES' ability to:
 - access data
 - modify configurations
 - integrate with other systems
- Any proprietary component essential for operation must include assignable or transferable rights, or direct licensing in RITES' name, for the required term;
- Bidder must support transition to RITES or a new agency without additional lock-in fees.

6.7.6. Source Code and Documentation

- The bidder shall provide complete source code, configuration files, database schemas, APIs, and technical documentation
- Source code shall be handed over to RITES at defined milestones and upon project completion
- RITES shall have unrestricted rights to use, modify, and extend the solution

6.7.7. Restrictions on Bidder

The bidder shall not:

- Retain any ownership rights over the platform or its components developed under this project
- Use, reproduce, or commercialize the solution, in whole or in part, without prior written approval of RITES
- Share or disclose any project-related data or artifacts without authorization

6.7.8. Indemnity

The bidder shall ensure that the solution does not infringe upon any third-party intellectual property rights and shall indemnify RITES against any such claims.

6.7.9. Continuity and Transition

Upon completion or termination of the contract, the bidder shall ensure:

- Full transfer of all intellectual property, source code, and documentation to RITES
- Smooth transition to RITES or any agency nominated by RITES

In the event of any claim asserted by a third party of infringement of copyright, patent, trademark or industrial design rights arising from the use of the Goods/software or any part thereof, the contractor shall expeditiously extinguish such claim. If the contractor fails to do so and RITES LTD is required to pay compensation to a third party resulting from such infringement, the contractor shall be responsible for the compensation including all expenses, court costs and lawyer fees. RITES LTD will give notice to the contractor of such claim, if it is made, without delay.

6.7.10 Third-Party Libraries and Components

No third-party or open-source component shall be used in a manner that imposes an obligation on RITES to disclose, license, or distribute RITES-owned source code, or otherwise restricts RITES' rights to use, modify, maintain, host, or transition the platform. Any such component, if proposed, shall require prior written approval of RITES

Use of Third-Party Libraries and Components

a. Permitted Use

The bidder may use third-party libraries, frameworks, tools, and components (including open-source and proprietary software) as part of the solution, provided that such use does not restrict the ownership, control, and usability of the platform by RITES.

b. Disclosure Requirement

The bidder shall provide a comprehensive list of all third-party components proposed to be used in the solution, including:

- Name and version of the component
- Type (open-source / proprietary)
- Licensing model and terms
- Intended usage within the platform

This shall be submitted as part of the technical proposal and updated during implementation, if required.

c. Compliance with Licensing Terms

- The bidder shall ensure full compliance with all applicable license terms of third-party components
- The bidder shall ensure that no component is used in a manner that imposes restrictive obligations on RITES, including but not limited to:

- mandatory disclosure of proprietary source code
- restrictive distribution or usage conditions

d. Preference for Open-Source Technologies

- The use of open-source technologies is encouraged, subject to compliance with licensing terms and suitability for enterprise deployment
- The bidder shall ensure that selected open-source components are stable, secure, and actively supported

e. Proprietary Components

Where proprietary or licensed components are used:

- The bidder shall provide valid licenses for the duration specified in the RFP
- All associated costs shall be clearly indicated
- Such components shall not create long-term dependency on the bidder
- RITES shall have continued access to the system and data independent of the bidder

f. No Vendor Lock-in

The bidder shall ensure that:

- The use of third-party components does not result in vendor lock-in
- The platform remains interoperable and extensible
- RITES or its nominated agency can maintain, modify, or enhance the system without restrictions

g. Security and Risk Responsibility

- The bidder shall be responsible for ensuring that all third-party components are secure, free from vulnerabilities, and regularly updated
- The bidder shall monitor and address any security risks associated with such components during the project and AMC period

Bidder shall provide transition support for a minimum of 60 days, including knowledge transfer, documentation handover, and onboarding of RITES or its nominated agency, at no additional cost.

h. Indemnity

The bidder shall indemnify RITES against any claims arising from improper use, licensing violations, or intellectual property infringement related to third-party components used in the solution.

6.7.11 Patent Rights and Protection of Innovation

- All inventions, discoveries, innovations, processes, algorithms, methodologies, system designs, and technical solutions developed under this project, whether patentable or not, shall be the **exclusive property of RITES**.
- The bidder shall:
 - Not file or attempt to file any **patent, utility model, or similar protection** for any component developed under this project without prior written approval of RITES.
 - Disclose any patentable invention arising during the course of the project to RITES promptly.
- In case any patentable innovation arises:
 - The **rights to file and own such patents shall vest exclusively with RITES**

- The bidder shall provide all necessary support, documentation, and technical inputs required for patent filing
- If the bidder uses any **pre-existing patented technology**:
 - It must be disclosed upfront
 - A **perpetual, royalty-free license** must be granted to RITES for usage within the platform
- The bidder shall indemnify RITES against:
 - Any claims of **patent infringement**
 - Any unauthorized use of patented technologies in the solution

6.8. Methodology, Documentation, and Approval Framework

6.8.1 Definition of Methodology

For the purpose of this project, “Methodology” shall include, but not be limited to:

- DPR validation frameworks and rule engines
- Scoring and appraisal methodologies
- Ontology, dependency models, and parameter relationships
- Benchmarking approaches and comparison logic
- AI/ML models, algorithms, and decision-support mechanisms
- Data processing, transformation, and interpretation logic
- Risk assessment and compliance evaluation frameworks

6.8.2 Requirement for Documented Methodology

The bidder shall:

- Develop and maintain **comprehensive documentation** covering all methodologies used in the platform, including:
 - rule definitions and validation logic
 - scoring criteria and weightages
 - benchmarking models and reference datasets
 - AI/ML model design, training approach, and assumptions
 - data flows, transformations, and dependencies
- Ensure that:
 - all methodologies are **transparent, explainable, and auditable**
 - no “black-box” logic is deployed without adequate documentation

6.8.3 Submission and Approval by RITES

- The bidder shall submit all methodologies and related documentation to RITES at defined stages, including:
 - design phase
 - pre-development approval
 - pre-UAT validation
 - post-implementation updates
- **RITES shall act as the final approving authority** for:
 - validation frameworks
 - appraisal methodologies
 - scoring logic
 - ontology and dependency structures
 - AI/ML models and their application
- No methodology shall be deployed in production without **prior written approval from RITES**.

6.8.4 Iterative Review and Refinement

- Methodologies shall be subject to:
 - periodic review
 - refinement based on feedback, data insights, and evolving standards
- The bidder shall:
 - incorporate feedback from RITES SMEs
 - update documentation accordingly
 - maintain version history and change logs

6.8.5 Explainability and Auditability

The platform shall ensure that:

- all outputs (validation results, scores, recommendations) are:
 - traceable to underlying rules, standards, and data inputs
- AI/ML outputs include:
 - explanation of reasoning
 - source data and assumptions
- audit trails are maintained for:
 - methodology changes
 - rule updates
 - model revisions

6.8.6 Ownership of Methodology

- All methodologies developed under this project, including:
 - validation logic
 - scoring frameworks
 - benchmarking models
 - ontology structures
 - AI/ML models (excluding bidder's pre-existing generic libraries)shall be the **exclusive intellectual property of RITES**.

6.8.7 Restrictions on Bidder

The bidder shall not:

- deploy undocumented or unapproved methodologies
- modify approved methodologies without prior approval
- reuse or commercialize RITES-specific methodologies in other projects without written consent

6.8.8 Deliverables

The bidder shall provide:

- Methodology Design Documents
- Rule Books / Validation Framework Documents
- AI/ML Model Documentation
- Benchmarking Framework Documentation
- Version-controlled updates and change logs
- User-facing explanation modules (where applicable)

The platform shall function as a transparent, methodology-driven system where all appraisal outcomes are explainable, governed, and approved by RITES.

6.9 Security, Vulnerability Assessment, and OWASP Compliance

6.9.1 Secure-by-Design Principles

The bidder shall design, develop, and deploy the platform following **secure-by-design and secure-by-default principles**, ensuring protection against application, data, and AI-specific vulnerabilities.

6.9.2 Compliance with OWASP Standards

The solution shall comply with:

- **OWASP Top 10 (latest version)** for web application security
- **OWASP API Security Top 10** for all exposed APIs
- **OWASP ASVS (Application Security Verification Standard)** – minimum Level 2 compliance for enterprise-grade applications
- **OWASP SAMM (Software Assurance Maturity Model)** – recommended for development lifecycle maturity

The bidder shall ensure that all modules, including web, mobile, APIs, and integrations, adhere to these standards.

6.9.3 AI and Agentic AI Security (Emerging OWASP AI Risks)

For AI/ML and agentic AI components, the platform shall address vulnerabilities aligned with emerging OWASP AI risk frameworks (including OWASP Top 10 for LLMs), including but not limited to:

- Prompt injection attacks
- Data poisoning and training data manipulation
- Model inversion and data leakage
- Unauthorized model access or misuse
- Over-reliance on AI outputs without validation
- Sensitive data exposure through AI responses
- Insecure plugin / tool integration in agentic systems
- Supply chain risks in AI models and datasets
- Lack of output validation and hallucination risks

The bidder shall implement:

- guardrails and input/output validation
- role-based access and control over AI actions
- explainability and audit mechanisms
- human-in-the-loop controls for critical decisions

6.9.4 Vulnerability Assessment and Penetration Testing (VAPT)

The bidder shall:

- Conduct **Vulnerability Assessment and Penetration Testing (VAPT)** at:
 - pre-UAT stage
 - pre-Go-Live stage
 - periodic intervals during AMC (at least annually or as specified)
- Cover:
 - application layer
 - APIs
 - data layer
 - infrastructure (cloud/on-premise)
 - AI/ML components
- Testing shall be aligned with:
 - OWASP testing guidelines
 - CERT-In empanelled standards (for India)

- API security testing and interoperability validation shall be conducted as part of VAPT and integration testing
- All external integrations shall be validated against defined interoperability and data exchange standards

6.9.5 Remediation and Compliance

- All identified vulnerabilities shall be:
 - classified by severity (Critical / High / Medium / Low)
 - remediated within defined timelines
- No **Critical or High vulnerabilities** shall remain open at the time of Go-Live.
- The bidder shall submit:
 - VAPT reports
 - remediation reports
 - compliance certificates

6.9.6 Secure Development Lifecycle (SSDLC)

The bidder shall adopt a **Secure Software Development Lifecycle**, including:

- secure coding practices
- code reviews and static code analysis
- dependency and vulnerability scanning (aligned with SBOM)
- patch and update management

6.9.7 API and Integration Security

The platform shall ensure:

- authentication and authorization for all APIs
- encryption (in transit and at rest)
- rate limiting and abuse prevention
- protection against:
 - injection attacks
 - broken authentication
 - excessive data exposure

6.9.8 Data Security and Privacy

The bidder shall ensure:

- encryption of sensitive data
- role-based access control (RBAC)
- audit logging and monitoring
- compliance with applicable data protection guidelines

6.9.9 Third-Party and Supply Chain Security

- All third-party components (including open-source and AI models) shall be:
 - assessed for vulnerabilities
 - monitored for updates and patches
- The bidder shall:
 - ensure no known critical vulnerabilities exist
 - update components during implementation and AMC

6.9.10 Continuous Monitoring and Incident Response

The platform shall support:

- logging and monitoring of security events
- alerting mechanisms for suspicious activity
- incident response procedures

The bidder shall:

- report security incidents to RITES within defined timelines
- support investigation and remediation

The platform shall be compliant with OWASP standards and resilient against both traditional application vulnerabilities and emerging AI/agentic AI security risks.

6.10 Delivery Schedule

Milestones & Delivery Schedule (From Date of LOA – D)

Milestone	Timeline	Key Deliverables
Milestone 1 – Inception, Architecture & Framework Design	D + 1 Month	Finalization of SRS covering all scope areas modular, API-driven architecture; Data Lake design; DPR framework for all sectors AI/ML strategy; implementation roadmap
Milestone 2 – Draft Tool Version 1 (Core Platform with AI)	D + 3 Months	Functional core platform including DPR preparation module, document management, basic workflows, initial Data Lake setup, rule-based validation engine, dashboards; AI-enabled capabilities (document parsing, summarization, basic validation using NLP/RAG) ; API framework (integration-ready)
Milestone 3 – Tool Version 2 (Enhanced Multi-Sector AI Platform)	D + 5 Months	Enhanced platform with sector-specific modules across DPR sectors; advanced validation and appraisal engine (cross-validation, scoring); expanded Data Lake and dataset integration; advanced AI capabilities (anomaly detection, assisted appraisal, pattern recognition) ; initial integration with GIS/external systems; RBAC and audit logs
Milestone 4 – Final Version of Tool (Production Deployment)	D + 7 Months	Fully integrated and production-ready platform covering all modules under complete integrations (GIS/APIs as applicable); analytics and decision-support dashboards; scenario analysis security, performance validation; UAT completion and acceptance
Milestone 5 – Stabilization &	D + 9 Months	Stabilized system with issue resolution, performance tuning, refined AI models/rules, training,

Operational Readiness		documentation, knowledge transfer, and operational support readiness
Milestone 6 - Monitoring, Correction and Maintenance	15 months from completion of Milestone 5	Monitor the deployed solution for performance, rectify wherever non-conformance is observed and Maintenance of the entire deployed solution

6.11 Payment Terms

Milestone	Deliverable	Payment (%)
Milestone 1 (D+1 Month)	Submission & Acceptance of SRS, architecture, DPR framework, Data Lake design, AI/ML strategy	10%
Milestone 2 (D+3 Months)	Submission & Acceptance of Draft Tool Version 1 (core platform with AI-enabled basic capabilities)	25%
Milestone 3 (D+5 Months)	Submission & Acceptance of Tool Version 2 (enhanced platform with sector modules, integrations, and advanced AI capabilities)	25%
Milestone 4 (D+7 Months)	Submission & Acceptance of Final Version of Tool (fully integrated, UAT approved system)	20%
Milestone 5 (D+9 Months)	Submission & Acceptance of Stabilized System (post-deployment support, training, documentation)	10%
Milestone 6 (D+24 months)	Maintenance of the entire deployed solution . Payments of 2% per quarter for total of 5 quarters (15 months)	10%
Total		100%

Payments shall be released only upon **successful submission, demonstration, and formal acceptance of deliverables** by RITES, based on compliance with scope requirements, performance benchmarks, and milestone-specific acceptance criteria.

As per RBI guidelines, payments are to be made through ECS/EFT. Vendor is required to give its consent in a mandate form (as per Annexure-10) for receipt of payment

through ECS/EFT. It includes party A/c No, Bank name, Branch address, Account type, Bank and Branch code as appears on MICR cheque issued by bank. Payment will be released through EFT only. In case of non payment through EFT or where EFT facility is not available then payment may be released through cheque.

It may be noted that Government of India has implemented Trade Receivable Discounting System (TReDS) to address challenges faced by MSMEs in delayed payments (after receipt/acceptance of services) from Government buyers leading to shortfall of Working capital. TReDS is an online electronic institutional mechanism for facilitating the financing of trade receivables of MSMEs through multiple financiers. RITES is already registered on the following TReDS platforms :

- M/s Receivable Exchange of India (RXIL)
- M/s Mynd Solutions (M1 Xchange)
- M/s TreDS Ltd (Invoicemart)

RITES shall prefer to process payments of MSME vendors through TreDS platform . In order to facilitate the same, MSME vendors may register on any one or more of the above TreDS platforms.

6.12 Deduction

Payments shall be subject to deductions such as TDS etc. or levies regulated by Govt. of India/ State Govt.

6.13 Location of Work

Please refer Annexure 13.

6.14 Failure to agree with the Terms & Conditions of the Tender/Contract

Failure of the bidder to agree with the Terms & Conditions of the Tender/Contract shall constitute sufficient grounds for the annulment of the award and forfeiture of EMD / encashment of PBG as the case be.

6.15 Data Rights

The ownership of any data related to the various services under the purview of this project lies with RITES LTD, during the entire duration of the project. Upon completion of the project and/or termination of the contract, the contractor needs to ensure that complete data is made available to RITES LTD.

6.16 Termination of contract

RITES Ltd, at its sole discretion and for any reason whatsoever, can terminate this contract at any time by giving **30 days written** notice to agency, in any of the following events :-

- i. If contractor's services are not found satisfactory with respect to time bound completion of work & nonperformance, product quality, failure to meet

warranty/service commitments, inadequate technical expertise then RITES has right to terminate the contract without assigning any reason thereof at any time after giving one-month notice to the agency and will make the alternate arrangement at the risk & cost of contractor.

- ii. The contract may also be curtailed / terminated before the contract period owing to deficiency in service or substandard quality delivery by the selected Firm /Agency. This office, however, reserves right to terminate the contract without assigning any reason thereof at any time after giving 30 days notice to the selected service providing Firm/ Agency.
- iii. RITES Ltd. reserves the absolute right to terminate the Vendor's services and this contract at any time, at its sole discretion, without assigning any reason thereof. In the event of such termination, RITES shall provide thirty (30) days' prior written notice to the selected Service Provider/Firm/Agency and payments shall strictly be released only up to the stage of services rendered and accepted to the full satisfaction of RITES prior to the date of notice. The Vendor shall immediately hand over all code, data, and documentation upon notice
- iv. Contractor is adjudged as insolvent.
- v. Contractor has abandoned the contract.
- vi. Contractor fails to proceed with the work with due diligence as per requirements of the contract.
- vii. Any major deviations from contractual terms and conditions including quality of job.
- viii. When the successful bidder is found to have made any false or fraudulent declaration or statement to get the contract, or he is found to be indulging in unethical or unfair practices.
- ix. On termination of contract for whatever be the reason bidder will ensure that knowledge transfer has been completed to the person identified by RITES. The bidder will also provide complete document of the work.

6.17 **Exit Clause**

RITES shall have the option to terminate the contract at any time during the tenure of the contract without payment of compensation for the balance period to the successful bidder by giving notice of 30 days, in writing to the successful bidder of its intention to do so. In case the exit clause is executed, payments shall strictly be released only up to the stage of services rendered and accepted to the full satisfaction of RITES

On termination of contract for whatever be the reason the consultant will ensure that knowledge transfer has been completed to the person identified by RITES. The consultant will also provide complete document of the work.

6.17 Force Majeure

- i. The contractor shall not be liable for penalty, liquidated damages or termination for default, if and to the extent that, his delay in performance or other failure to perform his obligations under the contract is the result of an event of Force Majeure.
- ii. For purposes of this clause, "Force Majeure" means an event beyond the control of the contractor and not involving the contractor and not involving the Bidders fault or negligence and not foreseeable. Such events may include, but are not restricted to, instances of wars or revolutions, fires, floods, epidemics, quarantine restrictions and freight embargoes which would have an impact on RITES LTD.
- iii. If a Force Majeure situation arises, the contractor shall promptly notify RITES LTD in writing of such conditions and the cause thereof. Unless otherwise directed by RITES LTD, the contractor shall seek all reasonable alternative means for performance not prevented by the Force Majeure event.

6.18 Resolution of Conflicts

- i. RITES LTD and the contractor shall make every effort to resolve amicably by direct informal negotiation, any disagreement or dispute, arising between them under or in connection with the contract.
- ii. If the dispute can't be resolved amicably by direct informal negotiation between RITES and the contractor, then the same shall be referred to a Sole Arbitrator to be appointed by the Chairman and Managing Director, RITES Ltd. The Arbitration and Conciliation Act, 1996 and the rules there under and any statutory modification or re-enactments thereof made till the date of signing of contract, shall apply to the arbitration proceedings.

6.19 Laws

The terms of this Tender and the contract shall be governed by the Indian laws under the exclusive jurisdiction of the competent courts in NCT of New Delhi

6.20 Notices

Any notice by one party to the other pursuant to the contract shall be sent in writing by registered post only to the addresses as defined under this contract.

6.21 Indemnity

Contractor shall indemnify and at all times keep RITES Ltd. fully indemnified and hold harmless against any and all loss, damages, costs and expenses including Attorney's fees, which may be incurred as a result of any claim or action associated with such infringement. Further the Contractor shall also

indemnify RITES Ltd. against any such claims arising at any time either during the contract period or subsequently.

6.22 Liability

- i. The selected vendor will be liable for all the deliverables.
- ii. The contractors aggregate liability in connection with obligations undertaken as part of the Project regardless of the form or nature of the action giving rise to such liability (whether in contract, tort or otherwise), shall be at actual and limited to the value of the contract.
- iii. Indemnity would be limited to court awarded damages and shall exclude indirect, consequential and incidental damages. However, indemnity would cover damages, loss or liabilities, compensation suffered by RITES Ltd. arising out of claims made by its customers and/or regulatory authorities.

6.23 Confidentiality

- i. The entire Request for Proposal (RFP) document, including the Terms of Reference, Scope of Work, conceptual architecture, PARAKH appraisal methodology, formats, processes, specifications, and any related information shared by RITES, shall remain the exclusive property of RITES Ltd. The Bidder shall use the same solely for the purpose of understanding the requirement and submitting its bid against this RFP. No part of the RFP document or any proprietary information contained therein shall be copied, reproduced, disclosed, repurposed, commercialized, or utilized by the Bidder for any other project, client, tender, commercial engagement, or governmental assignment.
- ii. Contractor understands and agrees that all data, materials and information marked and identified by RITES Ltd. as "Confidential" are valuable assets of RITES Ltd. and are to be considered RITES Ltd. proprietary information and property. **A Non- disclosure agreement has been to be executed by the successful bidder.**
- iii. Contractor will treat all confidential data, materials and information provided by RITES Ltd. with the highest degree of care necessary to insure that unauthorized disclosure does not occur.
- iv. Contractor will not use or disclose any data, materials or information provided by RITES Ltd. without RITES Ltd. 's prior written approval. Contractor shall not be liable for disclosure or use of any materials or information provided by RITES Ltd. or developed by Contractor which is:
 - a. Possessed by Contractor prior to receipt from RITES Ltd., other than through prior disclosure by RITES Ltd., as documented by Contractor 's written records;

- b. Published or available to the general public otherwise than through a breach of Confidentiality;
- c. Obtained by Contractor from a third party with a valid right to make such disclosure, provided that said third party is not under a confidentiality obligation to RITES Ltd.
- d. Developed independently by the contractor.
- v. In the event that Contractor is required by judicial or administrative process to disclose any information or materials required to be held confidential hereunder, Contractor shall promptly notify RITES Ltd. and allow RITES Ltd. a reasonable time to oppose such process before making disclosure.
- vi. Contractor understands and agrees that any use or dissemination of information in violation of this Confidentiality Clause will cause RITES Ltd. irreparable harm, may leave RITES Ltd. with no adequate remedy at law and RITES Ltd. is entitled to seek to injunctive relief.
- vii. RITES Ltd. does not wish to receive any Confidential Information of Contractor, and Contractor agrees that it will first provide or disclose information which is not confidential. Only to the extent that RITES Ltd requests Confidential Information from Contractor, will the contractor furnish or disclose Confidential Information.
- viii. Nothing herein shall be construed as granting to either party any right or license under any copyrights, inventions, or patents now or hereafter owned or controlled by the other party.
- ix. The requirements of use and confidentiality set forth herein shall survive the expiration, termination or cancellation of this contract.
- x. Confidential Information disclosed under this contract shall be subject to confidentiality obligations for a period of three years following the initial date of disclosure.

6.24 Transfer of Contractor's Rights

The Contractor shall not either wholly or partly sell, transfer assign otherwise dispose of the rights, liabilities and obligations under the contract between him and RITES Ltd. without prior consent of the RITES in writing.

6.25 Integrity Pact

- (i) All the Users in GeM i.e. Seller as well as Buyer agree not to indulge in any corrupt practices including without limitation any activity or action to influence the transaction on any aspect of contract and commit to take all

measures necessary to prevent corruption maintaining complete transparency and fairness in all activities related to GeM. Users agree to follow and adhere with the Integrity Pact guidelines provided on GeM Portal. Code of Integrity (as detailed in Rule 175 and Rule 151 of the General Financial Rules, 2017 (and its amendments, if any)) and penalties for violating the Govt of India, Ministry of Finance, Department of Expenditure shall apply to this tendering process. Procuring authorities, consultants, suppliers, contractors, and consultants should observe the highest standard of integrity and not indulge in prohibited practices or other misdemeanours, either directly or indirectly, during the entire procurement Process or the execution of resultant contracts. Bidders shall have to sign the Integrity Pact with the Procuring Entity as per 'Form T-10: Integrity Pact'. Entering into this Pact would be a preliminary qualification. Any bid not accompanied by Integrity Pact duly signed by the bidder shall be considered to be a non-responsive bid and shall be rejected straight away.

- (ii) The Bidder/Consultant is required to enter into an Integrity Pact (Enclosed as Form T-10/1, T-10/2 & T-10/3) with the Employer/Principal. The Integrity Pact (Form T-10/1 & T-10/3) duly signed by the Authorized representative of RITES Ltd. for and on behalf of 'Employer/Principal' as its Agent/Power of Attorney Holder is enclosed herewith. Bidder/Consultant shall take out the printout of the Integrity Pact (Form T-10/1, T-10/2 & T-10/3), sign on each page by authorized signatory of Bidder and duly witnessed, scan and submit/upload the same along with the Bid. In case of failure to submit the Integrity Pact, as prescribed herein, duly signed and witnessed, along with the Bid, the Bid is likely to be rejected.
- (iii) Names, addresses and e-mail ids of the Independent External Monitors(IEM) as well as the Name, designation, address and e-mail id of the official nominated by the RITES to act as the Liaison Officer between the Independent External Monitors and Procuring Officer as well as the Bidder/Consultant is mentioned below.

1. **Sh. Jitendra Sharma, IFoS (Retd.)**, House No. 85, Sector 5-A, Ecocity 1, P. O. Mullanpur Garibdas, New Chandigarh, Distt. SAS Nagar, Punjab — 140901,

Email ID: sharmaj7@gmail.com

- 2 **Mrs. Gauri Kumar, IAS (Retd.)**, Kalypso Court, Tower-1/502, JP Wishtown, Sector — 128, Noida —201304

Email ID: gkck1955@vahoo.co.in

- 3 **Sh. Cadaba Devnath Balaji, Indian Navy (Retd.)**, D-429, Jal Vayu Vihar, Kammanhalli Main Road, Bengaluru-560043

Email ID: cdbalaji@gmail.com

4. **ITES Liaison Officer: Sh. S B Samanta, Vertical Head (Finance)**, Shikhar, Plot No. 1, Sector 29, Gurgaon-122001 (Haryana), Email ID: ed.fin@rites.com

Annexure 1 : Financial Capability Statements

(On Consultant's Letter-head)

Tender No.:

Services Consultant's Name_____

[Address and Contact Details]

Consultant's Reference No._____ Date.....

Form should be CA Certified.

Annual Turnover Data (Rs Crores) for the Last Three (3) Financial Years i.e 2024-25, 2023-24 and 2022-2023		
Year	Total Turnover Amount (Total certified payments received for contracts in progress or completed)	Turnover from IT Services (Total certified payments received for contracts in progress or completed)
Average Annual Turnover		

.....

(Signature with date)

.....

(Name and designation)

Duly authorised to sign tender for and on behalf of

.....

.....

[Name, address, and seal of the Consultant]

“ Certificate of CA with UDIN

This is to certify that the information given in the above table is derived from audited B/S and P/L account and are true and correct reflection of company's financial performance and capability”.

Annexure-02 - Bidder's Details

Tender Reference No.:

Please ensure that your response clearly answers all the questions. If you use additional schedules or documentation to support your response, make sure that they are clearly cross referenced to the relevant question.

Bidder's Details

1.	Name of Company / Firm	
2.	Address of the Company (for communication)	
3.	Date of Incorporation (with document evidence for Certificate of Incorporation)	
4.	Location of Registered Office/ Corporate Office along with mailing address	
5.	E-mail address	
6.	Telephone Number(s)	
7.	Website	
8.	Address of Office in Delhi/ NCR	
9.	GST Registration Number (with document evidence)	
10.	PAN NO (with document evidence)	

Name and Details of the person to whom all references shall be made regarding the tender.

Name:

Telephone:

Mobile:

E-mail:

(Signature of the Bidder/Authorized Signatory)

Name:

Designation:

Seal of firm/ Company:

Date:

Pl. Note:

1. In-adequate information could lead to disqualification of the bid.
2. All items should be supported by proper documents.

Annexure-03- Format for Power of Attorney to Authorised Signatory

POWER OF ATTORNEY (To be executed on non-judicial stamp paper of the appropriate value in accordance with relevant Stamp Act. The stamp paper to be in the name of the firm/company who is issuing the Power of Attorney) We, M/s..... (name of the firm/company with address of the registered office) hereby constitute, appoint and authorize Mr./Ms..... (Name and residential address) who is presently employed with us and holding the position of and whose signature is given below as our Attorney to do in our name and our behalf all or any of the acts, deeds or things necessary or incidental to our bid for the Services (name of Services), including signing and submission of application/proposal, participating in the meetings, responding to queries, submission of information/documents and generally to represent us in all the dealings with RITES or any other Government Agency or any person, in connection with the services until culmination of the process of bidding, till the Contract Agreement is entered into with RITES and thereafter till the complete discharge of all the contractual obligations. We hereby agree to ratify all acts, deeds and things lawfully done by our said Attorney pursuant to this Power of Attorney and that all acts, deeds and things done by our aforesaid Attorney shall always be deemed to have been done by us.

Dated this the day of 20.....

..... (Signature and name of authorized signatory being given Power of Attorney)

..... (Signature and name in block letters of *All the partners of the firm, *Authorized Signatory for the Company) (*Strike out whichever is not applicable)

Seal of firm/ Company

Witness 1:

Name:

Address:

Occupation:

Witness 2:

Name:

Address:

Occupation:

Notes :

The mode of execution of the Power of Attorney should be in accordance with the procedure, if any, laid down by the applicable law and the charter documents of the executant(s) and when it is so required the same should be under common seal affixed in accordance with the required procedure.

Annexure-04 - Tender Offer Form (TOF)

Dated:

Tender Reference No.: RITES/ITES/APR/2026

To,

GGM/ITS
RITES Ltd
RITES Bhawan,
Plot No 1, Sector 29
Gurgaon,
Haryana - 122001

Subject: Design, Development and Implementation of Artificial Intelligence (AI) Based Detailed Project Report Appraisal Platform

Dear Sir,

Having examined the tender documents including all Annexures, I/ we, the undersigned, offer to work for subject matter above for the period of one (1) year as mentioned in Section 3 - "Scope of work" in conformity with the said tender documents.

- i. I/ We have enclosed an EMD in the form of a Demand Draft No. _____ issued by the branch of the _____ Bank, for the sum of **Rs. 18,00,000 /- (Rupees Eighteen Lakhs Only)**. This EMD is as required by clauses 5.9(vii) of the Instructions to the Bidder of the above referred tender.
- ii. I/ We undertake that the Prices are in conformity with the specifications prescribed. The quote is inclusive of all costs likely to be incurred for executing this work.
- iii. I/ we undertake, if our tender offer is accepted, to perform in accordance with the time schedule specified in the Tender.
- iv. Our offer is open for acceptance for the period of 90 days after the date fixed for bid opening and it shall remain binding upon us and may be accepted at any time before the expiry of that period.
- v. Until a formal Purchase order is prepared and executed, this bid offers, together with your written acceptance thereof and your notification of award shall constitute a binding contract between us.

vi. The scope of work has been clearly understood by us.

vii. I/We also certify that information data/ particulars furnished in our bid are factually correct. We also accept that in the event of any information / data/ particulars proving to be incorrect, RITES will have a right to disqualify us from the bid.

We agree to the terms and conditions mentioned in the Tender Document without any deviations.

Dated this ____ day of _____

Signature:

(In the Capacity of :)

Duly authorized to sign the tender offer for and on behalf of

Name:

Designation:

Seal:

Date:

Annexure-05- Self-Declaration

Dated:

Tender Reference No.: RITES/ITES/APR/2026

To,
GGM/ITS
RITES Ltd
RITES Bhawan,
Plot No 1, Sector 29
Gurgaon,
Haryana – 122001

Subject: Design, Development and Implementation Of Artificial Intelligence (AI) Based Detailed Project Report Appraisal Platform.

In response to tender No. **RITES/ITES/APR/2026** as an employee of-----
----- in the capacity of -----. I/We hereby declare :

1. That our organization/firm <organization/firm name> is having unblemished past record and was not declared ineligible for corrupt and fraudulent practices either indefinitely or for a particular period of time.
2. That we are not currently on holiday list, debarred or blacklisted by any Government entity, PSU, Bank or corporate institution in India or abroad.
3. That our organization/firm <organization/firm name> does not have any connection with employees of RITES Ltd.

The declaration submitted by us is true and correct to the best of our knowledge and if it is proved otherwise at any stage of execution of the contract, RITES has the right to summarily reject the bid and disqualify us from the process.

(Signature of the Bidder/Authorized Signatory)

Name:

Designation:

Seal:

Date:

Annexure 6: Bank Guarantee Format for Performance Security

(The Bank Guarantee shall be on a Stamp Paper of appropriate value in accordance with Stamp Act and shall be purchased in the name of the guarantee issuing Bank or the Party on whose behalf the BG is being issued.)

Ref
No.....

Bank Guarantee

Date.....
.....

To

GGM/ITS

(RITES Ltd.)

Shikhar, No. 1
Sector 29
Gurugram-122001
Haryana

Whereas M/s.....with its Registered/ Head Office at.....

(name and address of the Consultant, hereinafter called "the Consultant", which expression shall, unless repugnant to the context or meaning thereof, include its successors, administrators, executors and assigns) has undertaken, in pursuance of contract no date..... to delivery
(description of Services) (hereinafter called "the contract").

And Whereas you (unless repugnant to the context or meaning thereof, including your successors, administrators, executors and assigns) have stipulated in the said contract that the Consultant shall furnish you with a bank guarantee by a Commercial bank for the sum specified therein as security for compliance with its obligations as per the contract;

And Whereas we with our Head Office at..... (name and address of the

Bank, hereinafter referred to as the 'Bank', which expression shall, unless repugnant to the context or meaning thereof, include its successors, administrators, executors and assigns) have agreed to give the Consultant such a bank guarantee.

Now, Therefore, we hereby affirm that we are guarantors and responsible to you, on behalf of the Consultant, up to a total of (amount of the guarantee

in words and figures), and we undertake to pay you, upon your first written demand declaring the Consultant to be in default under the contract and without cavil or argument, any sum or sums within the limits of (amount of guarantee) as aforesaid, without your needing to prove or to show grounds or reasons for your demand or the sum specified therein, notwithstanding any difference between you and the Consultant or any dispute pending before any Court, Tribunal, Arbitrator or any other authority.

We hereby waive the necessity of your demanding the said debt from the Consultant before presenting us with the demand.

The Bank undertakes not to revoke this guarantee during its currency without your previous consent and further agrees that the guarantee herein contained shall continue to be enforceable till you discharge this guarantee. This guarantee will not be discharged due to a change in the constitution of the Bank or the Consultant's.

We further agree that no change or addition to or other modification of the terms of the contract to be performed thereunder or of any of the contract documents which may be made between you and the Consultant shall in any way release us from any liability under this guarantee, and we hereby waive notice of any such change, addition, or modification.

This guarantee shall be valid until theday of20..... ***[Insert date i.e. 60 days after the date of completion of all contractual obligations including warranty obligations by the Consultant]***

Our.....branch at.....*(Name & Address of the*(branch) is
liable to pay the guaranteed amount depending on the filing of a claim and any part thereof under this Bank Guarantee only and only if you serve upon us at our *
branch a written claim or demand
and received by us at our* branch on or before Dt otherwise, the bank shall be
discharged of all liabilities under this guarantee after that.

(Signature of the authorized officer of the Bank)

.....
.....

Name and designation of the officer

.....

Seal, name & address of the Bank and address of the Branch

*Preferably at the authority's headquarters competent to sanction the expenditure for procurement of goods/services or at the concerned district headquarters or the state headquarters.

Annexure-07 - Financial Offer

(Financial Bid to be submitted on GeM Portal)

(Rates are generally quoted inclusive of GST on the GeM Portal. However, bidders are advised to refer to the prevailing GeM guidelines at the time of quoting rates. No changes shall be permitted in the financial quote after opening of the bids.)

Annexure-08 - List of Documents to be submitted in Original

Tender Reference No.: RITES/ITES/APR/2026

Tender Description : Design, Development and Implementation Of Artificial Intelligence (AI) Based Detailed Project Report Appraisal Platform.

List of Documents to be submitted in Original (To be submitted within One Week of Closing Date of RFP)

S.No	Documents to be Submitted in Original
1.	Draft for EMD <i>If claiming exemption ,then document in support of Exemption</i>
2.	Bidder's Authorization Certificate (Power of Attorney on Stamp Paper Duly Notarized) as per Annexure-03

Annexure-09 - Support Information Towards Eligibility Criteria

Tender Reference No.: RITES/ITES/APR/2026

Tender Description : Design, Development and Implementation Of Artificial Intelligence (AI) Based Detailed Project Report Appraisal Platform.

Support Information Towards Eligibility Criteria

(to be uploaded as part of the bid)

Bidder shall upload only those documents that are required for this RFP (Quantity and Quality). Only the works/assignments specifically mentioned by the bidder in the table below shall be considered for eligibility/evaluation, subject to submission and verification of requisite supporting documents.

A. Performa	
Name of Work	Design, Development and Implementation Of Artificial Intelligence (AI) Based Detailed Project Report Appraisal Platform
Tender No.	
Name of Bidder	
Details of EMD	

B. Credentials & Work Experience											
S. No	Client's Name and Address	Name of Work and	Scope of Work	Agreement / Letter of Award	Contract Value	Date of Start	Date of Completion	Reason for Delay in completion	Ref. of document (with page no.) in	Document Placed At	Remarks

	ss with PIN, Phone no. and E-mail Id	Locati on	carried by the Bidder	No. and date	(Awarded)	(Actual on Completion)		(As per LOA/ Agreement)	(Actual)	on if Any	support of meeting Qualification Criterion		

Copy of the following documents:

- Purchase Order/ Work Order /Agreement signed with the client
- Client certificate mentioning satisfactory completion of assignments must be enclosed.
- Works carried out by another Contractor on behalf of the Bidder on a back to back basis will not be considered for satisfaction of the Qualification Criterion by the Bidder.
- The cutoff date shall be calculated backwards from the last stipulated date for submission/ opening of Tender.
- **ITES reserves the right to verify the bidder's submitted credentials/documents from the concerned client/authority at any stage, and any false, misleading, or unverifiable information may lead to rejection of bid and/or action as per tender conditions.**

C. Other Documents (to be uploaded along with the bid)

Document		Document Placed At	Remarks
D.1	Name of Company / Firm (Copy of the Certificate of Incorporation from the office of the Registrar of Companies / Certification of registration of the firm showing details pertaining to place of registration, principal place of business of the firm, nature of business etc. and Memorandum /Articles of Association / Partnership Deed / Proprietorship Deed/ Declaration of Proprietorship etc. as the case may be.)		

D.2	Annexure 1 certified by CA, self-attested copy of Auditor's Report along with audited Balance Sheet and Profit and Loss Statement along with Schedules for the relevant Financial Years i.e 2024-25, 2023-24 and 2022-23. Provisional Audit Reports or certified statements will not be accepted.		
D.3	GST Registration No. (copy of GST certificate)		
D.4	PAN No. (Copy of PAN Card)		
D.5	Self Declaration as per Annexure-05		
D.8	Copy of Address giving details of Location of office/support center		
D.9	Bidder's Details as per Annexure-02		
D.10	Scanned copy of Power of Attorney to Authorised Signatory duly notarized as per Annexure-03		
D.11	Scanned copy of Tender Offer Form as per Annexure-04		
D.12	Mandate Form duly authorised by the bank (Annexure-10).		

D.13	Signed and scanned Integrity Pact forms : T-10/1,T-10/2,T-10/3		
D.14	Proposed Technology Stack – Annexure 16		

Annexure-10 - Mandate Form

Tender Reference No.: RITES/ITES/APR/2026

Tender Description : Design, Development and Implementation Of Artificial Intelligence (AI) Based Detailed Project Report Appraisal Platform

MANDATE FORM

To
RITES Ltd.

Authorization for payments through Electronic Fund Transfer System (RTGS/NEFT)

Dear Sir,

We hereby authorize RITES Ltd. to make all our payments, including refund of Earnest Money, through Electronic Fund Transfer System (RTGS/NEFT). The details for facilitating the payments are given below:

(TO BE FILLED IN CAPITAL LETTERS)

1	NAME OF THE BENEFICIARY	
2	ADDRESS WITH PIN CODE	
3	(A) TELEPHONE NO. WITH STD CODE	
	(B) MOBILE NO.	
4	BANK PARTICULARS	
A	BANK NAME	
B	BANK TELEPHONE NO. WITH STD CODE	
C	BRANCH ADDRESS WITH PIN CODE	
D	BANK FAX NO. WITH STD CODE	
E	11 CHARACTER IFSC CODE OF THE BANK (EITHER ENCLOSE A CANCELLED CHEQUE OR OBTAIN BANK CERTIFICATE AS APPENDED)	
F	BANK ACCOUNT NUMBER AS APPEARING IN THE CHEQUE BOOK	
G	BANK ACCOUNT TYPE (TICK ONE) SAVING CURRENT LOAN CASH CREDIT OTHERS	SAVING CURRENT LOAN CASH CREDIT OTHERS
H	IF OTHERS, SPECIFY	
5	PERMANENT ACCOUNT NUMBER (PAN)	
6	E-MAIL ADDRESS	

I / We hereby declare that the particulars given above are correct and complete. If the transaction is delayed or credit is not effected at all for reasons of incomplete or incorrect information, I / We would not hold RITES Ltd. responsible. Bank charges for such transfer will be borne by us.

Date: _____

SIGNATURE
(AUTHORISED SIGNATORY)
Name _____

BANK CERTIFICATION

It is certified that the above mentioned beneficiary holds bank account No.
_____ with our branch and the Bank particulars above are correct.

Date: _____

SIGNATURE
(AUTHORISED SIGNATORY)
Name _____

Annexure-11
Guidelines on Banning of Business Dealings

1. Introduction

- 1.1 RITES, being a Public Sector Enterprise and 'State', within the meaning of Article 12 of Constitution of India, has to ensure preservation of rights enshrined in Chapter III of the Constitution. RITES has also to safeguard its commercial interests. It is not in the interest of RITES to deal with Agencies who commit deception, fraud or other misconduct in the execution of contracts awarded / orders issued to them. In order to ensure compliance with the constitutional mandate, it is incumbent on RITES to observe principles of natural justice before banning the business dealings with any Agency.
- 1.2 Since banning of business dealings involves civil consequences for an Agency concerned, it is incumbent that adequate opportunity of hearing is provided and the explanation, if tendered, is considered before passing any order in this regard keeping in view the facts and circumstances of the case.

2. Scope

- 2.1 The procedure of (i) Suspension and (ii) Banning of Business Dealing with Agencies, has been laid down in these guidelines.
- 2.2 It is clarified that these guidelines do not deal with the decision of the Management not to entertain any particular Agency due to its poor/inadequate performance or for any other reason.
- 2.3 The banning shall be with prospective effect, i.e., future business dealings.

3. Definitions

In these Guidelines, unless the context otherwise requires:

- i) 'Consultant/Bidder/Contractor/Supplier' in the context of these guidelines is indicated as 'Agency'.
- ii) 'Competent Authority' shall mean the Executive Director of the Division.
- iii) 'First Appellate Authority' shall mean Director in charge of the Division and
- iv) "Second Appellate Authority" shall mean the CMD, RITES.
- v) 'Investigating Department' shall mean any Department, Division or Unit investigating into the conduct of the Agency and shall include the Vigilance

Department, Central Bureau of Investigation, the State Police or any other department set up by the Central or State Government having powers to investigate.

- vi) 'Banning Committee' shall mean a Committee constituted for the purpose of these guidelines by the **CMD**. The members of this Committee shall not, at any stage, be connected with the tendering process under reference.

4. Initiation of Banning/Suspension

Action for banning/suspension business dealings with any Agency should be initiated by the department/unit having business dealings with them after noticing the irregularities or misconduct on their part.

5. Suspension of Business Dealings

- 5.1 If the conduct of any Agency dealing with RITES is under investigation by any department, the Competent Authority may consider whether the allegations under investigation are of a serious nature and whether pending investigation, it would be advisable to continue business dealing with the Agency. If the Competent Authority, after consideration of the matter including the recommendation of the Investigating Department/Unit, if any, decides that it would not be in the interest to continue business dealings pending investigation, it may suspend business dealings with the Agency. The order to this effect may indicate a brief of the charges under investigation. The order of such suspension would operate for a period not more than six months and may be communicated to the Agency as also to the Investigating Department.

The Investigating Department/Unit may ensure that their investigation is completed and whole process of final order is over within such period.

- 5.2 As far as possible, the existing contract(s) with the Agency may be continued unless the Competent Authority, having regard to the circumstances of the case, decides otherwise.
- 5.3 If the Agency concerned asks for detailed reasons of suspension, the Agency may be informed that its conduct is under investigation. It is not necessary to enter into correspondence or argument with the Agency at this stage.
- 5.4 It is not necessary to give any show-cause notice or personal hearing to the Agency before issuing the order of suspension. However, if investigations are not complete in six months' time, the Competent Authority may extend the period of suspension by another three months, during which period the investigations must be completed.

6. Grounds on which Banning of Business Dealings can be initiated

- 6.1 If the security consideration, including questions of loyalty of the Agency to the

State, so warrants;

- 6.2 If the Director/Owner of the Agency, proprietor or partner of the firm, is convicted by a Court of Law for offences involving moral turpitude in relation to its business dealings with the Government or any other public sector enterprises or RITES, during the last five years;
 - 6.3 If there is strong justification for believing that the Directors, Proprietors, Partners, owner of the Agency have been guilty of malpractices such as bribery, corruption, fraud, substitution of tenders, interpolations, etc.;
 - 6.4 If the Agency employs a public servant dismissed/removed or employs a person convicted for an offence involving corruption or abetment of such offence;
 - 6.5 If business dealings with the Agency have been banned by the Govt. or any other public sector enterprise;
 - 6.6 In case any information/document which may result in the tenderer's disqualification is concealed by the Tenderer or any statement/information/document furnished by the Tenderer or issued by a Bank/Agency/third party and submitted by the tenderer, is subsequently found to be false or fraudulent or repudiated by the said Bank/Agency/Third Party.
 - 6.7 If the Agency uses intimidation/threatening or brings undue outside pressure on the Company (RITES) or its official in acceptance/performance of the job under the contract;
 - 6.8 If the Agency indulges in repeated and/or deliberate use of delay tactics in complying with contractual stipulations;
 - 6.9 Based on the findings of the investigation report of CBI/Police against the Agency for mala fide/unlawful acts or improper conduct on his part in matters relating to the Company (RITES) or even otherwise;
 - 6.10 Established litigant nature of the Agency to derive undue benefit;
 - 6.11 Continued poor performance of the Agency in several contracts;
- (Note: The examples given above are only illustrative and not exhaustive. The Competent Authority may decide to ban business dealing for any good and sufficient reason).

7. Banning of Business Dealings

- 7.1 A decision to ban business dealings with any Agency shall apply throughout the Company.
- 7.2 If the Competent Authority is prima-facie of view that action for banning business dealings with the Agency is called for, a show-cause notice may be issued to the Agency as per paragraph 8.1 and an enquiry held accordingly.

8. Show-cause Notice

- 8.1 In case where the Competent Authority decides that action against an Agency is called for, a show-cause notice has to be issued to the Agency. Statement containing the imputation of misconduct or misbehaviour may be appended to the show-cause notice and the Agency should be asked to submit within 30 days a written statement in its defence. If no reply is received, the decision may be taken ex-parte.
- 8.2 If the Agency requests for inspection of any relevant document in possession of RITES, necessary facility for inspection of documents may be provided.
- 8.3 On receipt of the reply of the Agency, or in case no reply is received within the prescribed time, the Competent Authority shall refer the case along with relevant details to the Banning Committee, which shall examine the reply of the Agency and other facts and circumstances of the case and submit its final recommendation to the Competent Authority for banning or otherwise. In case the action contemplated against the Agency includes forfeiture of EMD also besides Banning of Business Dealings. A final decision on forfeiture or otherwise of the EMD and for Company-wide banning or otherwise shall be taken by the Competent Authority. The Competent Authority may consider and pass an appropriate speaking order:
- a) For completely exonerating the Agency; or
 - b) For forfeiture of EMD but for not banning of business dealings with the Agency; or
 - c) For forfeiture of EMD and banning the business dealing with the Agency.
- 8.4 The decision should be communicated to the Agency concerned along with a reasoned order. If it is decided to ban business dealings, the period for which the ban would be operative may be mentioned.

9. Appeal against the Decision of the Competent Authority

- 9.1 The Agency may file an appeal against the order of the Competent Authority banning business dealing, etc. to the First Appellate Authority.
- 9.2 In the event the Agency is dissatisfied with the decision of the First Appellate Authority, it shall file a second appeal to the Second Appellate Authority.
- 9.3 The Appeals shall be preferred within one month from the receipt of the order of the Competent Authority or First Appellate Authority, as the case may be.
- 9.4 Appellate Authorities would consider the appeal and pass appropriate order which shall be communicated to the Agency as well to the Competent Authority.

10. Review of the Decision by the Competent Authority

Any petition/application filed by the Agency concerning the review of the banning order passed originally by Competent Authority under the existing guidelines either before or after filing

of appeal before the Appellate Authority or after disposal of appeal by the Appellate Authority, the review petition can be decided by the Competent Authority upon disclosure of new facts/circumstances or subsequent development necessitating such review.

11. Circulation of the names of Agencies with whom Business Dealings have been banned.

- 11.1 Depending upon the gravity of misconduct established, the Competent Authority of RITES may circulate the names of Agency with whom business dealings have been banned, to the Ministry of Railways and PSUs of Railways, for such action as they deem appropriate.
- 11.2 If Ministry of Railways or a Public Sector Undertaking of Railways request for more information about the Agency with whom business dealings have been banned a copy of the report of Inquiring Authority together with a copy of the order of the Competent Authority/Appellate Authority may be supplied.

12. Restoration

- 12.1 The validity of the banning order shall be for a specific time & on expiry of the same, the banning order shall be considered as "withdrawn".
- 12.2 In case any agency applies for restoration of business prior to the expiry of the ban order, depending upon merits of each case, the Competent Authority which had passed the original banning orders may consider revocation of order of suspension of business/lifting the ban on business dealings at an appropriate time. Copies of the restoration orders shall be sent to all those offices where copies of Ban Orders had been sent.

Annexure-12 -

Details of the AI based APR Appraisal Platform

Introduction

RITES operates in following sectors but not limited to :-

- Rail Infrastructure (Sidings / Corridors / FLS)
- Bridges
- Tunnels
- Metro
- Mobility / Logistics Plans
- Highways
- Ports
- Airports

RITES provides Detailed Project Reports (DPRs) across all the above sectors. DPR forms the foundation of project planning and investment decisions. The quality of a DPR directly influences project cost, timelines, risk exposure, and long-term sustainability. However, current DPR appraisal practices remain largely manual, fragmented, and document-centric, relying on checklists, spreadsheets, and siloed workflows. These approaches often lead to incomplete inputs, inconsistencies across sections, limited traceability, and weak institutional memory.

The indicative details of DPR are as follows

- (i) DPR size is typically in the range of 500–1000 pages.
- (ii) DPRs may extend up to ~1000 pages or more.
- (iii) DPR formats are not fully standardized and vary across sectors.
- (iv) DPRs are available in digital, scanned, and physical formats.
- (v) DPRs are primarily in English, with possible multi-lingual content. The repository consists of ~1000+ DPRs.

PARAKH (AI based DPR Appraisal platform); at its core, transforms DPR appraisal from a document review exercise into a data-driven, intelligence-enabled decision-support system. The platform leverages Artificial Intelligence and advanced analytics to enable intelligent validation, anomaly detection, benchmarking, and predictive insights across DPRs, thereby enhancing the speed, consistency, and quality of appraisal processes.

It is a configurable, data-driven digital system designed to transform the way Detailed Project Reports (DPRs) are prepared, evaluated, and continuously improved across infrastructure, consultancy, and technology projects.

PARAKH addresses these challenges by introducing a **structured, integrated, and lifecycle-driven digital environment** for DPR preparation, validation, appraisal, and knowledge management.

At its core, the platform transforms DPR appraisal from a **document review exercise** into a **data-driven, intelligence-enabled decision-support system**. It integrates structured data capture, engineering logic, geospatial intelligence, and historical benchmarking within a unified architecture supported by a centralized Data Lake.

The platform enables:

- Structured preparation and submission of DPRs using configurable templates
- Context-aware validation using engineering dependencies, site conditions, and standards
- Integration with GIS systems and national platforms such as PM GatiShakti for spatial and multi-modal analysis
- Evaluation of alternative scenarios, including route alignment and design options
- Multi-scenario analysis combining technical, financial, and spatial parameters
- Detection of inconsistencies, anomalies, and unrealistic assumptions
- Configurable appraisal workflows, scoring frameworks, and decision dashboards
- Creation of an institutional knowledge base for benchmarking and continuous learning

A key differentiator of the platform is its ability to incorporate **site-specific intelligence**, including terrain, soil characteristics, hydrology, environmental conditions, and infrastructure context, enabling **context-aware and realistic appraisal outcomes**.

The platform is designed as a **closed-loop system**, where insights from validation, appraisal, and benchmarking are continuously fed back into DPR preparation. This ensures progressive improvement in DPR quality, consistency, and decision-making over time.

Artificial Intelligence and advanced analytics are incorporated as **assistive and advisory components**, supporting anomaly detection, benchmarking, and pattern recognition, while ensuring that final decisions remain with designated experts and authorities.

PARAKH is not intended to replace expert judgment. Instead, it enhances transparency, consistency, and efficiency in appraisal processes while enabling organizations to transition from **experience-based evaluation to evidence-based, data-driven decision-making**.

This document presents the conceptual architecture, functional modules, and operational principles of the platform, providing a foundation for detailed design, implementation, and stakeholder engagement.

Current methods of Creation and DPR appraisal

1. Manual and Spreadsheet-Based Appraisal

This is still the dominant practice.

Typical workflow:

- DPR submitted in PDF/Word
- Reviewers use:
 - checklists
 - comments
 - Excel sheets
 - internal notes

Even structured scoring systems (like those used by agencies for evaluating DPR consultants) are generally applied manually or semi-manually using templates and scoring sheets rather than software platforms. The evaluation logic exists, but not digitized as a system.

2. Document Management Systems (DMS)

Some organizations use:

- eOffice
- SharePoint
- PMIS systems

These provide:

- document storage
- workflow routing
- approvals

But they do **not** provide:

- technical validation
- rule engines
- engineering consistency checks

They are repositories, not appraisal tools.

3. Domain-Specific Engineering Software

There are tools for:

- traffic modelling
- geotechnical analysis
- cost estimation
- BIM
- project planning

But these tools:

- analyze one discipline at a time
- do not evaluate the DPR as a whole
- do not detect missing sections or inconsistencies across domains

So they are analytical tools, not appraisal platforms.

What Does Not Exist (Gap in the Ecosystem)

There is no system that can:

- check completeness of a DPR
- verify stakeholder inputs
- detect missing utilities or plots

- flag unrealistic technical inputs
- cross-check quantities and costs
- generate appraisal summaries
- maintain audit trails of technical review

All of these activities are currently done manually.

Platform Architecture Overview

1. Architectural Philosophy

PARAKH is designed as a **modular, data-driven, and configurable digital system** that supports the complete lifecycle of DPR preparation, validation, appraisal, and institutional learning.

The architecture is guided by the following principles:

- **Separation of concerns** between preparation, appraisal, and knowledge layers
- **Engineering logic-driven validation**, supported by structured data and dependencies
- **Scalability across sectors and project types**
- **Human-in-the-loop decision support**, with explainable and auditable outputs
- **Extensibility** to incorporate additional datasets, rules, and analytics over time

Integration with GIS systems and national data platforms such as PM GatiShakti to enrich the Data Lake and support spatial and multi-modal analysis.

The platform shall support integration with route and alignment optimization tools, enabling evaluation of alternative alignments based on terrain, cost, environmental constraints, and multi-modal connectivity.

The platform shall incorporate site-specific datasets including soil characteristics, hydrology, terrain, and environmental conditions to enable context-aware validation and appraisal.

The system shall implement a standardized data model and ontology framework to ensure semantic interoperability across modules and external systems

All datasets ingested shall be mapped to canonical schemas with metadata tagging

The platform shall support master data management (MDM) and data harmonization across sectors (Railways, Highways, Ports, etc.)

Data lineage and traceability shall be maintained for all exchanged datasets

The platform shall adhere to open interoperability standards and protocols, including:

- API standards such as REST / GraphQL / gRPC with OpenAPI-compliant documentation
- Standard data exchange formats such as JSON, XML, GeoJSON, and IFC (for BIM where applicable)

- Schema-based data exchange with version control and backward compatibility
- API gateway-based architecture for secure, scalable, and managed integration
- Support for event-driven architecture (publish-subscribe / streaming) for real-time data

2. Three-Layer Architecture

The platform is organized into three logical layers that together support the DPR lifecycle:

A. DPR Preparation Layer

This layer enables structured creation and submission of DPRs.

Key functions include:

- Configurable DPR templates (chapters, sections, tables)
- Structured data entry for engineering and financial parameters
- Upload / Import (or through API) and management of supporting documents and drawings
- Guided data entry and basic validation checks
- Version control and submission workflows

Objective:

To ensure that DPRs are **complete, consistent, and digitally structured at the point of creation**, reducing downstream appraisal effort.

B. Appraisal and Evaluation Layer

This layer performs systematic validation, review, and scoring of DPRs.

Key functions include:

- Completeness validation (sections, inputs, stakeholder submissions)
- Technical validation using rules, standards, and contextual checks
- Cross-section consistency verification
- Anomaly detection using patterns and comparative analysis
- Structured review workflows and issue tracking
- Configurable scoring frameworks (internal and external)

Objective:

To enable **transparent, structured, and repeatable appraisal processes** supported by engineering logic and validation frameworks.

C. Knowledge and Data Layer

This layer provides the data backbone for benchmarking, analytics, and institutional learning. This layer is implemented through a centralized Data Lake architecture, which enables ingestion, storage, structuring, and analysis of multi-source datasets including DPRs, engineering parameters, GIS layers, and socio-economic data. The Data Lake serves as the foundational data backbone supporting validation, benchmarking, analytics, and decision-support functions across the platform.

Key functions include:

- Storage of historical DPRs and appraisal outcomes
- Management of structured datasets including engineering parameters, cost data, and project characteristics
- Integration of external datasets such as GIS layers, demographic and socio-economic data

- Benchmarking and comparative analysis across projects
- Support for analytics, anomaly detection, and decision-support insights

Objective:

To transform DPR appraisal from a **document-centric process to a data-driven system**, enabling continuous improvement and evidence-based decision-making.

3. Data-Centric Architecture

The platform adopts a **data-centric architecture**, where structured and unstructured data are systematically captured, organized, and reused across modules.

Key aspects include:

- Integration of **structured inputs** (tables, parameters) and **unstructured content** (documents, drawings)
- Creation of a **centralized data repository** for DPRs and related datasets
- Use of **metadata and classification frameworks** for tagging and retrieval
- Enablement of **cross-project benchmarking and analytics**
- Support for **future AI and advanced analytics capabilities**

This approach ensures that data captured during DPR preparation and appraisal can be reused for validation, benchmarking, and continuous learning.

The platform **should support** advanced analytical and simulation capabilities, including scenario analysis and optimization techniques (such as route alignment optimization, cost optimization, and resource allocation), based on available data and through integration with relevant systems where required.

The platform shall support a configurable engineering computation layer to derive quantities, costs, and key design parameters based on input data, rules, and reference standards.

4. Modular and Extensible Design

The platform is composed of independent but interlinked functional modules.

- Each module performs a **specific role** (e.g., template management, validation, workflow, reporting)
- Modules interact through **well-defined data flows and interfaces**
- New modules or capabilities (e.g., advanced analytics, additional sector templates) can be added without redesigning the entire system

This modular approach enables:

- phased implementation
- flexibility in configuration
- adaptability across sectors and organizations

5. Integration, Interoperability, and Standards Compliance

The platform shall be designed as an **interoperability-first, open, and standards-compliant system**, enabling seamless integration across internal modules, external systems, and future digital ecosystems. The architecture shall adhere to **open standards, Digital Public Infrastructure (DPI) principles, and vendor-neutral design**.

5.1. Integration Architecture

- The system shall adopt a **modular, API-first architecture** enabling seamless integration with enterprise and external systems.
- Integration shall be supported through:
 - **RESTful APIs, GraphQL, and gRPC services**
 - **Event-driven architecture using streaming platforms (e.g., Apache Kafka or equivalent)**
- All APIs shall:
 - Be documented using **OpenAPI/Swagger specifications**
 - Support **versioning, backward compatibility, and defined deprecation policies**
- APIs shall be exposed through a **managed API Gateway** with:
 - Authentication and authorization
 - Throttling and rate limiting
 - Logging and monitoring
- The platform shall support both:
 - **Synchronous (API-based)** and
 - **Asynchronous (event-based / publish-subscribe)** integration patterns

5.2. Data Exchange and Semantic Interoperability

- The platform shall support standard data exchange formats including:
 - **JSON, XML**
 - **GeoJSON (GIS)** and **IFC/CityGML (BIM/infrastructure)** where applicable
- The system shall implement:
 - **Canonical data models and standardized schemas**
 - **Metadata-driven data exchange mechanisms**
 - **Ontology-based data relationships (approved by RITES)**
- All integrations shall ensure:
 - **Schema versioning**
 - **Data lineage and traceability**
 - **Auditability of data exchange**

5.3. Event-Driven and Real-Time Interoperability

- The platform shall support **event-driven architecture** using publish-subscribe mechanisms and streaming platforms (e.g., Kafka or equivalent)
- Event data shall:
 - Follow standardized message formats
 - Use schema registry mechanisms for consistency
- The system shall support:
 - **Real-time and near real-time data exchange**
 - Fault-tolerant message handling
 - Stream processing for analytics

5.4. Integration with External and National Systems

The platform shall support seamless integration with:

- GIS platforms
- BIM and engineering systems (e.g., Autodesk, Bentley, Revit)
- National platforms such as **PM GatiShakti** (where applicable)
- Other enterprise and third-party systems

Additional capabilities:

- Provision of a **sandbox / developer environment** for integration testing and onboarding
- Support for **federated data access mechanisms**, enabling integration without mandatory data ingestion

5.5. Security and Access Interoperability

- The platform shall support:
 - **OAuth2.0, SAML, and token-based authentication mechanisms**
- API security shall comply with:
 - **OWASP API Security Top 10 and industry best practices**
- All integrations shall ensure:
 - Secure data exchange
 - Encryption in transit
 - Role-based access control

5.6. Mobile, Edge, and Lightweight Interoperability

- The platform shall be **device-agnostic**, supporting:
 - Desktops
 - Tablets (including iPad)
 - Mobile devices
- APIs and payloads shall be:
 - Optimized for **low bandwidth and edge environments**
- Where required, the system may support:
 - **Offline or low-connectivity synchronization mechanisms**

7. Interoperability Testing and Compliance Validation

The bidder shall demonstrate interoperability through:

- API exposure and testing using standard tools (e.g., Postman/OpenAPI)
- Integration with:
 - At least one external system or
 - A simulated environment
- Demonstration of:
 - Structured data exchange via APIs
 - Event-driven workflows (Kafka-based or equivalent)
- Validation of:
 - Schema compliance
 - Data integrity across systems

Interoperability compliance shall be validated through:

- API conformance testing
- Schema validation
- Cross-system data exchange testing prior to Go-Live

Failure to demonstrate interoperability may result in non-acceptance of deliverables.

5.8. Standards Compliance

The platform shall comply with relevant **Indian and global standards**, including but not limited to:

Infrastructure & Engineering Standards

- RDSO (Railways)

- IRC / MoRTH (Highways & Bridges)
- BIS / NBC (Buildings)
- ICAO / AAI (Airports)
- Port and Maritime standards (as applicable)

Technology & Data Standards

- ISO 27001 (Information Security)
- ISO 9001 (Quality)
- ISO 19650 (BIM, where applicable)
- OpenAPI Specification (API standards)
- JSON / XML data standards
- OGC standards (GIS interoperability, where applicable)

5.9. Vendor Neutrality and DPI Alignment

- The platform shall be:
 - **Vendor-neutral**
 - Based on **open standards**
 - Free from restrictive dependencies
- The architecture shall align with **Digital Public Infrastructure (DPI) principles**, including:
 - Openness
 - Interoperability
 - Scalability
 - Federated architecture
 - Minimal vendor lock-in

5.10. Performance and Optimization Considerations

- Integration and API layers shall be:
 - **Lightweight and efficient**
 - Optimized for **low latency and minimal resource consumption**
- AI/ML integrations shall:
 - Avoid unnecessary compute overhead
 - Be optimized for **scalable and cost-efficient deployment environments**

6. Role of AI and Advanced Analytics

Artificial Intelligence and advanced analytics are envisaged as **assistive components** within the architecture.

They may be used to support:

- anomaly detection and outlier identification
- benchmarking against historical datasets
- pattern recognition across projects
- generation of risk indicators and decision-support insights

All outputs shall be:

- explainable
- auditable
- advisory in nature

Final appraisal decisions shall remain with designated authorities.

7. Deployment and Hosting Approach

The platform shall support a **scalable and secure deployment model**, including:

- Cloud-based or hybrid infrastructure
- Modular deployment of components
- Secure access and role-based controls

The **Selected Bidder** shall provision, manage, and maintain the required infrastructure, tools, environments, software components, and associated resources for development, configuration, integration, testing, demonstration, and User Acceptance Testing (UAT) of the DPR Evaluation Tool.

The **Selected Bidder** shall clearly inform RITES of the detailed hosting and infrastructure requirements for production deployment. Post successful completion of UAT and acceptance by RITES, the system shall be deployed on the production infrastructure designated/provided by RITES. The **Selected Bidder** shall be responsible for installation, configuration, migration, hardening, integration, and handholding support required for successful deployment and operationalization of the system on the RITES-designated infrastructure.

The **Selected Bidder** shall ensure that all development, testing, and UAT environments comply with applicable data security, cyber security, access control, backup, confidentiality, and data protection requirements **in line with applicable MeitY/Government of India guidelines, policies, and advisories issued from time to time**.

Any test data, DPR documents, credentials, logs, or project information used during development/UAT shall not be stored, transferred, or processed on any unauthorized third-party/public platform.

Data Lake, Data Ingestion and Data Management Architecture

1. Overview

PARAKH shall be supported by a **centralized Data Lake architecture** designed to enable ingestion, storage, structuring, management, and analysis of diverse datasets relevant to DPR preparation, appraisal, benchmarking, and decision support. The Data Lake shall serve as the **foundational data backbone** of the platform, enabling transition from a document-centric approach to a **data-driven appraisal ecosystem**. The Data Lake functions as an intelligence backbone, enabling continuous learning, benchmarking, and AI-assisted decision support across the DPR lifecycle.

1A. Role within Platform Architecture

The Data Lake forms the core of the **Knowledge and Data Layer** of the platform architecture and supports both the DPR Preparation Layer and Appraisal Layer through structured data access and reuse.

- The **DPR Preparation Layer** utilizes the Data Lake for reference data, benchmarks, and guided inputs
- The **Appraisal Layer** leverages the Data Lake for validation, consistency checks, benchmarking, and analytics

This integration ensures that all stages of the DPR lifecycle are supported by a consistent and reusable data foundation.

2. Objectives

The Data Lake architecture is intended to:

- Enable **systematic ingestion and organization of historical and ongoing DPR datasets**
- Support **structured storage of engineering, financial, and project-related parameters**
- Integrate **external datasets** such as GIS, spatial, demographic, and socio-economic data
- Provide a **reliable data foundation for validation, benchmarking, and analytics**
- Facilitate **institutional learning through reuse of historical project knowledge**

3. Data Sources

The Data Lake shall support ingestion of data from multiple sources, including but not limited to:

A. DPR and Project Data

- Historical DPRs (PDF, Word, Excel formats)
- Ongoing DPR submissions through the platform
- Drawings, reports, and supporting documents

B. Engineering and Cost Data

- Engineering parameters and design assumptions
- Cost estimates and bill of quantities
- Standard schedules and benchmarks

C. Spatial and GIS Data

- Base maps and spatial layers
- Location-specific project data
- Environmental and terrain-related datasets

D. Socio-Economic and External Data

- Census and demographic datasets
- SECC and related socio-economic indicators
- Traffic, demand, and planning datasets
- Any other relevant external datasets

E. Geotechnical and Environmental Data

- Soil investigation data and classifications
- Bearing capacity and geotechnical parameters
- Groundwater conditions
- Hydrology, flood levels, and river characteristics

Environmental and ecological datasets, including wind zones, seismic zones, temperature ranges, rainfall patterns, flood risk, and other relevant environmental parameters

4. Data Ingestion Framework

The platform shall provide a structured mechanism for **data ingestion and onboarding**, including:

- Upload and ingestion of DPRs in multiple formats (PDF, Word, Excel)

- Integration with external systems and datasets through APIs
- Batch ingestion of legacy DPR repositories
- Assisted extraction and structuring of data from unstructured documents
- Provision for future real-time or incremental data ingestion where required

All ingested data shall be subject to validation and classification processes.

5. Data Structuring and Transformation

Given the heterogeneous nature of DPR data, the Data Lake shall support:

- Conversion of **unstructured documents into structured datasets**, where feasible
- Mapping of DPR contents to standardized templates and data models
- Extraction of key parameters (technical, financial, spatial)
- Creation of **sector-specific structured datasets**
- Tagging and classification based on:
 - sector
 - project type
 - geography
 - project size and cost range

The system shall allow **human validation and correction** to ensure data accuracy.

6. Metadata and Data Catalog Management

The platform shall maintain a **comprehensive metadata and cataloging framework**, including:

- Dataset classification and tagging
- Searchable data catalog for discovery and reuse
- Standardized naming conventions and identifiers
- Version control and lineage tracking of datasets

This enables efficient retrieval, traceability, and governance of data assets.

7. Data Quality, Validation and Governance

To ensure reliability of data used for appraisal and analytics, the Data Lake shall incorporate:

- Data validation mechanisms during ingestion and transformation
- Identification of incomplete, inconsistent, or duplicate data
- Audit trails for data updates and corrections
- Role-based access control and data security measures
- Data governance policies for usage and updates

8. Data Access and Integration Layer

The Data Lake shall expose data through controlled access mechanisms, including:

- APIs for consumption by:
 - validation engines
 - benchmarking modules
 - analytics and AI components
 - reporting dashboards
- Role-based access for users and stakeholders
- Secure integration with internal and external systems

9. Support for Benchmarking and Analytics

The Data Lake shall enable:

- Creation of **benchmark datasets across sectors and project types**
- Comparative analysis of project parameters
- Identification of trends, patterns, and recurring issues
- Support for anomaly detection and risk identification
- Generation of inputs for decision-support dashboards

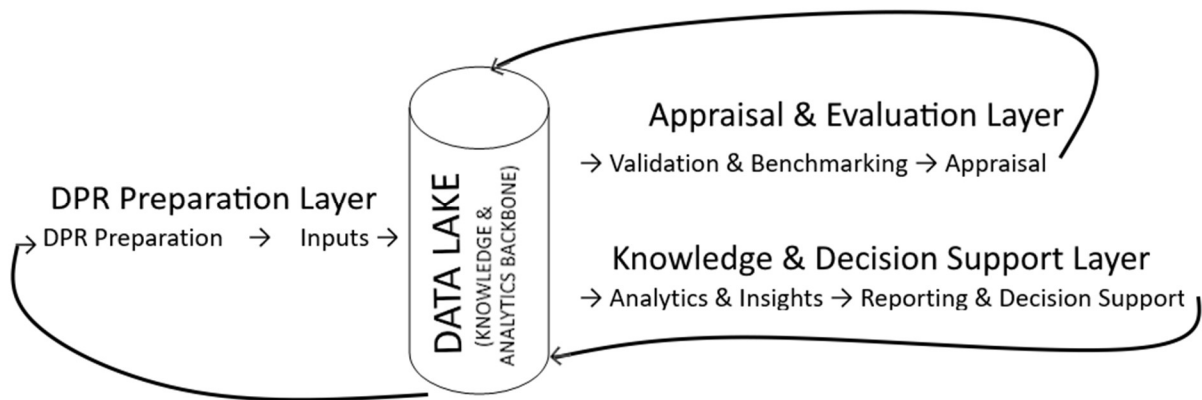
This capability is central to improving DPR quality and appraisal outcomes over time.

10. Relationship with Platform Modules

The Data Lake architecture shall function as a **shared foundational layer** supporting multiple modules, including:

- Validation and consistency engines
- Scoring and appraisal modules
- Historical DPR repository
- Reporting and decision-support systems
- AI and analytics components

All modules shall leverage datasets managed within the Data Lake to ensure consistency and reuse.



The platform operates as a closed-loop system, where data captured during DPR preparation and appraisal is continuously leveraged for validation, benchmarking, analytics, and feedback, enabling improved quality and consistency of future DPRs. The platform operates as a continuous closed-loop system where insights from appraisal, validation, and benchmarking are fed back into DPR preparation to improve quality, consistency, and decision-making over time.

10. Global Knowledge Ingestion and Standards Layer (Data Architecture)

10.1 Knowledge-Driven Data Architecture

The platform shall be designed as a **knowledge-centric data architecture**, wherein structured and unstructured data, standards, codes, and guidelines form an integral part of the data ecosystem.

The Data Architecture shall include a dedicated **Knowledge Layer** to support ingestion, structuring, harmonization, and utilization of both **Indian and global standards, frameworks, and best practices**.

10.2 Ingestion of Global and National Knowledge Sources

The bidder shall design and implement data pipelines to ingest:

a. Indian Standards and Regulatory Sources

- RDSO (Railways)
- IRC, MoRTH (Highways, Bridges, Tunnels)
- BIS (IS Codes)
- NBC, CPWD manuals (Buildings)
- AAI (Airports)
- MoPSW / Port Authorities (Ports & Harbours)
- NDMA, MoEFCC (Environment & Safety)
- Any other sector-specific regulatory frameworks

b. Global Standards and Knowledge Sources

- ISO, IEC, ASTM, EN standards
- BIM standards (ISO 19650)
- Asset management standards (ISO 55000)
- Multilateral frameworks (World Bank, ADB, AIIB)
- Sectoral bodies (UIC, ICAO, PIANC, etc.)
- Global project datasets, benchmarks, and case studies

10.2 Knowledge Structuring and Ontology Integration

The platform shall:

- Transform ingested content into:
 - structured datasets
 - rule libraries
 - validation parameters
- Integrate with:
 - ontology frameworks
 - DPR templates and schema
- Enable mapping between:
 - DPR inputs
 - applicable standards (Indian/global)

10.3 Harmonization Engine

The Data Architecture shall include a **Standards Harmonization Engine** that:

- aligns global standards with Indian regulatory requirements
- applies configurable precedence rules (e.g., Indian standards override global standards where applicable)
- maintains traceability of:
 - source standards applied
 - conflict resolution logic

10.4 Continuous Data and Knowledge Updates

The platform shall support:

- periodic ingestion and updating of:
 - revised standards
 - new guidelines
 - emerging technologies
- configuration-driven updates without major code changes
- version control and audit trails for all standards and rules

10.5 Continuous Data and Knowledge Updates

The platform shall support:

- periodic ingestion and updating of:

- revised standards
 - new guidelines
 - emerging technologies
- configuration-driven updates without major code changes
- version control and audit trails for all standards and rules

10.6 Benchmarking and Reference Data Layer

The Data Architecture shall include:

- a **benchmarking repository** containing:
 - historical DPR datasets
 - global cost and performance benchmarks
- capability to:
 - compare DPR inputs against global datasets
 - identify deviations and optimization opportunities

10.7 AI-Enabled Knowledge Processing

The data layer shall support:

- AI/ML models for:
 - document parsing and knowledge extraction
 - pattern recognition and anomaly detection
 - recommendation of best practices
- explainability features indicating:
 - source datasets
 - applied standards and rules

10.8 Governance and Control

- RITES shall:
 - approve standards, ontologies, and knowledge frameworks
 - define validation and benchmarking logic
- The bidder shall:
 - implement ingestion pipelines and knowledge repositories
 - ensure data quality, integrity, and traceability

10.9 Compliance and Risk Flagging

The platform shall:

- automatically flag:
 - deviations from applicable standards
 - gaps against global benchmarks
- generate:
 - structured compliance reports
 - risk scores for decision support

The Data Architecture shall incorporate a knowledge-driven layer capable of ingesting, structuring, and harmonizing both Indian and global standards to enable intelligent, benchmark-driven DPR appraisal.

11. Scalability and Extensibility

The Data Lake shall be designed to:

- Scale across multiple sectors and large volumes of DPR data
- Support addition of new datasets and data sources over time
- Accommodate evolving validation rules, benchmarks, and analytics models
- Enable future integration with digital engineering systems and external platforms

PARAKH Modules

Module No.	Module Name	One-Line Objective / Outcome
1	Governance, Users & Roles	Establish accountability, access control, and auditability across DPR preparation and appraisal workflows.
2	Master Data Management	Define standardized reference data, classifications, and site/context parameters to support structured appraisal and validation.
3	DPR Template Creation Engine	Configure DPR structures, chapters, and mandatory inputs to ensure uniform and complete submissions.
4	Role-to-DPR Mapping	Assign responsibility for preparation, submission, and review of specific DPR sections to relevant stakeholders.
5	Dictionary and Standards Repository	Maintain controlled terminology, units, and reference standards to support consistent interpretation and validation.
6	Ontology and Dependency Framework	Define relationships among parameters and sections to enable context-aware validation and logical consistency checks.
7	DPR Authoring / Submission Interface	Provide structured tools for preparing, uploading, revising, and submitting DPRs in an organized and traceable manner.
8	Completeness Validation Engine	Identify missing sections, incomplete inputs, and absent stakeholder submissions before appraisal progresses.
9	Technical Validation Engine	Detect incorrect, unrealistic, or non-compliant technical inputs using rules, references, and contextual checks.
10	Consistency Engine	Cross-verify related parameters and sections to identify mismatches and logical inconsistencies across the DPR.
11	Anomaly Detection Engine	Identify unusual values, patterns, or outliers that may indicate errors or unrealistic assumptions.
12	Review Workflow and Observations	Enable structured review, issue tracking, iterative revisions, and configurable scoring using internal or external appraisal frameworks.
13	Reporting and Appraisal Dashboard	Provide consolidated dashboards, summaries, and decision-support views of DPR quality, risks, and appraisal status.

14	Simulation and Sensitivity Analysis	Support scenario modelling and what-if analysis to assess the robustness of assumptions and key project parameters.
15	Security, Audit and Monitoring	Ensure secure access, activity logging, data protection, and traceability across all platform operations.
16	Historical DPR Repository and Benchmarking	Build an institutional knowledge base to compare DPRs, benchmark parameters, and improve appraisal quality over time.

The following sections describe each module in detail, including its purpose, scope, key functions, and role in the DPR appraisal lifecycle.

Module 1: Admin, Users, Roles & Governance

Purpose: Establish governance, accountability, and access control across the platform to ensure that DPR submissions, appraisal activities, configurations, and decisions are traceable, appropriately authorized, and auditable.

In simple terms: **This module answers: “Who can do what, and who is responsible for which part of the appraisal process?”**

Scope

This module provides the foundational governance layer for the platform and includes:

- Creation and management of user accounts
- Definition of user roles and responsibilities
- Role-based access control across modules
- Mapping of users to departments, projects, or appraisal teams
- Audit logging of key actions such as:
 - DPR submissions
 - review observations
 - approvals and closures
 - configuration changes

The module ensures that all actions within the system are attributable to specific users and roles.

Typical Roles (Illustrative)

Examples of roles that may be defined include:

- System Administrator
- Project Administrator
- DPR Author / Consultant
- Department Reviewer
- Technical Expert
- Financial Reviewer
- Appraisal Committee Member
- Approver
- Viewer (read-only access)

Roles are configurable and may vary by organization or project type.

Key Concepts

Governance Layer

A cross-cutting layer that defines authority, accountability, and traceability across all modules of the platform.

Role-Based Access Control (RBAC)

Permissions to view, modify, review, or approve information are determined by role rather than individual users.

Auditability

All significant actions are logged to maintain transparency, support review, and enable compliance or investigation if required.

Key Characteristics

Traceable

All submissions, observations, and decisions are linked to users, roles, and timestamps.

Configurable

Organizations can define roles and responsibilities according to their internal structures and workflows.

Non-intrusive

This module governs access and responsibility but does not control appraisal logic, validation rules, or technical evaluation.

Foundational

All other modules depend on the governance structure defined here for access control and workflow routing.

Module 2: Master Data Management

Purpose: To establish a structured and controlled reference framework that enables consistent configuration, validation, and appraisal of DPRs across sectors and project types.

In simple terms: **This module answers: “What are the standardized reference elements used across the platform?”**

Without master data, the system cannot:

- create structured templates
- assign responsibilities
- validate inputs consistently
- maintain comparability across projects

This module forms the **foundation layer** of the platform.

Scope

The Master Data Management module provides the capability to define, configure, and maintain reference datasets that are reused across multiple DPRs and projects.

It includes:

- Organizational masters
- Functional masters
- Project classification masters
- DPR classification masters
- Standards and reference libraries
- Units and measurement systems

These masters support configuration, validation, reporting, and workflow routing.

Master Data Management shall include configurable Site & Context masters (topography, climate zone, soil classification/strength, hydrology/river regime, groundwater conditions, and site constraints) to enable ontology-driven dependency checks and context-aware technical validation.

Types of Masters (Illustrative)

1. Organizational Masters

Defines the institutional structure involved in DPR preparation and appraisal.

Examples:

- Departments (Civil, Mechanical, Electrical, Finance, Environment, Traffic, Planning)
- Divisions or zones
- External stakeholders or consultants

These masters are used for:

- Role mapping
- Responsibility assignment
- Workflow routing

2. Functional Masters

Defines functions or domains contributing to DPR preparation.

Examples:

- Survey and investigations
- Design and engineering
- Cost estimation
- Traffic and demand assessment
- Environmental and social assessment
- Financial appraisal

These masters help classify sections of a DPR by technical domain.

3. Sector and Project Type Masters

Defines the classification of projects handled by the platform.

Examples:

- Sector: Transport, Buildings, Irrigation, Energy, Digital Systems
- Project type: Construction, Consultancy, Technology implementation

These masters allow:

- Template selection
- Rule configuration
- Reporting and analytics

4. DPR Type Masters

Defines different types of DPRs supported by the platform.

Examples:

- Infrastructure DPR
- Consultancy DPR
- Digital Systems DPR
- Feasibility Study DPR

Each DPR type may be linked to:

- Templates
- Validation rules

- Required sections

5. Standards and Reference Libraries

Defines reference standards used for appraisal.

Examples:

- Technical codes and norms
- Design standards
- Cost schedules
- Classification codes

This supports validation and compliance checks in later modules.

6. Units and Measurement Masters

Defines standardized units used across the platform.

Examples:

- Length (mm, m, km)
- Area (sqm, hectare)
- Volume (cum)
- Load, capacity, cost units

This helps prevent:

- unit mismatches
- inconsistent entries
- conversion errors

7. Site & Context Masters (New subsection)

These masters allow the platform to reason about “where the asset is” and apply context-specific rules.

A. Terrain / Topography Master

- Plain / Rolling / Hilly / Mountainous
- Coastal / Inland
- Floodplain / Riverine / Valley
- Urban / Rural / Forest / Desert (as applicable)

B. Climate / Environmental Zone Master

- Temperature band / heat zone
- Rainfall category
- Humidity / corrosion exposure category
- Wind zone / cyclone-prone (where relevant)
- Seismic zone (important for bridges/structures)

C. Soil & Geotechnical Master

- Soil type/classification (e.g., clay/silt/sand/gravel/rock; plus standard classification fields)
- Bearing capacity / SBC ranges
- N-value / strength indicators (if used)
- Groundwater table characteristics
- Liquefaction susceptibility / expansive soil indicator (where applicable)

D. Hydrology / River & Waterbody Master

- “On river / over river / near river” classification
- Scour risk category
- Flood return period category
- River regime / HFL references (as applicable)
- Bank stability / erosion category

E. Site Constraints Master

- Land type: solid land / reclaimed / marshy
- Settlement/encroachment sensitivity category
- Utility density category (high/medium/low)
- Access constraints (remote/limited access)

These should be **configurable** (not hardcoded), with the ability to add agency-specific classifications later.

Key Concepts

Master Data

Reference information that remains relatively stable but is used repeatedly across multiple projects and modules.

Controlled Vocabulary

Standardized terminology that ensures uniform interpretation of technical and administrative inputs.

Configurability

Masters can be updated or expanded without modifying the core system architecture.

Key Characteristics

Reusable

Masters are defined once and used across multiple DPRs and projects.

Configurable and Extensible

New sectors, departments, or project types can be added as needed.

Independent of Appraisal Logic

This module defines reference structures but does not perform validation or scoring.

Foundational for Other Modules

The following modules depend heavily on master data:

- Template engine
- Role mapping
- Dictionary and ontology
- Validation engines

Module 3: DPR Template Creation Engine

Purpose: To enable the creation and configuration of structured DPR formats that define how project reports are organized, submitted, and appraised across different sectors and project types.

In simple terms: **This module answers: “What should a DPR contain, and how should it be structured?”**

Since DPR formats vary widely across sectors, agencies, and project categories, this module ensures that the platform remains configurable rather than rigid.

Scope

The DPR Template Creation Engine provides tools to design and manage structured templates that define:

- Chapters
- Sections
- Subsections

- Tables
- Mandatory inputs
- Attachments
- Supporting documents

Templates serve as the structural backbone for:

- DPR authoring
- Completeness validation
- Role mapping
- Technical and consistency checks

Key Functions

1. Template Definition

Users with appropriate permissions can create and configure DPR templates for different project types.

Templates may be defined for:

- Infrastructure projects
- Consultancy projects
- Digital systems or technology projects
- Sector-specific variants

Each template can be customized without altering the core platform.

2. Chapter and Section Configuration

Templates allow definition of:

- Chapter hierarchy
- Section titles
- Subsections
- Descriptions or guidance notes

This helps ensure that DPRs are organized consistently and comprehensively.

3. Mandatory and Optional Inputs

Each section or table can be configured as:

- Mandatory
- Optional
- Conditional (required under specific circumstances)

This capability is critical for detecting missing inputs during appraisal.

Examples:

- Land acquisition details
- Utility shifting plans
- Traffic or demand projections
- Cost schedules

4. Table and Data Structure Definition

Templates allow configuration of structured tables, including:

- Field names
- Data types
- Units
- Validation constraints (basic level)

Structured tables enable automated completeness and technical checks in later modules.

5. Attachment Requirements

Templates may specify required supporting documents such as:

- Drawings
- Survey reports
- Design calculations
- Environmental clearances
- Photographs or maps

This ensures that essential evidence and technical documentation are submitted along with the DPR.

Key Concepts

Template

A configurable blueprint that defines the structure and required content of a DPR.

Hierarchy

Templates support structured organization into chapters, sections, and subsections to reflect real-world DPR formats.

Configurability

Templates can be created or modified without requiring changes to system code.

Key Characteristics

Sector-Agnostic

Templates can be configured for multiple sectors and project types, allowing the platform to remain generic.

Foundation for Validation

Completeness checks, role mapping, and technical validation depend on the structure defined in templates.

Flexible and Extensible

New templates can be created as project requirements evolve.

Separation of Structure and Logic

This module defines structure only; validation rules and appraisal logic are handled in later modules.

How This Module Supports Appraisal

By enforcing structured templates, the system ensures that:

- Required sections are not omitted
- Data is captured in comparable formats
- Reviewers can assess DPRs systematically
- Missing items such as plots, utilities, or schedules can be detected reliably

Structured templates significantly reduce ambiguity in DPR preparation and review.

Module 4: Role-to-DPR Mapping

Purpose: To define and manage the mapping between DPR sections, functional domains, and responsible stakeholders so that preparation, submission, and appraisal activities are clearly assigned and traceable.

In simple terms: **This module answers: "Who is responsible for which part of the DPR?"**

DPRs are inherently multi-disciplinary, and effective appraisal depends on ensuring that each domain has provided its inputs and that the appropriate reviewers examine them.

Scope

This module provides the capability to:

- Map DPR sections and subsections to departments or functional roles
- Assign responsibility for preparation, review, and approval
- Track pending inputs from stakeholders
- Generate alerts or notifications for incomplete or overdue submissions
- Support accountability and traceability during appraisal

The module operates on the structures defined in:

- Master Data Management (departments, functions)
- DPR Template Creation Engine (sections and tables)

Key Functions

1. Section-to-Department Mapping

Each section or subsection of a DPR template can be mapped to one or more responsible departments or functional domains.

Examples:

- Alignment and survey → Survey / Civil
- Traffic projections → Traffic / Planning
- Cost estimates → Finance
- Environmental impact → Environment

This mapping helps ensure that subject-matter experts contribute to the relevant portions of the DPR.

2. Responsibility Assignment

For each mapped section, the system may assign:

- Primary author or contributor
- Reviewer or technical expert
- Approver (if applicable)

This allows a structured workflow and avoids ambiguity regarding ownership of inputs.

3. Input Tracking and Status Monitoring

The system tracks the status of each mapped section:

- Not started
- In progress
- Submitted
- Reviewed
- Closed

This enables dashboards showing:

- Pending inputs by department
- Sections awaiting review
- Bottlenecks in preparation or appraisal

4. Alerts and Notifications

Where inputs are incomplete or delayed, the platform can generate notifications to responsible users or departments.

Examples:

- Environmental section not submitted
- Financial analysis incomplete
- Mechanical inputs pending

This improves coordination among stakeholders.

Key Concepts

Responsibility Mapping

A structured linkage between DPR content and accountable stakeholders.

Multi-disciplinary Coordination

Ensuring that multiple technical domains contribute appropriately to a single project report.

Traceability of Contributions

Every input and revision can be associated with a responsible user or department.

Key Characteristics

Configurable

Mappings can be defined or modified based on project type, organization structure, or sector.

Template-Driven

Role mapping operates on the sections and structures defined in DPR templates, ensuring consistency.

Supports Appraisal Readiness

Before a DPR proceeds to technical validation, the system can verify that all responsible parties have submitted required inputs.

Independent of Technical Validation

This module ensures completeness of responsibility and submission but does not evaluate technical correctness; that is handled in later modules.

How This Module Supports Appraisal

Effective appraisal depends not only on what is written in a DPR but also on:

- whether all relevant domains have contributed
- whether inputs have been reviewed by appropriate experts
- whether any stakeholder inputs are missing

By structuring responsibility and tracking submissions, this module reduces the risk of incomplete or poorly coordinated DPRs entering the appraisal stage.

Module 5: Dictionary and Standards Repository

Purpose: To establish a structured reference base of technical terminology, units, classifications, and applicable standards that support consistent interpretation, validation, and appraisal of DPR inputs.

In simple terms: **This module answers: “What do the terms, units, and standards used in the DPR mean, and what references should they comply with?”**

Without a controlled dictionary and standards base, validation becomes inconsistent and prone to ambiguity.

Scope

This module provides the capability to define and manage:

- Technical terminology and abbreviations
- Units of measurement and conversions
- Material classifications and engineering parameters
- Codes, manuals, and reference standards
- Cost and classification heads
- Domain-specific nomenclature

These references are used by later modules for:

- technical validation
- anomaly detection
- consistency checks
- reporting and analytics

Key Functions

1. Technical Dictionary

The system maintains a repository of commonly used technical terms and abbreviations to ensure consistent usage across templates and DPR submissions.

Examples:

- BoQ
- ESIA
- FIRR / EIRR
- Pavement layers
- Formation width

This helps standardize terminology and reduces ambiguity.

2. Units and Measurement Standards

The module defines accepted units and, where necessary, conversion rules.

Examples:

- Length: mm, cm, m, km
- Area: sqm, hectare
- Volume: cum
- Load, capacity, and cost units

This supports:

- consistent data entry
- detection of unit mismatches
- reliable comparisons

3. Material and Parameter Definitions

Reference ranges or descriptions for commonly used engineering parameters may be maintained.

Examples:

- Reinforcement diameters
- Concrete grades
- Pavement layer classifications

This information is used by technical validation modules to detect unrealistic or incorrect entries.

4. Standards and Codes Repository

The system may maintain references to applicable technical standards and guidelines.

Examples:

- Design standards
- Investigation requirements
- Survey norms
- Environmental and safety requirements

The platform does not replace the standards themselves but provides a structured way to reference and link them to relevant sections of the DPR.

5. Classification and Cost Head Definitions

To ensure consistency in financial and quantity data, the system may define standard classifications for:

- Cost components
- Work items
- Major heads and subheads

This improves consistency in reporting and cross-verification.

Key Concepts

Controlled Vocabulary

A standardized set of terms and definitions that ensures uniform interpretation across users and projects.

Reference Standards

Authoritative technical or procedural guidelines that form the basis for compliance checks.

Semantic Consistency

Ensuring that terms and units are used consistently so that automated checks and analytics remain reliable.

Key Characteristics

Configurable and Extensible

New terms, units, and standards can be added as new sectors or project types are incorporated.

Reusable Across Projects

Dictionary and standards repositories are shared across templates and DPRs, ensuring consistency.

Supports Validation and Appraisal

Later modules rely on these references to detect:

- incorrect units
- unrealistic values
- non-standard terminology

Independent of Templates

While templates define structure, this module defines the meaning of terms and parameters used within that structure.

How This Module Supports Appraisal

In practical DPR appraisal, reviewers frequently encounter:

- inconsistent terminology
- incorrect units
- ambiguous parameter descriptions
- references to outdated or inappropriate standards

By providing a structured dictionary and standards base, this module helps ensure that:

- inputs are interpreted consistently
- technical validation is reliable
- appraisal decisions are better informed

Module 6: Ontology and Dependency Framework

Purpose : To define structured relationships among technical, financial, and procedural parameters in a DPR so that logical dependencies, missing inputs, and inconsistencies can be identified systematically.

In simple terms: **This module answers: “How are different parameters in a DPR related to each other?”**

A DPR is not just a collection of sections—it is a network of interdependent inputs. This module enables the platform to understand and use those relationships during appraisal.

Scope

This module provides the capability to:

- Define relationships between parameters, sections, and domains
- Capture dependencies among technical inputs
- Identify missing prerequisite information
- Support logical and contextual validation
- Provide a structured knowledge layer for appraisal engines

The ontology and dependency framework operates on:

- Structures defined in the Template Engine
- Terms and standards defined in the Dictionary and Standards Repository
- Inputs provided in DPR submissions

Key Functions

1. Definition of Entities and Relationships

Soil type + soil strength → feasible foundation families

- e.g., soft clay + low SBC → pile foundation likely
- rock + high SBC → open foundation / shallow options more plausible

Bridge on river → additional dependency requirements

- hydrology inputs required (HFL, scour depth category, flood risk)
- design narrative must include protection works, bank protection, etc.

Topography + climate zone → constructability / schedule risk indicators

- hilly terrain → additional earthwork/retaining structures likely
- high rainfall / floodplain → seasonality impacts schedule

The system allows configuration of entities such as:

- Design parameters
- Cost components
- Surveys and investigations
- Environmental inputs
- Schedule elements

Relationships among these entities can be defined.

Examples:

- Traffic projections influence pavement design
- Soil investigation influences foundation design
- Land acquisition influences project cost and timeline

These relationships help the system detect missing or inconsistent inputs.

2. Dependency Mapping

Dependencies between inputs can be defined so that:

- Certain parameters require prior inputs
- Certain sections cannot be considered complete without prerequisite data

Examples:

- Pavement design requires traffic data and soil data
- Cost estimates require quantities and rates
- Construction schedules require scope and resource assumptions

This allows the system to flag incomplete logical chains.

3. Logical Consistency Support

By understanding dependencies, the platform can assist later modules in identifying:

- Inconsistent assumptions
- Missing prerequisite investigations
- Parameters that do not logically align

For example:

- Design elements provided without corresponding survey data
- Cost components included without quantities

4. Configurable Domain Ontologies

Different sectors may require different ontologies.

Examples:

- Infrastructure projects: engineering parameters and construction elements
- Consultancy projects: process flows and institutional components
- Digital systems: architecture and integration components

The framework allows domain-specific configuration while maintaining a common structure.

Key Concepts

Ontology

A structured representation of entities and their relationships within a domain.

Dependency

A condition in which one parameter or section relies on another to be meaningful or complete.

Contextual Validation

Validation based not only on values but also on relationships and logical structure.

Key Characteristics

Configurable

Relationships and dependencies can be defined and extended without altering the core system.

Sector-Agnostic Framework

While specific ontologies vary, the framework for defining relationships remains consistent.

Supports Advanced Validation

Later modules use dependency information to detect:

- missing inputs
- logical inconsistencies
- unrealistic assumptions

Non-Intrusive to Authoring

This module defines relationships in the background and does not alter the DPR content itself.

How This Module Supports Appraisal

In real DPR appraisal, reviewers often identify issues such as:

- design based on incomplete investigations
- cost estimates lacking supporting inputs
- sections that appear complete but lack prerequisite data

By formalizing dependencies, the system assists reviewers in identifying such issues more systematically and earlier in the appraisal process.

Module 7: Historical DPR Repository and Benchmarking

Purpose: To store, classify, and analyze previously appraised DPRs in order to enable benchmarking, pattern recognition, and institutional learning.

In simple terms: **This module answers: “How does this DPR compare to past projects?”**

Scope

This module provides the capability to:

- Store completed DPRs and appraisal records
- Tag DPRs by characteristics
- Classify DPRs by quality outcomes
- Compare new DPRs with historical benchmarks
- Support learning for anomaly detection and scoring

Key Functions

1. Historical DPR Repository

The system maintains a structured repository of:

- DPR documents
- Appraisal results
- Scores and observations
- Project characteristics

This repository becomes a knowledge base over time.

2. Classification of DPR Quality

DPRs may be tagged as:

- High quality
- Acceptable
- Required major revisions
- Led to implementation issues (optional advanced tagging)

This classification helps benchmarking.

3. Benchmark Comparison

The system may compare:

- Cost per unit
- Quantities
- Investigation scope
- Design parameters

Examples:

- Cost/km vs similar terrain projects
- Foundation types used in similar soil conditions
- Investigation depth vs project type

This strengthens anomaly detection and reviewer confidence.

4. Pattern Identification

The system may identify recurring issues such as:

- Missing utility identification
- Underestimated earthwork
- Incomplete geotechnical investigations

This helps improve DPR preparation and review practices.

5. Support for Learning Models (Future Capability)

Over time, the system may:

- Improve anomaly detection thresholds
- Improve risk indicators
- Suggest typical parameter ranges

The above is a **future enhancement**, not an immediate development

Module 8: DPR Authoring / Submission Interface

Purpose : To provide a structured environment for preparation, entry, and submission of DPR content so that data is captured in a consistent and appraisable format. The platform should support pre-population of DPR sections using historical data, templates, and Data Lake insights to improve efficiency and consistency.

In simple terms: **This module answers: “How is a DPR prepared and submitted into the platform?”**

This module ensures that DPR inputs are organized, structured, and traceable before appraisal begins.

The platform should support generation and comparison of alternative design scenarios (e.g., alignment options, configuration alternatives) to enable evaluation of optimal solutions during DPR preparation). The platform should support integration with design and drawing systems (e.g., BIM/GIS/alignment tools) and enable linkage between design outputs and DPR inputs.

Scope

This module provides the capability to:

- Enter DPR data through structured forms and tables
- Upload supporting documents and drawings
- Manage multiple revisions of DPR submissions
- Track submission status

- Ensure required sections and inputs are provided before submission
- The module operates using:
- Templates defined in the Template Creation Engine
 - Masters and dictionaries defined in earlier modules

Key Functions

1. Project Context & Site Conditions"

- topography category
- climate zone category
- soil class + strength indicators
- groundwater/hydrology categories
- river/solid land classification

2. Structured Data Entry

The platform allows users to enter DPR information in structured formats based on predefined templates.

Examples:

- Tables for quantities and costs
- Fields for technical parameters
- Sections for narrative descriptions

Structured entry improves:

- completeness checks
- consistency validation
- reporting and analytics

3. Document and Drawing Upload

Users can upload supporting material such as:

- Survey reports
- Drawings and maps
- Design calculations
- Environmental and statutory documents
- Photographs or site records

The system links these documents to relevant sections of the DPR for easy review.

4. Version Control

DPR preparation is iterative. This module maintains:

- Version history
- Submission timestamps
- Revision records
- Comparison between versions

This ensures traceability of changes during appraisal.

5. Submission Workflow

Once preparation is complete, the DPR or individual sections can be formally submitted for review.

Submission may be configured to:

- Require mandatory sections to be completed
- Ensure required attachments are provided
- Validate basic structural completeness

This prevents incomplete DPRs from entering the appraisal stage prematurely.

6. Status Tracking

The module provides visibility into the status of each submission, such as:

- Draft
- Submitted
- Under Review
- Returned for Revision
- Resubmitted

This helps both authors and reviewers monitor progress.

Key Concepts

Structured Authoring

Capturing DPR data in defined fields and tables rather than relying solely on unstructured documents.

Version Traceability

Maintaining records of revisions and submissions to support accountability and review.

Submission Readiness

Ensuring that minimum structural requirements are satisfied before appraisal begins.

Key Characteristics

Template-Driven

All authoring and submission follow structures defined in the Template Engine.

Traceable and Auditable

All submissions and revisions are recorded with timestamps and user information.

Supports Multiple Formats

Both structured inputs and document uploads are supported.

Non-Evaluative

This module captures and organizes information but does not perform technical validation or appraisal; those functions are handled by later modules.

How This Module Supports Appraisal

In practice, many appraisal delays arise from:

- incomplete submissions
- missing attachments
- inconsistent formatting
- difficulty locating supporting documents

By structuring authoring and submission, this module ensures that DPRs entering appraisal are better organized and easier to review.

Module 9: Completeness Validation Engine

Purpose : To assist reviewers and project teams in identifying missing sections, incomplete inputs, absent stakeholder submissions, and missing supporting documents before and during the appraisal process. The platform should support assisted computation of quantities, costs, and derived parameters based on input data and configurable formulas.

In simple terms: **This module answers: “Is the DPR complete?”**

Completeness validation is often the first stage of appraisal, and addressing gaps early prevents delays and repeated review cycles.

Scope

This module provides the capability to:

- Check whether all mandatory sections defined in templates are present
- Verify that required tables and fields have been completed
- Detect missing stakeholder inputs based on role mappings
- Identify missing attachments or supporting documents
- Highlight incomplete logical dependencies where prerequisite inputs are absent

The module operates using:

- Structures defined in the Template Creation Engine
- Role mappings defined in the Role-to-DPR Mapping module
- Dependencies defined in the Ontology Framework

Key Functions

1. Section Completeness Checks

The system verifies whether all required chapters, sections, and subsections have been provided.

Examples:

- Survey section not submitted
- Financial analysis missing
- Environmental section incomplete

Incomplete sections are flagged for review.

2. Field and Table Completeness Checks

Within structured tables and forms, the system checks for:

- Missing fields
- Empty mandatory values
- Partially completed tables

Examples:

- Cost tables missing rate or quantity
- Investigation results missing key parameters

3. Stakeholder Input Verification

Based on role mapping, the system identifies whether required departments or contributors have submitted their inputs.

Examples:

- Mechanical inputs pending
- Finance review not submitted
- Survey data not provided

This helps ensure that all technical domains are represented before appraisal proceeds.

4. Attachment and Supporting Document Checks

The system verifies whether required attachments have been uploaded where specified by templates.

Examples:

- Drawings not attached

- Survey reports missing
- Environmental clearance documents not provided

This prevents appraisal from proceeding without supporting evidence.

5. Dependency-Based Completeness Checks

Using the Ontology and Dependency Framework, the system can detect missing prerequisite inputs.

Examples:

- Design provided without corresponding survey data
- Cost estimates provided without quantities
- Schedule provided without scope definition

These checks go beyond structural completeness to logical completeness.

Key Concepts

Structural Completeness

Ensuring that all required sections, tables, and fields are present.

Stakeholder Completeness

Ensuring that all responsible departments have contributed required inputs.

Logical Completeness

Ensuring that prerequisite inputs exist for dependent parameters.

Key Characteristics

Template-Driven

Completeness rules are derived from template definitions and dependency mappings.

Non-Intrusive

The module flags issues but does not modify the DPR content.

Supports Iterative Preparation

Authors can correct gaps and resubmit without disrupting workflow.

Pre-Appraisal Gate

Completeness checks can be configured as a prerequisite for entering technical appraisal stages.

How This Module Supports Appraisal

In real-world DPR appraisal, a significant portion of review time is spent identifying:

- missing plots or drawings
- absent survey data
- incomplete cost details
- sections prepared by some departments but not others

By automatically highlighting such gaps, this module:

- reduces appraisal time
- improves submission quality
- helps reviewers focus on technical and analytical aspects rather than administrative completeness

Module 10: Technical Validation Engine

Purpose : To assist reviewers in identifying incorrect, unrealistic, or inconsistent technical inputs in a DPR by applying structured validation rules, reference standards, and contextual checks. The platform should support assisted computation

of quantities, costs, and derived parameters based on input data and configurable formulas.

In simple terms: **This module answers: “Are the technical inputs reasonable and correct?”**

Errors in technical inputs—whether due to typographical mistakes, incorrect assumptions, or misinterpretation of standards—can significantly affect project cost, safety, and feasibility. This module helps detect such issues early in the appraisal process.

Scope

This module provides the capability to:

- Validate technical parameters against reference ranges or norms
- Detect unit mismatches and order-of-magnitude errors
- Identify unrealistic or inconsistent engineering values
- Verify alignment with applicable standards and codes
- Flag inputs requiring reviewer attention

The module operates using:

- Dictionaries and standards repositories
- Ontology and dependency mappings
- Structured data captured during DPR authoring

Key Functions

1. Range-Based Validation

The system checks whether entered values fall within defined acceptable ranges or thresholds.

Examples:

- Steel reinforcement diameter outside acceptable limits
- Pavement thickness below typical engineering ranges
- Cost rates significantly outside expected ranges

Such values are flagged for review rather than automatically rejected.

2. Unit and Format Validation

The system verifies consistency of units and formats.

Examples:

- Quantity entered in meters where millimeters are expected
- Area expressed without units
- Mixing of incompatible units in calculations

This helps prevent errors arising from inconsistent measurement systems.

3. Contextual Validation

Using ontology and dependency mappings, the system evaluates technical inputs in context.

If soil = “low bearing capacity” but foundation selected = “shallow isolated footing”
→ **flag**

If bridge marked “over river” but hydrology section missing → **flag**

If seismic zone high but structural detailing assumptions absent → **flag/advisory**

Foundation type inconsistent with soil bearing capacity → **flag**

Bridge design without hydrology/flood inputs → **flag**

Pavement design without soil classification → **flag**

Examples:

- Reinforcement size inconsistent with structural component type
- Design assumptions not aligned with investigation results
- Capacity assumptions inconsistent with demand projections

Contextual validation helps detect technically unrealistic combinations of inputs.

4. Standards-Based Validation

Where applicable, inputs can be checked against reference standards or guidelines maintained in the standards repository.

Examples:

- Required investigation steps missing
- Design parameters inconsistent with recommended practices
- Required safety considerations not addressed

These checks assist reviewers in ensuring compliance with accepted norms.

5. Typographical and Order-of-Magnitude Error Detection

The system identifies values that may be technically plausible in isolation but unrealistic in context.

Examples:

- Steel diameter entered as 200 mm instead of 20 mm
- Traffic volume or cost values differing by an order of magnitude from expected ranges
- Decimal placement errors in quantities or costs

Such anomalies are flagged for reviewer attention.

Key Concepts

Technical Validation

Verification that engineering and financial parameters are reasonable, consistent, and aligned with accepted norms.

Contextual Reasonableness

Assessing whether values make sense in relation to other parameters rather than in isolation.

Advisory Flagging

The system highlights potential issues but does not override human judgment.

Key Characteristics

Rule-Driven

Validation rules are configurable and can be adapted for different sectors or project types.

Non-Intrusive

The system flags potential issues but does not automatically alter submitted data.

Supports Reviewer Judgment

Flags and indicators are intended to assist reviewers rather than replace expert appraisal.

Extensible

Additional validation rules can be added as experience with the platform grows.

How This Module Supports Appraisal

In practice, many DPR appraisal challenges arise from:

- incorrect technical assumptions
- typographical errors in critical parameters
- unrealistic quantities or rates
- inconsistencies between design and investigation data

By systematically highlighting such issues, this module:

- reduces the risk of technical errors passing unnoticed
- improves the reliability of appraisal decisions
- allows reviewers to focus on higher-level technical evaluation

Module 11: Consistency Engine

Purpose : To assist reviewers in verifying that information presented in different sections of a DPR is internally consistent and logically aligned across technical, financial, and scheduling components.

In simple terms: **This module answers: “Do different parts of the DPR agree with each other?”**

Even when individual inputs appear correct, inconsistencies across sections can lead to inaccurate appraisal or downstream implementation issues. This module helps identify such discrepancies systematically.

Scope

This module provides the capability to:

- Cross-verify related parameters across sections and tables
- Identify mismatches between quantities, costs, and schedules
- Detect inconsistencies between design assumptions and supporting data
- Flag discrepancies between narrative descriptions and structured inputs

The module operates using:

- Structured data captured during DPR authoring
- Dependency mappings defined in the Ontology Framework
- Reference classifications defined in the Masters and Dictionary modules

Key Functions

1. Cross-Verification of Quantities and Costs

The system compares quantities, rates, and totals across relevant tables and summaries.

Examples:

- BoQ totals not matching cost summaries
- Quantities inconsistent across schedules
- Cost heads appearing in summaries but not in detailed tables

Such mismatches are highlighted for reviewer attention.

2. Design and Investigation Alignment

The system helps identify cases where design inputs are inconsistent with supporting investigations or assumptions.

Examples:

- Design parameters provided without corresponding survey results
- Soil classification inconsistent with foundation design assumptions
- Traffic assumptions inconsistent with capacity design

- Soil parameters in geotech report vs DPR design assumptions mismatch → **flag**
- River classification vs drawings/length vs number of spans mismatch → **flag**

These checks help ensure that technical decisions are supported by evidence.

3. Scope and Schedule Consistency

The system compares scope definitions with implementation timelines.

Examples:

- Construction schedule inconsistent with quantities or work scope
- Phasing plans not aligned with major activities

This helps reviewers assess feasibility.

4. Narrative and Structured Data Consistency

In many DPRs, narrative descriptions may not match structured tables or calculations.

Examples:

- Project length described differently in different sections
- Capacity figures differing between summary and detailed sections

The system flags such inconsistencies for review.

5. Cross-Section Logical Alignment

Using ontology and dependency relationships, the system can identify cases where related parameters appear inconsistent across domains.

Examples:

- Land acquisition area inconsistent with alignment length
- Utility shifting scope inconsistent with project description

These checks assist reviewers in identifying discrepancies that might otherwise require extensive manual cross-reading.

Key Concepts

Internal Consistency

Agreement among related parameters, tables, and sections within a DPR.

Cross-Verification

Comparison of information derived from multiple sources within the same document.

Logical Alignment

Ensuring that technical, financial, and scheduling assumptions do not contradict one another.

Key Characteristics

Non-Intrusive

The module highlights discrepancies but does not modify submitted data.

Dependency-Driven

Consistency checks are informed by relationships defined in the Ontology Framework.

Configurable

Cross-verification rules can be defined or extended based on sector and project type.

Supports Reviewer Judgment

The system provides indicators and flags; final interpretation remains with human reviewers.

How This Module Supports Appraisal

In practice, appraisal teams often spend considerable time manually reconciling:

- quantities and costs
- design assumptions and investigations
- scope definitions and schedules

This module reduces that effort by automatically identifying areas where information may not align, enabling reviewers to focus on resolving discrepancies rather than locating them.

Module 12: Anomaly Detection Engine

Purpose : To assist reviewers in identifying unusual, improbable, or outlier values and patterns in DPR inputs that may indicate errors, unrealistic assumptions, or data entry mistakes, even when those values do not strictly violate predefined rules.

In simple terms: **This module answers: “Does anything look unusual or suspicious?”**

While technical validation checks known rules and ranges, anomaly detection helps identify issues that may not be explicitly defined in rules but still warrant attention.

Soil, hydrology, or environmental parameters significantly deviating from expected ranges for similar geographies may be flagged.

Scope

This module provides the capability to:

- Detect outlier values in technical and financial parameters
- Identify order-of-magnitude or decimal-placement errors
- Highlight unusual patterns in quantities, costs, or assumptions
- Compare inputs with historical or reference data where available
- Flag entries that deviate significantly from expected trends

The module operates using:

- Structured DPR data
- Reference ranges and classifications
- Historical or comparative datasets (where available)

Key Functions

1. Outlier Detection

The system identifies values that differ significantly from typical or expected ranges.

Examples:

- Cost per unit significantly higher or lower than comparable items
- Quantities unusually large or small for similar project scope
- Traffic or demand projections inconsistent with historical patterns

Outliers are flagged for reviewer attention rather than automatically rejected.

2. Order-of-Magnitude and Decimal Error Detection

The system detects values that appear technically plausible but are inconsistent in scale.

Examples:

- Steel diameter entered as 200 mm instead of 20 mm

- Project cost entered one order of magnitude higher or lower than expected
- Quantity values with misplaced decimal points

Such anomalies are common in large data tables and can significantly affect appraisal outcomes.

3. Pattern-Based Anomaly Identification

The system examines patterns across related parameters to identify unusual relationships.

Examples:

- Identical values repeated across multiple rows where variation is expected
- Cost distributions inconsistent with project composition
- Resource assumptions inconsistent with activity durations
- SBC or N-value unusually high/low vs typical for stated soil class → **flag**
- Scour/flood category inconsistent with “river type” classification → **flag**

These checks help identify potential errors in data entry or calculations.

4. Historical or Comparative Analysis (Where Available)

If historical project data or benchmark datasets are available, the system may compare inputs to identify deviations.

Examples:

- Construction cost per km significantly different from comparable projects
- Investigation costs inconsistent with project scale

This capability improves over time as more data becomes available.

Key Concepts

Anomaly

A value or pattern that deviates significantly from expected norms, typical ranges, or historical trends.

Outlier Detection

Identification of values that are statistically or contextually unusual.

Advisory Flagging

The system highlights potential anomalies without making automated decisions.

Key Characteristics

Complementary to Validation Modules

This module works alongside technical validation and consistency checks, identifying issues that rule-based methods may not capture.

Non-Intrusive

Anomalies are flagged for review but do not prevent submission or appraisal unless configured otherwise.

Configurable and Extensible

Thresholds and comparison methods can be refined over time.

Improves with Data Availability

Effectiveness increases as the system accumulates historical project data and benchmarks.

How This Module Supports Appraisal

In practical DPR appraisal, reviewers often encounter:

- unrealistic quantities or costs

- values that appear technically valid but are improbable
- copy-paste errors or repeated values
- decimal placement mistakes

These issues can be difficult to detect manually, particularly in large tables. By highlighting anomalies automatically, this module:

- reduces the risk of oversight
- improves data reliability
- enables more focused technical review

Module 13: Review Workflow and Scoring

Purpose: To support structured appraisal by enabling reviewers to record observations, track issues, manage review workflows, and evaluate DPR quality through configurable scoring frameworks, including both internal organizational matrices and external assessment criteria.

In simple terms: **This module answers:**

1. What observations have reviewers made?
2. What issues remain unresolved?
3. How does the DPR rate against defined appraisal criteria?

This module forms the collaborative and evaluative layer of the platform, where outputs from validation and consistency checks are interpreted and formally recorded.

Scope

This module provides the capability to:

- Record reviewer observations and comments
- Track issues and their resolution status
- Route DPR sections for review and approval
- Maintain versioned records of appraisal actions
- Generate structured appraisal summaries
- Apply configurable scoring frameworks
- Support both internal and external appraisal matrices

The module operates using outputs from:

- Completeness Validation Engine
- Technical Validation Engine
- Consistency Engine
- Anomaly Detection Engine

Key Functions

1. Reviewer Observations

Reviewers can:

- Record comments at section, table, or parameter level
- Highlight deficiencies or risks
- Attach supporting notes or references

Observations are linked to:

- DPR sections
- Specific inputs or tables
- Users and timestamps

This ensures traceability and accountability.

2. Issue Tracking and Resolution

The system maintains structured records of issues, including:

- Open
- Under Review
- Resolved
- Closed

Issues may be assigned to responsible stakeholders and tracked until closure. This enables systematic follow-up and reduces the risk of unresolved observations.

3. Review Workflow Routing

The module supports configurable workflows for:

- Technical review
- Financial review
- Cross-disciplinary review
- Final appraisal or approval

Routing rules may be defined based on:

- DPR type
- Project category
- Organizational procedures

This ensures that appraisal follows a defined and transparent process.

4. Configurable Appraisal Scoring Framework

The platform supports scoring of DPR quality based on configurable criteria.

Scoring parameters may include:

- Completeness of inputs
- Adequacy of investigations
- Technical soundness
- Cost reliability
- Documentation quality
- Risk indicators

Scores may be:

- Section-wise
- Category-wise
- Overall DPR quality indicators

Scoring criteria, ranges, and weightages are configurable and may be defined in earlier modules.

5. Support for Multiple Appraisal Frameworks

The system should support multiple scoring frameworks, including:

- External or regulatory assessment formats
- Client-specific evaluation criteria
- Internal organizational appraisal matrices

Users may:

- Select a framework at project setup
- Apply multiple frameworks to the same DPR
- Reuse frameworks across projects

This ensures flexibility and applicability across sectors and organizations.

6. Internal Organizational Appraisal Matrix

The platform supports internal appraisal matrices that may be used for institutional decision-making.

Examples of internal assessment categories:

- Technical adequacy
- Risk exposure
- Cost realism
- Implementation readiness
- Documentation quality

Internal matrices may differ from external scoring systems but can operate in parallel.

7. Weighted and Category-Based Scoring

The system may support:

- Category-wise scoring
- Weightage assignment
- Rating bands or thresholds

These features help appraisal teams summarize complex reviews in a structured manner.

8. Appraisal Summary Generation

The module can generate structured summaries that include:

- Key observations
- Outstanding issues
- Section-wise status
- Scores or ratings
- Risk indicators

These summaries assist decision-makers in reviewing DPR quality efficiently.

Key Concepts

Structured Appraisal

A systematic method of recording observations, tracking issues, and summarizing findings.

Configurable Scoring Framework

A flexible scoring system that can be defined, modified, and reused without changes to core system logic.

Internal and External Assessment

Support for both organizational appraisal matrices and external evaluation criteria.

Human-in-the-Loop Evaluation

Scores and indicators assist expert judgment rather than replacing it.

Key Characteristics

Configurable

Scoring frameworks and workflows can be customized.

Transparent

Observations, actions, and scoring are traceable and auditable.

Reusable

Appraisal frameworks can be applied across projects.

Advisory

Scores assist appraisal but do not automatically determine approval decisions.

How This Module Supports Appraisal

In practical DPR appraisal processes, reviewers often face challenges such as:

- observations scattered across multiple documents
- lack of structured tracking of issues
- difficulty summarizing overall quality
- inconsistent scoring methods

By providing structured workflow, issue tracking, and configurable scoring, this module:

- improves transparency
- standardizes appraisal practices
- enables clearer decision-making

Module 14: Reporting and Appraisal Dashboard

Purpose : To provide consolidated, real-time visibility into the status, quality, risks, and appraisal outcomes of DPRs through structured reports and interactive dashboards that support technical reviewers, management, and decision-makers.

In simple terms: **This module answers: “What is the overall status and quality of the DPR, and what needs attention?”**

This module converts outputs from validation, scoring, and workflow modules into actionable insights.

Scope

This module provides the capability to:

- Generate structured reports and summaries of DPR appraisal
- Display dashboards showing completeness, validation findings, scores, and risks
- Track pending inputs, open issues, and review status
- Present internal and external appraisal scores
- Provide decision-support views for appraisal committees and management

The module operates using outputs from:

- Completeness Validation Engine
- Technical Validation Engine
- Consistency Engine
- Anomaly Detection Engine
- Review Workflow and Scoring Module

Key Functions

1. DPR Status Dashboard

The system provides a consolidated view of DPR progress and appraisal status.

Typical indicators may include:

- Sections completed vs pending
- Inputs pending by department
- Review stages completed
- Issues open or unresolved

This enables quick assessment of readiness for appraisal or approval.

2. Validation and Risk Indicators

The dashboard presents summarized outputs from validation modules, such as:

- Missing inputs or sections
- Technical validation flags
- Consistency issues
- Anomalies detected

These indicators help reviewers prioritize attention areas.

3. Appraisal Scoring and Ratings Display

The dashboard displays scoring outcomes, including:

- Section-wise scores
- Category-wise scores
- Internal appraisal matrix results
- External or regulatory scoring results (if applicable)

This allows comparison of results across different frameworks.

4. Issue and Observation Tracking View

The system provides a consolidated view of:

- Open issues
- Issues under review
- Recently resolved observations

Users may filter by:

- Section
- Department
- Severity
- Date

This helps appraisal teams track progress efficiently.

5. Comparative and Trend Reporting (Optional)

Where historical data is available, the system may provide:

- Comparison of DPR scores across projects
- Trends in common deficiencies
- Repeated issues by category

This helps organizations improve DPR preparation and appraisal practices over time.

6. Report Generation

The platform can generate structured reports for:

- Appraisal summaries
- Section-wise findings
- Risk indicators
- Scoring outcomes

Reports may be exported in standard formats suitable for documentation or presentation.

Key Concepts

Decision-Support Dashboard

A consolidated interface presenting key indicators needed for appraisal decisions.

Appraisal Summary

A structured synthesis of observations, scores, risks, and pending issues.

Comparative Insights

Ability to analyze patterns across multiple DPRs or projects.

Key Characteristics

Consolidated View

Brings together information from multiple modules into a single interface.

Configurable

Dashboards and reports can be customized based on organizational needs.

Real-Time Updates

Dashboards reflect the latest submissions, validations, and observations.

Non-Intrusive

This module presents information but does not alter DPR content or appraisal outcomes.

How This Module Supports Appraisal

In traditional appraisal processes, information about DPR quality and status is often:

- scattered across documents
- difficult to summarize
- dependent on manual compilation

By providing structured dashboards and reports, this module:

- improves situational awareness
- reduces effort in preparing appraisal summaries
- supports informed and timely decision-making

Module 15: Simulation and Sensitivity Analysis

Purpose : To assist reviewers and decision-makers in understanding how changes in key assumptions or parameters may affect project cost, timelines, risks, and feasibility by enabling scenario-based analysis and sensitivity evaluation. The platform should support generation and comparison of alternative design scenarios (e.g., alignment options, configuration alternatives) to enable evaluation of optimal solutions during DPR preparation.

In simple terms: **This module answers: “What happens if key assumptions change?”**

DPR appraisal often requires assessing the robustness of assumptions. This module provides tools to explore the potential impact of variations in critical inputs.

Scope

This module provides the capability to:

- Perform sensitivity analysis on selected parameters
- Simulate alternative scenarios based on revised assumptions
- Evaluate the potential impact of cost, quantity, schedule, or demand variations
- Support exploratory analysis during appraisal discussions

The module operates using:

- Structured data from DPR submissions
- Quantities and cost inputs
- Schedule assumptions
- Dependency relationships defined in earlier modules

Key Functions

1. Parameter Sensitivity Analysis

The system allows reviewers to vary selected parameters and observe the effect on related outcomes.

Examples:

- Variation in material cost affecting total project cost
- Change in traffic projections affecting revenue estimates
- Adjustment in project duration affecting phasing and resource requirements

Sensitivity analysis helps identify which parameters have the greatest influence on outcomes.

2. Scenario Modelling

Users may create and compare alternative scenarios based on different assumptions.

Examples:

- Revised cost assumptions
- Alternate implementation timelines
- Variations in demand or utilization

Scenarios can be stored and compared to support appraisal discussions and decision-making.

3. Impact Visualization

The module may provide graphical or tabular representation of the impact of changes in key inputs.

Examples:

- Cost variation summaries
- Timeline comparisons
- Risk indicators under different scenarios

Visualization helps communicate results effectively to decision-makers.

4. Exploratory Analysis Support

This module supports “what-if” analysis during appraisal meetings, allowing reviewers to examine:

- Feasibility under different assumptions
- Risk exposure under adverse conditions
- Sensitivity of major cost or demand drivers

This assists in evaluating project robustness.

Key Concepts

Sensitivity Analysis

Assessment of how variations in input parameters affect outputs.

Scenario Analysis

Evaluation of alternative sets of assumptions to compare potential outcomes.

Decision-Support Modelling

Providing analytical insights to assist expert judgment rather than replacing it.

Key Characteristics

Advisory in Nature

Simulation outputs are indicative and intended to support appraisal discussions.

Configurable

Parameters selected for sensitivity analysis can be defined based on project type and sector.

Non-Intrusive

Simulation does not alter the original DPR; it operates on copies or scenario datasets.

Incrementally Expandable

More advanced modelling techniques may be incorporated as the platform evolves.

How This Module Supports Appraisal

In real DPR appraisal, reviewers often need to assess:

- whether cost estimates are robust
- whether assumptions are realistic
- how sensitive outcomes are to key variables

Traditionally, such analysis requires manual recalculation or separate spreadsheets.

By enabling structured sensitivity analysis within the platform, this module:

- reduces manual effort
- improves transparency of assumptions
- supports informed decision-making

Module 16: Security, Audit and Monitoring

Purpose : To ensure the confidentiality, integrity, and traceability of DPR data, appraisal activities, and system configurations by providing secure access control, comprehensive audit trails, and monitoring of system usage and events.

In simple terms: **This module answers: “Who accessed or changed what, and is the system secure?”**

DPRs often contain sensitive technical, financial, and contractual information, and appraisal decisions must be fully traceable. This module safeguards both data and processes.

Scope

This module provides the capability to:

- Enforce secure authentication and role-based access control
- Maintain audit logs of key user actions and system events
- Monitor user activity and system usage patterns
- Detect and alert on suspicious or unauthorized access attempts
- Ensure secure storage and transmission of documents and data

The module operates across all other modules of the platform and functions as a cross-cutting layer rather than a standalone workflow.

Key Functions

1. Authentication and Access Control

The system ensures that only authorized users can access the platform and perform actions appropriate to their roles.

Functions may include:

- Secure login mechanisms
- Role-based access control (RBAC)
- Permission management by module, project, or document

This protects sensitive DPR information and appraisal records.

2. Audit Trail and Activity Logging

The system maintains a detailed record of significant actions, including:

- DPR submissions and revisions
- Reviewer observations and approvals
- Changes to templates, masters, or scoring frameworks
- Configuration updates

Each record includes:

- User identity
- Timestamp
- Nature of action

Audit trails support accountability and transparency.

3. User Activity Monitoring

The system monitors user activity to identify unusual or potentially risky patterns, such as:

- Repeated failed login attempts
- Access to multiple projects in rapid succession
- Unusual download or modification activity

Such monitoring helps detect potential misuse or security risks.

4. Alerts and Notifications

The module may generate alerts for:

- Unauthorized access attempts
- Critical configuration changes
- Suspicious system activity

Alerts may be sent to administrators or designated security personnel.

5. Data Protection and Integrity

The platform should ensure:

- Secure storage of documents and structured data
- Protection against unauthorized modification
- Controlled access to sensitive files

Where applicable, encryption and secure communication protocols may be used.

Key Concepts

Confidentiality

Ensuring that sensitive DPR data is accessible only to authorized users.

Integrity

Ensuring that data and records are not altered improperly.

Traceability

Maintaining a complete history of actions for accountability and review.

Monitoring

Observing system activity to detect anomalies or risks.

Key Characteristics

Cross-Cutting Module

Security and audit mechanisms apply across all modules of the platform.

Transparent and Traceable

All significant actions are recorded and can be reviewed when needed.

Configurable

Security policies and access controls can be adapted to organizational requirements.

Non-Intrusive

Security mechanisms operate in the background without affecting normal workflows.

How This Module Supports Appraisal

In DPR appraisal environments, it is critical to ensure that:

- submissions are authentic and unchanged
- reviewer comments and approvals are traceable
- sensitive project information is protected
- audit records are available for internal or external review

By providing structured security, audit, and monitoring capabilities, this module strengthens the reliability and credibility of the appraisal process.

Functional Modules of the Platform

The following is to provide an overview linking the modules to the respective layers:

A. DPR Preparation Layer Modules

These are modules that **create structured DPRs**

1. **Template Management Module**
2. **Master Data Management Module**
3. **User & Role Management Module**
4. **DPR Authoring Module**
5. **Document Management Module**

These modules support structured preparation, data capture, document management, and submission of DPRs.

B. Appraisal & Evaluation Layer Modules

6. **Ontology & Dependency Engine**
7. **Historical DPR / Data Integration Module** (*linked to Data Lake*)
8. **Completeness Check Module**
9. **Technical Validation Engine**
10. **Consistency & Cross-Validation Engine**
11. **Anomaly Detection Module**
12. **Scoring & Appraisal Module**
13. **Workflow & Review Management Module**

These modules enable structured validation, engineering checks, consistency analysis, and appraisal workflows.

C. Knowledge & Decision Support Layer Modules

14. **Reporting & Dashboard Module**
15. **Analytics & Insights Module**
16. **Simulation / Scenario Analysis Module (if included)**

These modules provide analytics, insights, reporting, and decision-support capabilities based on platform data.

Preferred Technology Stack and Architecture Guidelines

The bidder shall propose a modern, scalable, secure, and resource-efficient technology stack for PARAKH. Preference shall be given to **open-source and open-standard technologies** to avoid vendor lock-in, reduce long-term total cost of ownership, and ensure portability, maintainability, and extensibility. Any deviation from the preferred stack or use of proprietary components shall be clearly justified.

1. Front-End / User Interface

The platform shall support responsive, browser-based access across desktops, laptops, tablets, and mobile handheld devices, including iPad and Android-based field devices.

Preferred stack:

- React / Angular / Vue.js for web front-end
- HTML5, CSS3, JavaScript / TypeScript
- Responsive UI frameworks such as Bootstrap, Tailwind, or equivalent
- Progressive Web App (PWA) capability preferred for handheld and field usage

Requirements:

- touch-friendly interface for tablet/mobile use
- compatibility with major modern browsers
- optimized rendering for low-bandwidth and field environments
- support for role-based dashboards, forms, workflows, GIS views, and document interaction

2. Application / Backend Layer

The backend shall be modular, API-driven, secure, and scalable.

Preferred stack:

- Python (Django / FastAPI / Flask), Java (Spring Boot), or Node.js
- Microservices or modular service-oriented architecture preferred
- RESTful APIs as default; GraphQL may be proposed where justified
- asynchronous processing and queue-based architecture for heavy jobs

Requirements:

- lightweight and maintainable services
- separation of business logic, workflow logic, AI services, and integration services
- support for audit trails, approval workflows, and configurable validation engines

3. Database and Data Storage

The database architecture shall be optimized for performance, scalability, and efficient storage utilization.

Preferred stack:

- PostgreSQL / PostGIS for transactional, spatial, and structured data
- object storage / document repository for unstructured files and DPR documents
- Elasticsearch / OpenSearch for indexing and search, where required
- Redis or equivalent for caching
- NoSQL databases only where clearly required for specific use cases

Requirements:

- efficient schema design, indexing, query tuning, partitioning, and archiving
- minimization of data duplication
- optimization for structured, semi-structured, spatial, and document-heavy workloads
- support for metadata management and cataloging

4. Data Lake / Analytics Layer

The bidder shall propose a scalable and cost-efficient data management stack for analytics, benchmarking, and historical DPR ingestion.

Preferred stack:

- PostgreSQL/PostGIS + object storage for core architecture
- Apache Kafka / RabbitMQ for ingestion pipelines, where required
- Apache Spark / lightweight distributed processing tools only if justified by workload
- Apache Airflow or equivalent for orchestration of scheduled jobs
- open-source BI/dashboard tools or equivalent embedded analytics stack

Requirements:

- architecture shall be right-sized to the workload
- avoid unnecessary big data components where standard database and ETL pipelines are sufficient
- analytics infrastructure shall be designed for maintainability and efficient resource utilization

5. AI / ML and Rules Engine

AI/ML components shall be assistive, explainable, auditable, and resource-efficient.

Preferred stack:

- Python-based ML ecosystem
- Scikit-learn, XGBoost, LightGBM, spaCy, OCR/document parsing libraries, and equivalent lightweight frameworks
- rule engines / hybrid AI approaches preferred for validation-heavy use cases
- use of LLMs or compute-intensive deep learning models only where clearly justified

Requirements:

- preference for lightweight, task-specific models over large general-purpose models
- AI usage shall not be excessively resource-intensive
- GPU dependency shall be avoided unless specifically required and justified
- human-in-the-loop controls for critical outputs
- model inference, benchmarking, anomaly detection, and NLP pipelines shall be optimized for cost and performance

6. GIS / BIM / External Integration Layer

The platform shall support secure and standards-based integration with GIS, BIM, and enterprise systems.

Preferred stack:

- standards-based APIs
- PostGIS, GeoServer, OpenLayers / Leaflet / Mapbox-compatible frameworks for GIS
- integration-ready architecture for BIM tools, Autodesk ecosystem, Bentley, Revit, IFC-compatible systems, and other engineering platforms
- API gateway and middleware for interoperability

Requirements:

- interoperable, loosely coupled integrations
- clear API documentation
- support for structured import/export and federated access patterns where needed

7. Virtualization, Containerization, and Deployment

The platform shall be deployable in virtualized and containerized environments and shall support efficient infrastructure utilization.

Preferred stack:

- Docker / Podman
- Kubernetes / OpenShift or equivalent where justified
- Linux-based deployment environment preferred
- Nginx / Apache or equivalent reverse proxy and web server stack

Requirements:

- support for on-premise deployment and future portability
- efficient CPU, memory, and storage utilization
- horizontal scalability where needed
- no unnecessary infrastructure overhead

8. Security Stack

The bidder shall adopt secure development and deployment practices in line with the RFP's security, VAPT, and OWASP requirements.

Preferred components may include:

- SSO / IAM integration
- RBAC / ABAC access controls
- secrets management tools
- SIEM/logging integration
- dependency and container vulnerability scanning tools
- encryption at rest and in transit

Requirements:

- secure-by-design architecture
- continuous patching and dependency monitoring
- support for audit logs, security monitoring, and forensic traceability

9. Monitoring, Logging, and Performance Optimization

The bidder shall provide tools and architecture for continuous monitoring and optimization.

Preferred stack:

- Prometheus / Grafana or equivalent
- ELK / OpenSearch stack or equivalent for logging
- APM and database monitoring tools, where required

Requirements:

- monitoring of application, API, database, AI workload, and infrastructure performance
- periodic performance tuning during implementation and AMC
- optimization of database usage, caching, and compute consumption

10. Proprietary Components

Where proprietary software is proposed:

- it shall be clearly identified,
- licensing model and cost shall be separately disclosed,
- support validity for the required term shall be ensured,
- and the bidder shall demonstrate that the component does not create lock-in or restrict transition, modification, or continued use by RITES.

11. General Principle

The overall technology stack shall be:

- modular,
- open and interoperable,
- responsive across devices,
- secure,
- maintainable,
- and optimized for performance, storage, database efficiency, and prudent AI resource usage.
- Technology stack shall support containerization (Docker/Kubernetes) and microservices architecture
- Database systems shall support interoperability via standard connectors (JDBC/ODBC/API-based access)
- AI/ML components shall expose inference APIs for integration with external systems
- The UI layer shall be decoupled from backend services via API-first design to enable multi-device access (web, tablet, mobile)
- No component of the system shall restrict integration with future platforms, technologies, or third-party systems introduced by RITES or Government of India

The bidder shall avoid over-engineered or resource-intensive solutions that increase cost without corresponding functional benefit.

The proposed solution shall be based on a fit-for-purpose and resource-optimized technology stack; unnecessary use of heavy frameworks, excessive proprietary dependencies, or compute-intensive AI models without clear justification shall be discouraged.

The proposed solution should minimize dependency on proprietary or licensed components and avoid vendor lock-in. Where proprietary components are used, they should be clearly identified, justified, and capable of seamless integration within an open architecture.

The proposed solution shall be based on a fit-for-purpose and resource-optimized technology stack; unnecessary use of heavy frameworks, excessive proprietary dependencies, or compute-intensive AI models without clear justification shall be discouraged

The bidder shall provide all necessary licenses, subscriptions (including API etc), and support for proprietary components (if any) for a minimum period of 3 years from go-live in the name of RITES (and contact details). Bidders are required to clearly identify such components and associated costs.

Virtualization, Performance Optimization, and Efficient Resource Utilization

1. Virtualization and Infrastructure Optimization

The platform shall be designed to operate on a **virtualized and containerized architecture**, ensuring efficient utilization of compute, storage, and network resources.

The bidder shall:

- Use **containerization technologies** (e.g., Docker, Kubernetes) or equivalent
- Ensure **horizontal scalability and auto-scaling**
- Optimize resource allocation across:
 - CPU
 - Memory
 - Storage
- Avoid over-provisioning of infrastructure
- Ensure compatibility with:
 - On-premise virtualized environments
 - Cloud-native deployments (if applicable)

2. Software Performance Optimization

The application shall be designed for **high performance and efficiency**, including:

- Lightweight application architecture
- Efficient API design (low latency, minimal payload)
- Optimized data processing pipelines
- Use of **asynchronous processing where required**
- Caching strategies for frequently accessed data
- Response time benchmarks (to be defined during SRS stage)

3. Multi-Device and Cross-Platform User Interface

The platform shall support **multi-device accessibility**, ensuring usability across:

- Desktop systems
- Laptops
- Tablets (including iPad)
- Mobile handheld devices

The bidder shall ensure:

- **Responsive web design**
- Compatibility with major browsers and operating systems (iOS, Android, Windows)
- Touch-enabled interfaces for field usage
- Optimized UI/UX for:
 - low-bandwidth environments
 - field conditions (construction sites, inspections)

4. Database and Data Architecture Optimization

The platform shall implement an **efficient and scalable data architecture**, including:

- Optimized database design (normalized + performance-tuned where required)
- Use of appropriate database types:
 - Relational (PostgreSQL / MySQL)
 - NoSQL (for unstructured data, if needed)
- Query optimization and indexing strategies
- Data partitioning and archiving mechanisms
- Efficient storage of large datasets (DPRs, GIS, documents)
- Minimization of redundant data storage

5. Efficient AI/ML Utilization

AI/ML components shall be designed to be **efficient, explainable, and resource-conscious**, ensuring:

- Use of **lightweight and optimized models** where feasible
- Avoidance of unnecessarily large or compute-intensive models
- Preference for:
 - classical ML
 - hybrid AI (rule + ML)
 - domain-driven logic
- Controlled use of GPU/accelerated resources
- Batch processing or scheduled processing for heavy workloads
- Clear justification for any high-compute AI component

The platform shall ensure:

- AI does not degrade system performance
- AI outputs are **auditable and explainable**
- Human-in-the-loop remains for critical decisions.
- The AI/ML components of the platform shall be developed and operationalised in alignment with responsible AI principles and relevant guidance from ISO/IEC 42001:2023 and ISO/IEC 23894:2023, or equivalent AI governance and risk management practices. The bidder shall support RITES in documenting and implementing AI governance controls covering human oversight, AI risk identification, risk assessment, impact assessment, model validation, explainability, auditability, bias mitigation, performance monitoring, change control, and continual improvement.
- The system shall maintain traceability between input data, validation rules, AI/ML outputs, user review, overrides, approvals, and final appraisal decisions. AI-generated outputs shall remain assistive in nature and shall be subject to review and approval by competent RITES authorities.

6. Resource Efficiency and Cost Optimization

The bidder shall ensure that:

- The platform is designed for **cost-efficient operations**
- Infrastructure sizing is **justified and optimized**
- Resource consumption is monitored and reported
- No unnecessary dependency on:
 - high-cost proprietary tools
 - excessive cloud/GPU usage
- RITES has approximately 3,000 employees; however, the proposed system is not intended to be used by all employees. The bidder shall propose a scalable system architecture and clearly indicate the assumed concurrent-user capacity, simultaneous AI-processing capacity, infrastructure sizing, storage, compute, and performance assumptions. Since actual concurrency is not presently available, the proposed solution shall be appropriately sized for initial deployment and shall support scaling based on validated usage patterns within the accepted scope and proposed system capacity, without additional cost to the Purchaser and without major architectural redesign.

7. Performance Monitoring and Optimization

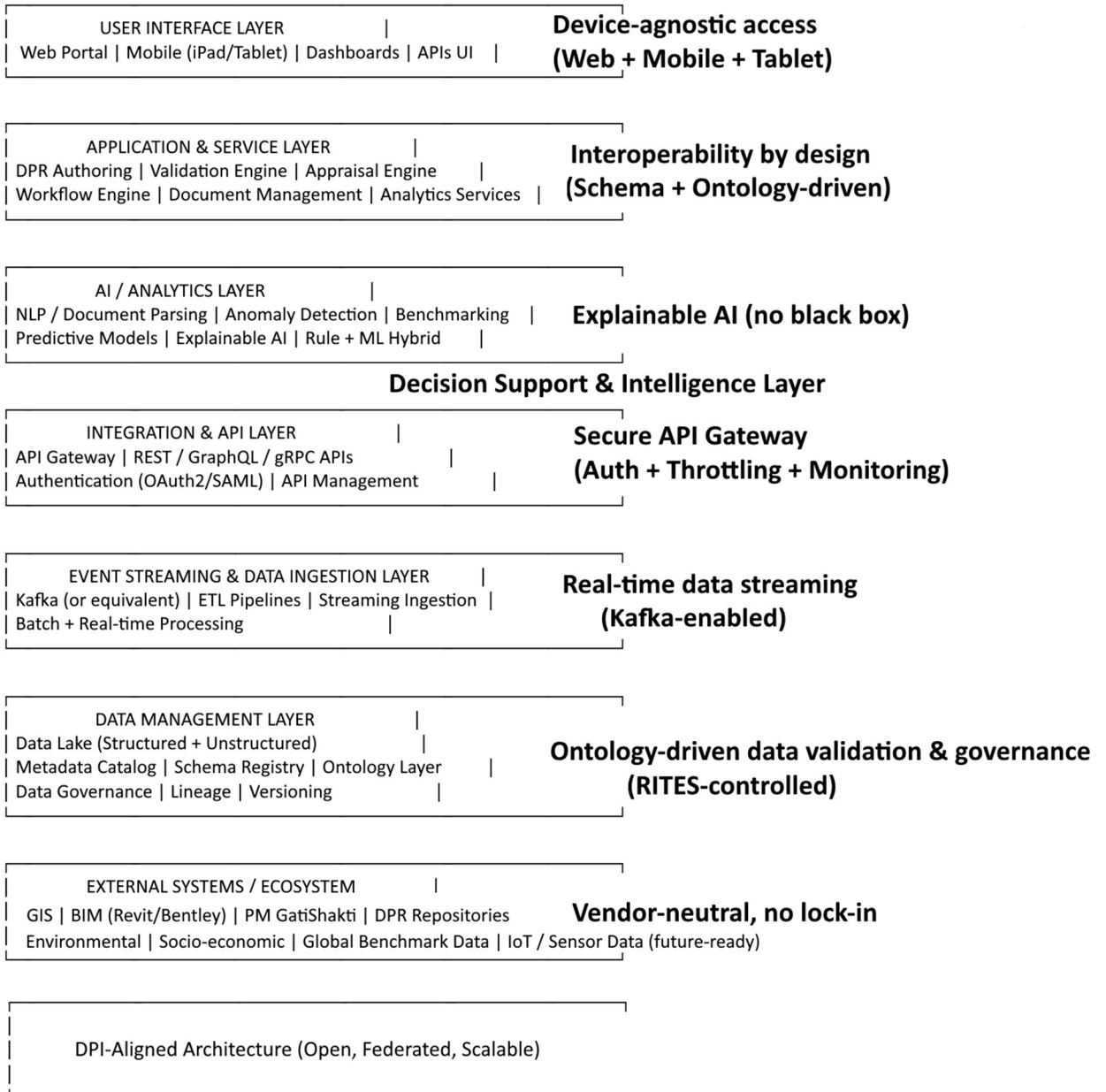
The platform shall include:

- System performance monitoring dashboards

- Logging and analytics for:
 - CPU/memory usage
 - API performance
 - database performance
- Mechanisms for continuous optimization during:
 - implementation
 - AMC phase

AI-Enabled, Interoperable DPR Appraisal Platform for Data-Driven Decision Making

Data → Intelligence → Decision Support → Insights



Annexure-13- Team Composition & Synergy

To effectively bridge the gap between IT system development and domain-specific (engineering/infrastructure) realities, a multidisciplinary team with strong synergy between AI, software engineering, and sectoral expertise is essential. The indicative team composition is as follows:

A. Leadership & Architecture

- Project Manager
- Solution Architect

B. AI, Data & Analytics

- AI/ML Engineer
- Data Scientist /Data Engineer
- NLP / Document Processing Specialist
- Data Analyst / BI & Analytics Specialist

C. Application Development

- Backend Developers
- Frontend Developers

D. Data & Knowledge Management

- Database Specialist
- Knowledge / Data Engineer
- Knowledge Library / Content Structuring Specialist

E. Domain & Appraisal Expertise

- Sector-Specific SMEs (*As per RITES Sectors Viz Railways, Highways, Ports/Marine etc*)
- Domain Expert (Multi-sector) - Urban Mobility/ Logistics Plan, Metro, Tunnels, Bridges etc

- Appraisal Logic / Rules Engine Specialist
- GIS / BIM Specialist

F. UX & Interface

- UI / UX Designer

G. DevOps & Quality Assurance

- DevOps / Infrastructure Engineer
- QA / Testing Engineers

Deployment Model & Engagement Approach:

The above roles are **indicative in nature** and shall be deployed **on a need basis** depending on project phase and requirements.

- **The Project Coordinator/Project Manager shall be deployed onsite at RITES Gurgaon (Haryana) Office throughout the project duration.**
- **Domain experts and specialized roles** to be onsite as when required by RITES.
- The bidder shall ensure **seamless coordination, knowledge transfer, and integration** between onsite and offsite teams to maintain continuity, quality, and timely delivery of the project.

Annexure-14- BANK GUARANTEE FORMAT FOR EARNEST MONEY DEPOSIT (EMD)

(On Non-Judicial Stamp Paper of Appropriate Value)

To,

RITES LTD.

1. WHEREAS RITES LTD*/RITES Ltd. Acting for and on behalf as Agent/Power of Attorney Holder of (hereinafter called 'The Employer') (which expression shall unless repugnant to the subject or context include its successors and assigns) has invited bids by way of E-tender Number-..... dated (hereinafter called 'the Tender') for (*description of the tender work*);

* Strike out which ever is not applicable

2. AND WHEREAS (*name of the bidder*) (hereinafter called 'the Bidder') has submitted a bid for the Tender to the Employer.
3. AND WHEREAS a Bank Guarantee for Rs..... (Rupees only) (*Amount of EMD*) in favour of RITES Ltd (hereinafter called the "Beneficiary") valid till (*the date of validity of this Guarantee from the date of the submission of Tender's offer*) is required to be submitted by the Bidder as EMD along with the Tender.
4. We (*Name of the bank*) having our office at (*Address of the bank*) (hereinafter called 'the Bank') hereby give this Bank Guarantee and hereby agree unequivocally and unconditionally to pay to RITES Ltd. immediately on demand and without demur to the extent of the said sum of Rs. (Rupees only) for which payment well and truly to be made to the said Beneficiary, the Bank binds itself, its successors and assigns by these presents on default of any of the conditions of the Tender as follows:

- i. if the Bidder after bid opening, but before expiry of bid validity or issue of Letter of Acceptance, whichever is earlier,

1. withdraws his tender; or

2. makes any modification in the terms and conditions of the tender which are not acceptable to the Employer
 3. impairs or derogates from the tender in any respect within the period of validity of the tender
 4. If the bidder does not accept the correction of his price during evaluation
- ii. in case any information/document which may result in the Bidder's disqualification is concealed by the Bidder or any statement/information/document furnished by the Bidder or issued by a Bank/Agency/Third Party and submitted by the Bidder, is subsequently found to be false or fraudulent or repudiated by the said Bank/Agency/Third Party.
 - iii. in the case of a successful Bidder, if the Bidder fails to furnish the Performance Guarantee within the period specified under Clause 5.8 of GCC.
5. We undertake to pay to the Beneficiary up to the above amount upon receipt of its first written demand, without the Beneficiary having to substantiate its demand, provided that in its demand the Beneficiary will note that the amount claimed by it is due to its owing to the occurrence of one or more of the conditions mentioned above, specifying the occurred condition or conditions.
 6. This guarantee will remain in force up to and any demand in respect thereof should reach the Bank not later than the above date.
 7. NOTWITHSTANDING ANYTHING CONTAINED HEREIN:
 - I. Our liability under this Bank Guarantee shall not exceed Rs. (Rupees only)
 - II. This Bank Guarantee shall be valid up to ***[Insert the date i.e. 45 days after the period of bid validity]***
 - III. It is a condition of our liability for payment of the guaranteed amount or any part thereof arising under this Bank Guarantee that we receive a valid written claim or demand for payment under this Bank Guarantee on or before failing which our liability under the guarantee will automatically cease.

Seal:

(Authorised Signatory of the Bank)

Date:

Annexure 15:- Bid Securing Declaration

Bidder exempted from submission of Bid Security are also required to submit this. (on Consultant's Letter-head)

Consultant's Name _____

[Address and Contact Details]

Consultant's Reference No. _____ Date.....

To

(insert Tender Inviting Authority (TIA) with Complete Address)

MITES Ltd.

[Complete address of the Procuring Entity]

Ref: RFP Document No. RFP No./ xxxx; Tender Title: Consultancy Services

Sir/ Madam

We, the undersigned, solemnly declare that

We understand that according to this RFP document's conditions, the Proposal must be supported by a Bid Securing Declaration in lieu of Bid Security.

We unconditionally accept the conditions of this Bid Securing Declaration . We understand we (all members individually and jointly in case of JV/C) shall stand automatically suspended from being eligible for bidding in any tender in Procuring Organisation for 2 years from the date of opening of this Proposal if we breach our obligation(s) under the tender conditions if we:

- 1) Withdraw/ amend/ impair/ derogate, in any respect, from our Proposal, within the Proposal validity; or
- 2) Being notified within the Proposal validity of the acceptance of our Proposal by the Procuring Entity:
 - (a) Refused or failed to produce the original documents for scrutiny or the required Performance Security within the stipulated time under the RFP document's conditions.
 - (b) Fail or refuse to sign the contract.

We know that this Proposal-Securing Declaration shall expire On receipt of the Procuring Entity's notification

- i) of cancellation of the entire tender process or rejection of all bids or ii) of the name of the successful bidder

- 1) If the contract is not decided - forty-five days after the expiration of the Proposal validity, any extension to it or
- 2) If the contract is not awarded to us - not later than thirty days after the conclusion of the resultant contract, or
- 3) If the contract is awarded to us - after furnishing performance security and signing of agreement.

(Signature with date)

.....

(Name and designation)

Duly authorized to sign Proposal for and on behalf of.....

[name, address, and seal of Consultant]

Dated on day of [insert date of signing]

Place.....[insert place of signing]

DA:.....

Annexure 16- Proposed Technology Stack

The bidder shall submit the proposed technology stack along with the Technical Bid, covering the following:

- Overall solution architecture and deployment model.
- Front-end, back-end, API and workflow technologies.
- AI/ML, NLP, OCR and document parsing tools/models.
- Database, data lake, search and knowledge repository technologies.
- Analytics, dashboard, benchmarking and report generation tools.
- Security framework including authentication, authorization, encryption, logging and audit trail.
- Integration approach with internal/external systems such as GIS, PM GatiShakti, DMS, ERP, etc., wherever applicable.
- Hosting and infrastructure requirements including server, storage, GPU/CPU and network sizing assumptions.
- Details of open-source/proprietary/licensed/subscription-based components and related recurring costs, if any.
- Monitoring, backup, recovery, patching and maintenance tools.
- Justification for selection of the proposed technology stack in terms of scalability, security, performance, interoperability and maintainability.

The proposed technology stack shall form part of the technical evaluation. Any major change in the approved technology stack during execution shall require prior written approval of RITES.

Form T-10/1:

The RITES authority competent to sign the contract agreement, will sign this Annexure before uploading the tender

To

All Bidders/Consultants for the Subject Tender

.....,

Subject: NIT/RFPL No.....: **RITES/ITES/MAR/2026**.....for the services of **Design, Development and Implementation Of Artificial Intelligence (AI) Based Detailed Project Report Appraisal Platform**

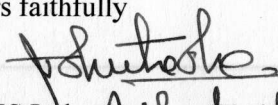
Dear Sir,

It is hereby declared that RITES LTD.,(The Principal / Employer) is committed to follow the principles of transparency, equity and competitiveness in public procurement.

The subject Notice Inviting Tender (NIT) is an invitation to offer made on the condition that the Consultant will sign the integrity Agreement, which is an integral part of tender/bid documents, failing which the tenderer /consultant will stand disqualified from the tendering process and the bid of the consultant would be summarily rejected.

This declaration shall form part and parcel of the Integrity Agreement and signing of the same shall be deemed as acceptance and signing of the Integrity Agreement on behalf of the RITES Ltd.

Yours faithfully


RITES Ltd *Ashutosh Namdev, GGM/ITS*

(Name, Signature and Designation of the Authority *competent to sign contract agreement on behalf of RITES*)

.....

Note- The RITES authority competent to sign the contract agreement, will sign this Annexure before uploading the tender.

Form T-10/2:

The consultant will download this document and affix his signatures and upload such signed document along with Integrity pact

To,

.....
.....

(Fill the Designation and Address of the Authority competent to sign contract agreement on behalf of RITES)

Name of Service: *(Fill in Name of the Service)*

Dear Sir,

I/We acknowledge that RITES Ltd. is committed to follow the principles thereof as enumerated in the Integrity Agreement enclosed with the tender/bid document.

I/We agree that the Notice Inviting Tender (NIT) is an invitation to offer made on the condition that I/We will sign the enclosed integrity Agreement, which is an integral part of tender documents, failing which I/We will stand disqualified from the tendering process. I/We acknowledge that SUBMISSION OF THE BID SHALL BE REGARDED AS AN UNCONDITIONAL AND ABSOLUTE ACCEPTANCE of this condition of the NIT.

I/We confirm acceptance and compliance with the Integrity Agreement in letter and spirit and further agree that execution of the said Integrity Agreement shall be separate and distinct from the main contract, which will come into existence when tender/ bid is finally accepted by RITES Ltd. and/or Client/*Strike out whichever is not applicable*). I/We acknowledge and accept the duration of the Integrity Agreement, which shall be in the line with Section 9 of the enclosed Integrity Agreement at Form T-10/3.

I/We acknowledge that in the event of my/our failure to sign and accept the Integrity Agreement, while submitting the tender/bid, RITES Ltd. and/or Client (*Strike out whichever is not applicable*), shall have unqualified, absolute and unfettered right to disqualify the tenderer / consultant and reject the tender/ bid in accordance with terms and conditions of the tender/bid.

Yours faithfully

**(Duly authorized signatory of the consultant/bidder)
(Official Seal)**

Place:.....

Date:

Witness 1:

(Name & Address).....
.....

Witness 2

(Name & Address).....
.....

Date:

*Note-1 The bidder/consultant will download this document and affix his signatures and upload such signed document, as explained in the tender document under para 6 (2) of RFPL. In case of **Failure to submit the Integrity Pact duly signed and witnessed, along with the bid, the bid is likely to be rejected.***

*Note-2 In case of a Joint Venture/Consortium, each member of JV/Consortium will submit this document duly signed and witnessed along with the bid. All authorized signatories of the members of JV/Consortium, can either sign on one document jointly or can sign on separate documents. In case of **Failure to submit the Integrity Pact duly signed and witnessed, along with the bid, the bid is likely to be rejected.***

FORM T-10/3

INTEGRITY PACT

(If stipulated in TIS, Ref Clause 6 (2) of RFPL)

(To be signed on Plain paper)

(To be submitted as part of Technical Proposal)

INTEGRITY PACT

Between

.....**MITES LTD.**

(Name of Principal/Employer)

(Hereinafter called the "Principal/Employer" and which expression shall unless repugnant to the meaning or context hereof including its successors and permitted assigns)

AND.....

(Name of the Bidder/Consultant)

(Hereinafter referred to as "The Bidder/Contractor" and which expression shall unless repugnant to the meaning or context hereof include its successors and permitted assigns)

PREAMBLE

'The Principal/Employer' intends to award, under laid down organizational procedures, contract/s for **_Design, Development and Implementation Of Artificial Intelligence (AI) Based Detailed Project Report Appraisal Platform**

, 'The Principal/Employer' values full compliance with all relevant laws of the land, rules, regulations, economic use of resources and of fairness/ transparency in its relations with its Bidder(s) and/ or Consultant(s).

In order to achieve these goals, the Principal/Employer shall appoint Independent External Monitors (IEMs) who shall monitor the tender Process and the execution of the contract for compliance with the principles mentioned above.

Section 1 - Commitments of the 'The Principal/Employer'

(1) 'The Principal/Employer' commits itself to take all measures necessary to prevent corruption and to observe the following principles: -

a. No employee of the Principal/Employer, personally or through family members, will, in connection with the tender or for the execution of a contract, demand, take a promise for or accept, for self or third person, any material or immaterial benefit which the person is not legally entitled to.

- b. The Principal/Employer will, during the Procurement Process, treat all bidders/Consultant(s) with equity and reason. The Principal/Employer will, in particular, before and during the Procurement Process, provide to all bidders/Consultant(s) the same information and will not provide to any bidders/Consultant(s) confidential/ additional information through which the bidders/Consultant(s) could obtain an advantage in relation to the tender Process or the contract execution.
- c. The Principal/Employer will exclude from the process all known prejudiced persons.
- (2) If the Principal/Employer obtains information on the conduct of any of its employees, which is a criminal offence under the IPC (Indian Penal Code)/ PC (Prevention of Corruption) Act, or if there be a substantive suspicion in this regard, the Principal/Employer will inform its Chief Vigilance Officer and in addition, can initiate disciplinary actions.

Section 2 - Commitments of the 'Bidder/ Consultant'

- (1) The 'Bidder/Consultant' commit himself to taking all measures necessary to prevent corruption. He commits himself to observe the following principles during participation in the tender Process and during the contract execution.
 - a. The 'Bidder/ Consultant' will not, directly or through any other person or firm, offer, promise, or give to any of the Principal/Employer's employees involved in the tender Process or the execution of the contract or to any third person any material or other benefit which he is not legally entitled to, in order to obtain in exchange any advantage of any kind whatsoever during the tender Process or during the execution of the contract.
 - b. The 'Bidder/Consultant' will not enter with other Bidders/Consultants into any undisclosed agreement or understanding, whether formal or informal. This applies in particular to prices, specifications, certifications, subsidiary contracts, submission or non-submission of bids or any other actions to restrict competitiveness or to introduce cartelisation in the bidding Process.
 - c. The 'Bidder/Consultant' will not commit any offence under the relevant IPC/ PC Act; further, the 'Bidder/ Consultant' will not use improperly, for purposes of competition or personal gain, or pass on to others, any information or document provided by the Principal/Employer as part of the business relationship, regarding plans, technical proposals, and business details, including information contained or transmitted electronically.
 - d. The 'Bidder/Consultant' will, when presenting his bid, disclose any and all payments he has made, is committed to, or intends to make to agents, brokers, or any other intermediaries in connection with the award of the contract.
 - e. Foreign bidders/Consultants shall disclose the name and address of agents and representatives in India .
 - f. Indian Bidders/Consultants shall disclose their foreign principals or associates.
- (2) The 'Bidder/Consultant' will not instigate third persons to commit offences outlined above or be an accessory to such offences.

Section 3 - Disqualification from Procurement Process and exclusion from future contracts

If the 'Bidder/Consultant', before award or during execution, has committed a transgression through a violation of Section 2, above or in any other form such as to put their reliability or credibility in question,

the Principal/Employer is entitled to disqualify the 'Bidder/Consultant' from the tender Process or terminate/determine the contract, if already executed, as per Clause 12 of GCC or take action as per the procedure mentioned in the "Guidelines on Banning of Business Dealings" placed at Annexure-T-10A.

Section 4 - Compensation for Damages

(1) If the Principal/Employer has disqualified in terms of the provisions in Section 3, the 'Bidder/Consultant' from the tender Process prior to the award of contract, the Principal/Employer is entitled to demand and recover the damages equivalent to Earnest Money Deposit/ Bid Security.

(2) If the Principal/Employer has terminated the contract during execution in terms of the provisions under Section 3, the Principal/Employer shall be entitled to demand & recover from the Bidder/Consultant the damages equivalent to Earnest Money Deposit, Security Deposits already recovered and Performance Guarantee which shall be absolutely at the disposal of the Principal/Employer.

Section 5 - Previous transgression

(1) Bidder/Consultant to disclose any transgressions with any other public/government organization that may impinge on the anti-corruption principle. The date of such transgression, for the purpose of disclosure by the bidders/consultants in this regard, would be the date on which cognizance of the said transgression was taken by the competent authority. The period for which such transgression(s) is/are to be reported by the bidders/consultants shall be the last three years to be reckoned from date of bid submission. The transgression(s), for which cognizance was taken even before the said period of three years, but are pending conclusion, shall also be reported by the bidders/consultants.

(2) If the Bidder/Consultant makes an incorrect statement on this subject, he can be disqualified from the tender Process, or action can be taken as per the procedure mentioned in "Guidelines on Banning of Business Dealings".

Section 6 - Equal treatment of all Bidders/Consultants/Sub-contractors

(1) The Bidder/Consultant undertakes to demand from all partners/sub-contractors (if permitted under the conditions/clauses of the contract) a commitment to act in conformity with this Integrity Pact and to submit it to the Principal/Employer before signing the contract.

(2) The Bidder/Consultant confirms that any violation by any of his partners/subcontractors to act in conformity with the provisions of this Integrity Pact can be construed as a violation by the Bidder/Consultant himself, leading to possible Termination of Contract in terms of Section 4.

(3) The Principal/Employer will disqualify from the tender Process all bidders/consultants who do not sign this Pact or violate its provisions.

Section 7 - Criminal charges against violating Bidders(s)/ Consultant(s)/ Subcontractor(s)

If the Principal/Employer obtains knowledge of the conduct of a Bidder, Consultant or Partners/ Subcontractor, or of an employee or a representative or an associate of a Bidder, Consultant or Subcontractor, which constitutes corruption, or if the Principal/Employer has substantive suspicion in this regard, the Principal/Employer will inform the same to the Chief Vigilance Officer.

Section 8 - Independent External Monitor

(1) The Principal/Employer shall appoint competent and credible Independent External Monitor for this Pact. The task of the Monitor is to review independently and objectively whether and to what extent the parties comply with the obligations under this agreement on receipt of any complaint by them from the bidder(s)/consultant(s). The IEMs shall examine all the representations/grievances/complaints received by them from the bidders/consultants or their authorized representative related to any discrimination on account of lack of fair play in modes of procurement and bidding systems, tendering method, eligibility conditions, bid evaluation criteria, commercial terms & conditions, choice of technology/specifications etc.

For ensuring the desired transparency and objectivity in dealing with the complaints arising out of the tendering process, the matter should be examined as far as possible by the full panel of IEMs jointly or by the sole IEM as the case may be, who would look into the records, conduct an examination, and submit their joint recommendations to the Management. In case the full panel is not available due to some unavoidable reasons, the available IEM(s) will conduct examination of the complaints. Consent of the IEM(s), who may not be available, shall be taken on record.

(2) The Monitor is not subject to instructions by the representatives of the parties and will perform his functions neutrally and independently. He will report to the CMD/RITES Limited.

(3) The Bidder/ Consultant accepts that the Monitor has the right to access, without restriction, to all Project documentation of the Principal/Employer, including that provided by the bidder/Consultant. The bidder/Consultant will also grant the Monitor, upon his request and demonstration of a valid interest, unrestricted and unconditional access to his project documentation. The same is applicable to Partners/Sub-contractors. The Monitor is under contractual obligation to treat the information and documents of the Bidder/Consultants/Partners/Sub-Contractor with confidentiality.

(4) The Principal/Employer will provide to the Monitor sufficient information about all meetings among the parties related to the Project, provided such meetings could have an impact on the contractual relations between the Principal/Employer and the Consultant. The parties offer the Monitor the option to participate in such meetings.

(5) As soon as the Monitor notices, or has reason to believe that violation of the agreement by the Principal/Employer or the Bidder/Consultant, has taken place, he will request the party concerned to discontinue or take corrective action, or to take any other relevant action. The monitor can, in this regard, submit non-binding recommendations. Beyond this, the Monitor has no right to demand from the parties that they act in a specific manner, refrain from action, or tolerate action. The role of the Monitor will be advisory and his advice will be non-binding on the Organization. However, as IEMs are invariably persons with rich experience who have retired as senior functionaries of the government, their advice would help in proper implementation of the IP.

(6) The Monitor will submit a written report to the CMD/RITES Ltd. at the earliest from the date of reference or intimation to him by the Principal/Employer and, should the occasion arise, submit proposals for correcting problematic situations.

(7) The word 'Monitor' would include both singular and plural.

Section 9 - Pact Duration

This pact begins when both parties have legally signed it. It expires for the Consultant 12 months after the last payment under the contract and for all other consultants 6 month after the Contract has been awarded.

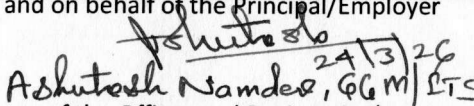
If any claim is made/lodged during this time the same shall be binding and continue to be valid despite the lapse of this pact specified above, unless it is discharged/determined by CMD/RITES Ltd.

Section 10 - Other provisions

- (1) This agreement is subject to Indian Law. The place of performance and jurisdiction shall be as stated in the Contract Agreement
- (2) Changes and supplements, as well as termination notices, need to be made in writing.
- (3) If the Bidder/Contractor is a Joint Venture/Consortium, this agreement must be signed by legally authorized signatories of each Member of JV/Consortium. The Memorandum of Understanding /Joint Venture Agreement will incorporate a provision to the effect that all Members of the Joint Venture/Consortium will comply with the provisions in the Integrity Pact. Any violation of Section 2 above by any of the Members will be construed as a violation by the Joint Venture/Consortium leading to possible Termination of Contract in terms of Section 3.
- (4) Should one or several provisions of this agreement turn out to be invalid, the remainder of this agreement remains valid. In this case, the parties shall strive to come to an agreement to their original intentions.
- (5) A person signing the IP shall not approach courts while representing the matters to IEMs and he/she will await their decision in the matter.
- (6) In case of sub-contracting, the principal consultant shall take the responsibility of the adoption of IP by the sub-contractor. It is to be ensured that all sub-contractors also sign the IP. In case of sub-contractors, the IP will be a tri-partite arrangement to be signed by the Organization, the contractor, and the sub-contractor.
- (7) In the event of any dispute between Principal/ Employer and the consultant/bidder, in case, both the parties are agreeable, they may try to settle dispute through mediation before the panel of IEMs in a time bound manner. However, not more than five meetings shall be held for a particular dispute resolution. The fees/expenses on dispute resolution shall be equally shared by both the parties.

In case, the dispute remains unresolved even after mediation by the panel of IEMs, the Principal/ Employer and consultant/bidder may take further action as per the terms & conditions of the contract.
- (8) The Bidder/Consultant undertakes (as part of Fall Clause) that they have not and will not sell the same material/equipment at prices lower than the bid price.

For and on behalf of the Principal/Employer


Ashutosh Namdeo, GGM/ITS
(Name of the Officer and Designation)

(Office Seal)

For and on behalf of 'Bidder/ Consultant.'

(Office Seal)

Place Gurgaon (Haryana)
Date 24/03/2026

Witness 1:

(Name & Address)

Witness 2:

(Name & Address)

Note-

1. RITES authority competent to sign the contract agreement, will sign this Annexure before uploading the tender. The bidder/consultant will download this pre-signed document and affix his signatures and those of witnesses and upload such signed document, as explained under para 6 (2) of NIT. In case of Failure to submit the Integrity Pact duly signed and witnessed, along with the Bid, the Bid is likely to be rejected.

2. In case of a Joint Venture/Consortium, each member of Joint Venture/Consortium will sign the Integrity Pact and submit the same with the Bid/Tender. *All authorized signatories of the members of JV/Consortium, can either sign on one document jointly or can sign on separate documents.* Failure to submit the Integrity Pact, as prescribed herein, duly signed and witnessed, along with the Bid, may lead to rejection of the Bid.